# Who Benefits from Remote Schooling? Self-Selection and Match Effects*

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**Abstract**

This paper studies the distributional effects of remote learning by combining preference data from a conjoint survey with administrative records. Our analysis uses a selection-on-observables framework that relies on choice experiments to characterize selection, allowing us to recover average and heterogeneous effects. We validate the approach using variation from school choice lotteries. On average, remote learning reduced test scores, but students with the highest estimated demand experienced gains. Bullying concerns strongly predict demand, and remote learning broadly improved bullying outcomes. For most enrollees, these improvements are a compensating differential for learning losses.

**JEL Classification:** I21, I24.
**Keywords**: Remote learning, COVID-19, school match effects, self-selection, school choice, virtual schooling

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# 1 Introduction

Classic theories of education markets predict that school choice can improve how students are allocated to schools (Hoxby, 2003). By improving match quality, choice policies hold the potential to engineer improved student outcomes. Yet, existing research in the U.S. fails to find meaningful evidence that student-school match effects exist at all (Abdulkadiroğlu et al., 2020). Even substantial changes to the choice environment can fail to produce meaningful improvements in student-school match quality (Campos and Kearns, 2024). Imperfect information is a leading hypothesis for explaining the gap between theory and data. Families may not know their match quality when choosing schools and only learn gradually through trial and error (Arcidiacono et al., 2016).

The coronavirus disease (COVID-19) pandemic provides a unique setting to explore allocative efficiency after families have more fully assessed their relative suitability for a particular schooling option: remote learning. Although mounting evidence shows that remote learning contributed to sizable learning losses during the pandemic (Goldhaber et al., 2022, Jack et al., 2022, Singh et al., 2022), many school districts now offer permanent remote options to satisfy ongoing parental demand (Musaddiq et al., 2022). Recent data from the National Center for Education Statistics (NCES) show that enrollment in exclusively virtual schools spiked at the onset of the pandemic and, after schools reopened, has continued to rise: as of 2025, enrollment remains roughly 50 percent above what the pre-pandemic trend would have predicted. Moreover, roughly 1.6 percent of all U.S. public school students are now enrolled in exclusively virtual schools and, at this moment in time, all of the 40 largest districts in the country offer a remote option or school.[1]

This paper studies the post-pandemic demand for remote learning and provides new evidence of its causal effect on student learning and non-cognitive outcomes. We focus on the second-largest school district in the United States, the Los Angeles Unified School District (LAUSD). At the onset of the pandemic, every student in the district had to participate in virtual learning and experienced a cycle of in-person and remote experiences in the following year. This unusual experience gave families direct exposure to remote instruction over an extended period, informing their subsequent schooling choices when the district made remote schooling a standing opt-in choice.[2] Demand for this option remained substantial: after the district returned to predominantly in-person instruction in 2021-2022, approximately 14,000 students, or 4.5 percent of district enrollment, chose to remain in remote schooling.

Our analysis of the impact of remote learning relies on a novel survey that we designed to learn about family experiences and preferences. The first component of the survey was descriptive and asked parents about their experiences with remote learning during the pandemic and their demand for remote options going forward. The second was a series of medium-stakes hypothetical choices to experimentally identify families' preferences for the remote option. Following previous research using choice experiments to understand preferences for workplace

[1] Many school districts offer remote options for medically fragile students. Excluding these school districts, 31 of the 40 largest school districts offer a remote option or school.

[2] The timeline of remote to in-person learning in L.A. is similar to that of other school districts across the U.S. (Jack et al., 2022), with perhaps a longer duration relative to other school districts in Southern California. Our setting provides a natural context for studying ongoing selection into remote learning.

characteristics and flexibility (Mas and Pallais, 2017, Wiswall and Zafar, 2018), we use these hypothetical choices to identify how families trade off academic quality, travel time, and remote offerings while holding remaining school attributes fixed.[3] The choice experiments allow us to characterize the preference heterogeneity that governs selection into remote schooling. As we detail below, we then use these experimentally elicited preferences in a new approach that accounts for selection and recovers both average and heterogeneous treatment effects.

The results from our descriptive survey point to substantial heterogeneity in both remote learning experiences and its perceived academic benefits. Although most respondents report having a negative experience with remote learning during the pandemic, one-third want expanded remote offerings, and a quarter expect to enroll their children in remote learning in the future. Moreover, 20 percent feel their children excelled in remote learning relative to traditional, in-person instruction. Existing evidence suggests several reasons why families may differ in their demand for remote schooling. Bacher-Hicks et al. (2022) document substantial declines in bullying during the remote-learning period, suggesting that safety and social environment may be important for some families, while Aucejo et al. (2020) find substantial heterogeneity in students' perceived remote-learning experiences in higher education, pointing to differences in academic fit. These findings suggest there is important scope for permanent, post-pandemic remote offerings to generate improvements in match quality. However, the results from our survey also reflect parental perceptions and stated preferences rather than rigorous estimates of demand. Moreover, the survey evidence does not provide an estimate of the causal impact of remote instruction.

To quantify household demand and identify heterogeneous causal effects, we develop a conceptual framework built on a discrete choice model of schooling enrollment decisions. The model summarizes each household's demand for remote learning using a scalar index that captures how much the household values remote enrollment relative to in-person schooling. This index serves two empirical goals. First, it allows us to address selection into remote learning using a selection-on-observables approach in the spirit of Rosenbaum and Rubin (1983) and Dale and Krueger (2002): under the framework's identifying assumption, conditioning on the index eliminates selection bias. An important testable implication is that conditioning on the index should also balance baseline observables. We verify empirically that it balances baseline achievement and non-cognitive outcomes (i.e., bullying and grit, a socio-emotional skill capturing perseverance and self-regulation). Second, the index also provides a measure of the intensity of household demand, which lets us examine how the causal effect of remote enrollment varies with demand—separating selection on levels from selection on gains in the spirit of Roy (1951).

The hypothetical choice component of our survey allows us to experimentally identify household demand. Importantly, these estimates constitute an empirical counterpart to the demand index in our conceptual framework, which we use to correct for selection and recover causal effects. Specifically, our design identifies preferences over three attributes—academic achievement, distance, and an omnibus preference for remote learning—that together characterize each family's relative valuation of remote versus in-person enrollment. Consistent with previous liter-

[3]Prior work finds that preference estimates from similar experiments contain a high degree of external validity (Wiswall and Zafar, 2018).

ature spanning several countries, we estimate that families have tastes for academic quality and distaste for distance (Abdulkadiroğlu et al., 2020, Ainsworth et al., 2023, Allende, 2019, Beuermann et al., 2022, Burgess et al., 2015, Campos and Kearns, 2024, Neilson, 2021). Reassuringly, we do *not* find a distaste for remote learning among families currently enrolled in remote offerings or among those who indicated they anticipate doing so in the future. This suggests that the survey data accurately capture the underlying preferences that drive parental decision making.[4] Baseline bullying outcomes also strongly predict increased demand for remote learning, with families of more heavily bullied students requiring substantially less compensation to switch to remote schooling.[5] Taken together, our survey analysis provides the first rigorous evidence of the diversity of preferences for remote learning in the post-pandemic landscape.

With the selection index in hand, we turn to estimates of the impact of remote learning. For the average student, remote learning has a large impact on both cognitive and non-cognitive outcomes. For example, remote-learning reduces reading and math scores by $-0.17\sigma$ and $-0.18\sigma$, respectively. These estimates differ from those of conventional value-added models that rely on lagged achievement and standard covariate controls and which generate estimates ranging between $-0.23\sigma$ and $-0.24\sigma$. Formal tests of equality between the two sets of estimates reject at the one percent level across our main outcomes, which highlights the importance of our preference estimates for recovering causal effects. For our index measure of bullying, we find that remote learning substantially improves outcomes: the average treatment effect is $0.14\sigma$. Overall, the results suggest that the mean improvement in bullying may function as a compensating differential for the negative learning effects, offering a potential explanation for why parental demand for remote instruction remains high despite mounting evidence of learning losses.

We next examine how these average impacts mask heterogeneity across families with different demand for remote learning. We find negative selection on achievement levels, indicating that students with high demand for remote learning perform poorly regardless of the type of school in which they enroll. In addition, there is positive selection on achievement gains, suggesting families choose remote learning, at least partly, using factors that correlate with their child's suitability for remote instruction. This has policy implications for understanding the efficiency of ongoing efforts to expand remote offerings. Taking our estimates at face value implies that students at the top of the distribution of demand for remote learning (e.g., 95th percentile and above) experienced improvements of at least $0.045\sigma$ in ELA and $0.035\sigma$ in math. For bullying, we find negative selection on levels and modest selection on gains that ultimately point to an across-the-board improvement in bullying outcomes for students who select into remote learning. The improvements are present for both in-person and online bullying, revealing that schooling environment affects students' well-being both at school and at home.

In sum, the heterogeneity analysis underscores the tradeoff families face when choosing between remote and in-person modalities. Some families appear willing to forego short-run achievement gains in exchange for guaranteed improvements in bullying-related outcomes. Families with the largest tastes for remote learning experience improvements along both margins.

[4]Empirically, we verify this by showing that the preferences uncovered by the survey imply choices that are forecast unbiased for the actual, real-world enrollment decisions subsequently made by parents.

[5]We measure bullying in a separate annual school experience survey linked to our data. See Section 3 for more detail.

These findings suggest that prior estimates of the impact of remote learning during the pandemic may not accurately predict the future effects that expanded remote offerings could have on the students who opt-in. These findings also show that bullying-related considerations are an important, and previously unexplored, factor that governs family decision making.

We conclude our analysis by providing additional evidence that supports a causal interpretation of our results. The ability to balance baseline characteristics using our propensity-score based approach constitutes a key validity check on the credibility of our main results. As a supplement, we provide further validation by studying a sample of students who apply to oversubscribed school choice programs with remote schooling as a fallback option. Our test relies on lottery-based admission offers as an instrument to examine whether outcomes predicted from our propensity-score model of achievement accurately forecast observed variation in test scores. We fail to reject that the estimates derived from the choice experiment are forecast unbiased for the causal effects of remote learning implied by the lottery variation. This is reassuring evidence that the choice experiments adequately characterize selection into learning modality. Additional exercises demonstrate the robustness of our empirical methods to alternative modeling choices.

**Contribution and Related Literature:** This paper contributes to four literatures. First, we contribute to a nascent but growing literature on the impact of remote or virtual learning by exploring selection and treatment effects in a post-pandemic environment. Evidence from higher education, where online instruction has been studied experimentally, finds that virtual formats reduce learning and persistence relative to in-person instruction, with losses concentrated among academically weaker students (Alpert et al., 2016, Bettinger et al., 2017, Figlio et al., 2013, Kofoed et al., 2024). In the K-12 setting in the United States, Bueno (2020) finds substantial negative effects of remote learning in the pre-pandemic era but also documents negative trends before the switch to remote. International evidence from prior to the pandemic is more positive, with a number of programs in developing countries yielding promising results (Bianchi et al., 2022, Borghesan and Vasey, 2024, Fabregas and Navarro-Sola, 2025, Johnston and Ksoll, 2022). More recent evidence estimates remote-learning effects during the pandemic, reaching a consensus that the pandemic caused sizable learning loss (Goldhaber et al., 2022, Jack et al., 2022, Singh et al., 2022). Jack et al. (2022) and Goldhaber et al. (2022) emphasize that remote-learning offerings exacerbated learning loss relative to in-person schools and districts. Our paper looks ahead by considering the post-pandemic landscape in the United States and explores the implications of expanded remote offerings on the selected group of families freely opting into remote schooling. To that end, we provide evidence about how the expansion and persistence of remote learning can affect educational inequality and efficiency going forward.

Second, we contribute to the literature on match effects in the education system by documenting evidence of selection on achievement gains with respect to remote offerings. The notion of academic mismatch has received considerable attention in the related affirmative action literature, with some evidence pointing to potential efficiency losses (Arcidiacono et al., 2016, Dillon and Smith, 2020) and more recent evidence pointing to the opposite (Bleemer, 2021, 2022, Otero et al., 2021). Student-school match quality has been more elusive in the K-12 space, with some evidence suggesting the importance of match quality based on observables (Bau, 2022, Bruhn,

2019) and some suggesting the contrary (Campos and Kearns, 2024). Other papers focus on match effects after accounting for preferences derived from observational choices and tend to find weak evidence of match quality (Abdulkadiroğlu et al., 2020, Mountjoy and Hickman, 2020). We complement this literature by using an experimentally derived, survey-based measure of preferences. This allows us to directly link parental demand to down-stream treatment effects and thereby assess the empirical relevance of match quality in this setting.

Third, we add to the literature on multi-dimensional learning by being the first to explore improvements in social-emotional well-being as a compensating differential (Beuermann et al., 2022, Campos and Kearns, 2024, Jackson, 2018, Jackson et al., 2020, Rose et al., 2022). Papers in the literature on multi-dimensional learning tend to find that schools and teachers impact both cognitive and non-cognitive domains. We find that learning modality, implicitly a schooling choice, affects achievement, bullying, and measures of student-level grit. Our holistic approach to evaluating the effects of remote learning demonstrates that students sort along these various dimensions in anticipation of gains, evidence consistent with multivariate Roy-style selection.

Fourth, this paper relates to the literature linking choice models to treatment effect estimation by exploring the potential for experimentally derived preferences to account for selection. Classic work in this space links choices to outcomes via an observed decision (Heckman, 1979, Heckman et al., 2006), while more recent advances leverage information on rank-ordered lists to account for selection bias (Abdulkadiroğlu et al., 2020, Einav et al., 2022, Otero et al., 2021). Our approach departs from this literature by using preferences derived from choice experiments as a direct measure of the selection-relevant heterogeneity. This extends canonical work in economics that has used hypothetical choice surveys to learn about preferences for workplace characteristics and flexibility (Mas and Pallais, 2017, Wiswall and Zafar, 2018). In that sense, we create an avenue for future work by bridging these two seemingly disconnected literatures and providing an empirical tool for estimating causal effects that we validate externally using school lotteries.

# 2 Background

The pandemic compelled all families to at least temporarily adopt remote schooling. Existing studies have shown that the disruption in learning modality has produced long-lasting changes to enrollment trends with increases in private and homeschooling (Bacher-Hicks et al., 2023, Musaddiq et al., 2022). A simultaneous and less-documented change in enrollment relates to remote schooling. In this section, we discuss national remote schooling trends, then zoom in on Los Angeles, and conclude with a discussion of remote schooling from an international perspective underscoring the broader relevance of remote schooling.

## 2.1 Remote Schooling from a National Perspective

The pandemic allows us to separate remote schooling into three distinct periods. The first corresponds to the pre-pandemic years, which have been the subject of numerous studies. These studies focus on the effectiveness of virtual schools, with nearly all finding negative selection into remote schooling and evidence of negative causal effects (Bueno, 2020, Cordes, 2023, Erickson

and Scriber, 2023, Kingsbury et al., 2022, Paul and Wolf, 2020). The second period corresponds to the pandemic years when families were compelled to adopt virtual schooling over varying durations. The research on this period emphasizes the learning losses induced by extended periods of remote instruction (Goldhaber et al., 2022, Jack et al., 2022). Although this period led to substantial learning loss with dire implications, it also compelled families to learn about their relative suitability for a learning modality they may not have otherwise tried. The third period is the post-pandemic era, when remote schooling again became a choice rather than a requirement. The information families gained from their prior compulsory exposure to remote instruction may partly explain the recent increase in virtual schooling.

To put this post-pandemic growth in context, remote enrollment was already on the rise in the years leading into the pandemic, increasing by roughly 51 percent between 2015 and 2019. Yet, the pandemic accelerated this growth substantially. Figure 1 shows that the number of students enrolled in exclusively virtual schools has more than doubled relative to 2015, with a clear trend break in 2020. As a result, the most recent enrollment numbers are nearly 50% larger than what would be expected based on the pre-pandemic trend, effectively accounting for a substantial share of the growth in this sector over the relevant time period. Overall, roughly 1.6 percent of public school enrollment is in exclusively virtual options.

Appendix Figure A.1 shows that the national trends mask substantial heterogeneity with remote enrollment shares exceeding 4 percent in 2025 in the top states. Alaska, Oklahoma, and Oregon have the largest reported remote schooling shares at 5.5, 5.1, and 4.5 percent, respectively. The states that grew the most, however, are largely distinct from those with the highest current levels: Oklahoma and Oregon led with increases of roughly 2.6 and 2.0 percentage points between 2019 and 2025, followed by Alabama, Pennsylvania, Utah, Arizona, and Arkansas, each rising between 1.3 and 1.8 points (Appendix Figure A.3). These magnitudes are much larger than recent documented increases in homeschooling rates around the United States, which appear to be less correlated with changes in remote schooling trends (see Appendix Figure A.4 and Appendix Figure A.5).

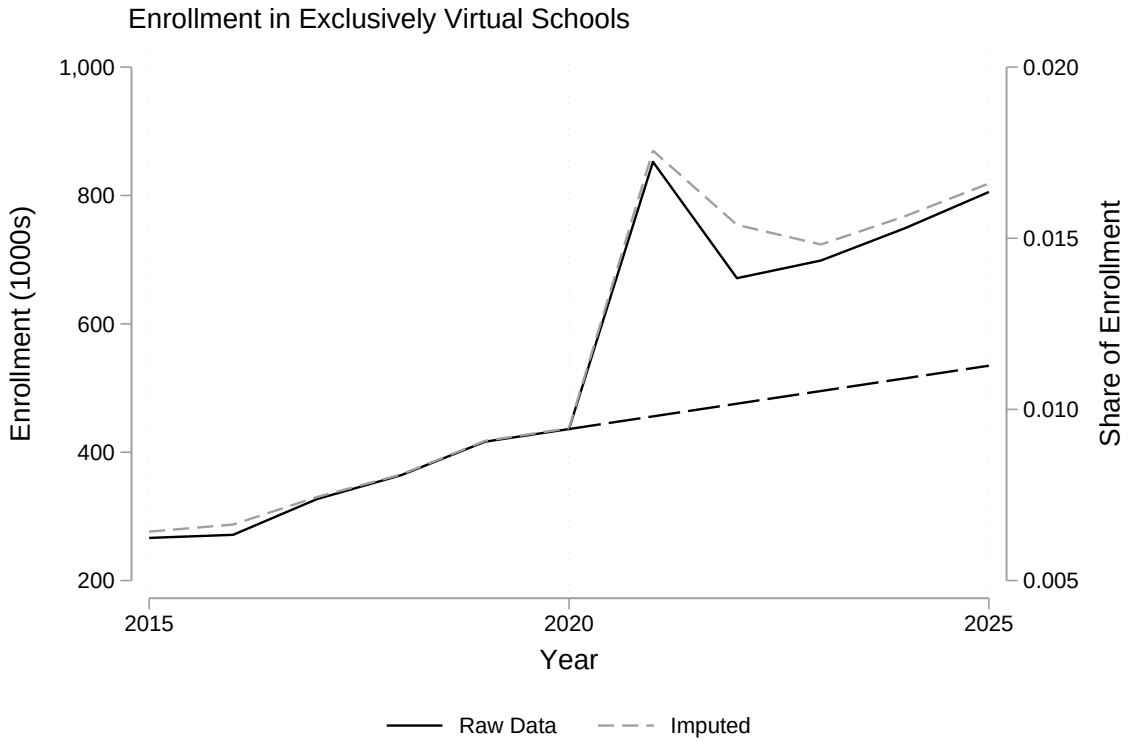
In California, the setting of our study, the data show that 2.5 percent of students in the public sector enrolled in remote schooling in 2025. California's remote schooling enrollment share has increased 100 percent from 2019 to 2025 and by 5.1 percent from 2024 to 2025, evidence that the sector continues to grow at a faster pace relative to the pre-pandemic trend. In fact, this high rate of enrollment places California as the state with the 13th largest reported virtual schooling share. Families in Los Angeles have expressed and continue to express persistent demand for remote learning above state-level averages.

### 2.2 Remote Schooling in Los Angeles

As in most U.S. school districts at the onset of the pandemic, the LAUSD closed its schools and transitioned to remote learning on March 16, 2020. To buffer the shock, the district took swift action by creating online videos, coordinating meal distribution, distributing laptops and tablets, and using private donations to provide broadband access and equipment for students. Students remained at home for the rest of the academic year.

The following academic year (2020–2021) started virtually, with a schedule that included

Figure 1: Remote Schooling Enrollment Trends, NCES 2015-2025




*Notes:* This figure reports enrollment trends in exclusively virtual schools as reported in the Common Core data provided by the National Center for Education Statistics (NCES). The figure reports three time-series plots. The first (solid black) corresponds to one derived by plotting the raw data in the Common Core data. Many school districts, however, under- or misreport their remote schooling numbers. The second (dashed gray) accounts for this measurement error by imputing additional enrollment for the fifteen largest school districts in the country. The third (dotted black) is a prediction-based series from a regression of enrollment on a linear time trend for the period 2015 to 2020. The left axis reports enrollment in 1000s and the axis on the right reports the corresponding share of public school enrollment.

daily interactions between teachers and students. While in-person tutoring services were offered, their provision ebbed and flowed with each COVID wave. LAUSD schools remained closed until the week of April 19, when the district commenced a staggered reopening and students slowly returned to in-person schooling, with some caveats. Elementary schools offered classes in three-hour blocks and adult supervision when students were not in classes. Middle and high school students reported to campus on alternating days, with similar adult supervision provided. However, all families had the option to continue with remote learning.

The LAUSD's response to the pandemic meant that, for roughly one year, students in the district remained at home and received instruction virtually. Anecdotal evidence suggests most families disliked the online experience, and mounting evidence suggests this contributed negatively to student learning.[6] However, there is also evidence that suggests *some* families may have preferred remote learning. For example, bullied students may excel without the mental health costs incurred from in-person schooling (Bacher-Hicks et al., 2022), and others may benefit from learning at their own pace and reduced disruption (Armstrong-Mensah et al.,

[6]For example, Williams (2022) discusses student and parental frustration with remote schooling.

2020).[7]

The pandemic therefore functioned as a large-scale, involuntary experiment in remote schooling. Although the experience was improvised and likely distinct from many post-pandemic remote offerings, it revealed a stable piece of match quality that families are unable to learn without trying: whether their child is able to learn semi-independently at home and whether the household can support this style of learning. Prior research on K–12 online learning emphasizes students' capacity for self-regulated and autonomous learning, as well as parental support, as important determinants of success in remote environments (Curtis and Werth, 2015, Johnson et al., 2023). Consistent with this interpretation, Berger et al. (2021) show that pre-pandemic measures of students' academic achievement and learning motivation strongly predicted their ability to cope with the self-regulatory demands of pandemic-era remote learning, while parental support was also positively associated with successful adaptation. Importantly, these dimensions of suitability are primarily stable characteristics of the student and household rather than of a particular remote program, and are therefore likely to be portable across remote-learning environments. Because remote schooling was compulsory during the pandemic, every family was able to learn about this idiosyncratic component of match quality, which likely contributes to the observed post-pandemic demand.

LAUSD returned to full in-person learning for the 2021–2022 academic year.[8] To accommodate a sizable share of families who continued to prefer remote learning, the district did not make in-person learning mandatory and integrated its virtual offering with their independent-study program, known as the City of Angels school. Families opting into the remote option had to disenroll their student from their current school and seek enrollment in City of Angels, though remote students could transition back to in-person learning at any time.

In terms of instruction, the post-pandemic remote option combined self-paced learning and regular interactions with virtual instructors. Fully credentialed teachers—a feature not guaranteed in all in-person options—offered daily synchronous instruction for at least 3 hours for elementary age students and 3.5 hours for middle and high school age students. This was complemented by a weekly one-on-one check-in with each student's assigned teacher and 20–30 hours of self-paced learning. All remote students could receive in-person tutoring at pre-specified locations throughout the year.

Internally, the district organized the remote learning option across its six regional districts, each managing its own school. Starting in the 2022-2023 academic year, the district created six distinct theme-based virtual options, a departure from their region-based options. This was a more formal expansion of the remote option, establishing it as a permanent option in the portfolio of offerings the district provides to families, but the day-to-day experience of remote students in the 2021-2022 academic year and 2022-2023 onwards is remarkably similar, with an identical breakdown of daily synchronous instruction, self-paced learning time, and one-on-one check-ins discussed above. On the cost side, per-pupil expenditures for the City of Angels option were roughly $12,000, an amount that is 36-50% lower than the median elementary, middle, or

[7] Media accounts also testified to remote-learning benefits for some students (Harris, 2020).

[8] California mandated that all school districts had to offer a remote option during 2021–2022 due to COVID-19-related concerns.

high school in the district.[9] Importantly, curriculum and graduation requirements are identical between the remote option and in-person options.

The heavy reliance on self-paced work at home invites comparison with homeschooling, but the two are legally and administratively distinct in California and elsewhere. Students enrolled in LAUSD's remote option remain public-school students enrolled in the district. They are taught and graded by a credentialed teacher, follow the district curriculum, generate average daily attendance for district funding, and earn coursework credit that counts as progress toward a regular diploma. Parents seeking to homeschool must instead exit the public system altogether, either by filing an annual Private School Affidavit establishing a home-based private school (Cal. Educ. Code § 33190) or by engaging a credentialed private tutor. Homeschooled students are therefore subject to no district oversight, receive no public funding, and disappear from public-school administrative data, whereas remote students remain fully visible in the enrollment records we use. Remote schooling in our setting is thus a publicly funded, teacher-supervised schooling option *inside* the public school system rather than an exit from it. While this arrangement is now common across U.S. states, it is only one point on a much wider international spectrum, to which we now turn.

### 2.3 Remote Schooling Internationally

The post-pandemic growth of remote schooling extends beyond the United States. The pandemic compelled many governments to build remote-instruction capacity, with 91 percent of countries delivering distance learning through online platforms during school closures (UNESCO, 2023). As schools reopened, many systems diverged: some countries dismantled the infrastructure they had built, while others repurposed it into a permanent feature. On balance, expansion appears to be the more common path. In a post-pandemic OECD survey, 24 of 28 responding education authorities reported expanding the provision of online materials and virtual classes (OECD, 2024).

These survey responses show that expansion is widespread, but its scale is harder to gauge, since systematic cross-country enrollment data do not exist. National case studies offer the best available window, and they point to sharp growth. The starkest case is Japan, where roughly one in ten high school students is enrolled in a *correspondence* school, a distance-learning option where students complete their coursework remotely via digital platforms (Statistics Bureau of Japan, 2025).[10] While correspondence school enrollment grew by 9.7% in the years leading into the pandemic, enrollment rose by more than 46% between 2019 and 2024, converting the sector from a niche to a mainstream educational option. Similar growth appears across other high-income systems: roughly 7% of Canadian K–12 students participated in distance or online learning in 2023–2024 (Barbour and LaBonte, 2024), full-time enrollment at New Zealand's national distance school doubled between 2019 and 2024 (Flexible Learning Association of New Zealand, 2024), and enrollment in Poland's school-supervised out-of-classroom education—a

[9] With the creation of new theme-based virtual academies in recent years, these numbers are roughly similar. On a per-pupil basis, there are far lower expenditures for remote students than in-person students.

[10] Correspondence schools in Japan (called tsushinsei) trace back to the mid-twentieth century when they were established as independent, diploma-granting institutions. They initially delivered textbooks and assignments to students by mail, which is the basis of their name. Today, the system is a predominantly online model.

sector dominated by online schools—grew fivefold between 2019 and 2024 (Fundacja Edukacji Domowej, 2023).[11]

Remote schooling has an even longer history in developing countries, where it has served as a supply-side tool for expanding access to schooling rather than a demand-driven alternative to it. Mexico's telesecundarias—secondary schools built around televised lessons—have operated since 1968 and currently enroll more than 20 percent of Mexican lower-secondary school students (Secretaría de Educación Pública, 2025). Existing work has shown that attending them yields substantial returns (Fabregas and Navarro-Sola, 2025), with returns for the marginal student comparable to those of traditional schools (Borghesan and Vasey, 2024). Similarly, a rural distance education program in China raised educational attainment and earnings (Bianchi et al., 2022), and experimental evidence from Ghana shows that interactive satellite-based instruction improved learning in remote primary schools (Johnston and Ksoll, 2022). Remote schooling is thus a growing and established feature of many school systems around the world. As education systems continue to convert pandemic-era infrastructure into permanent virtual options, the questions that we study—who selects into remote schooling when it is offered as a choice, and with what consequences—extend beyond the U.S. context.

# 3 Data

Our analysis combines administrative records from LAUSD with original survey data that we collected from district parents in April 2022. We describe each data source in turn and then summarize the key sample characteristics that motivate our empirical approach.

The administrative records contain student-level demographics, residential addresses, and standardized test scores. We use 2018–2019 test scores as baseline measures of lagged achievement and rely on 2021–2022 scores as outcomes.[12] A second source of administrative data comes from the School Experience Survey (SES), which LAUSD has administered to all students in the district since 2010. We use the SES to construct index measures of bullying and grit, following the approach of Jackson et al. (2020) and Campos (2023).[13] We interpret the former as a measure associated with school climate and safety and the latter as a core competency of the student, sometimes referred to as a socio-emotional skill. Although the two outcomes are inherently distinct, we refer to them as non-cognitive outcomes throughout the text for expositional ease.

Our original survey was distributed to a sample of parents with LAUSD students enrolled in grades 3–8 and grade 11. Invitations for the survey were distributed to a random sample of 100,000 families through LAUSD's internal communications system. Because messaging was on behalf of the district, incentives were forbidden; however, families were informed that their responses could affect future policy decisions made by the district. The survey had two primary

[11] In Ukraine, where war has closed schools across entire regions, roughly 600,000 students—more than one in ten—attended some 2,100 fully online schools in 2024–2025 (Ukrinform, 2025).

[12] The district did not administer standardized tests during the 2019-2020 pandemic year or the subsequent year.

[13] In the Student Experience Survey data, the bullying and grit indices are based on 6 and 13 questions, respectively. We standardize each question associated with the respective indices and average the available standardized components. Higher values of the indices indicate that students are bullied less or have more grit. Appendix B provides details related to how we constructed these indices.

Table 1: Summary Statistics for LAUSD Students

| | (1) In-Person in 2022 | (2) Remote in 2022 | (3) Mean Diff. | (4) Survey Respondents | (5) Conjoint Respondents |
|---|---|---|---|---|---|
| Baseline ELA Scores | 0.008 | -0.234 | -0.243 | 0.191 | 0.471 |
| | (0.992) | (0.955) | (0.028) | (1.045) | (1.045) |
| Baseline Math Scores | 0.012 | -0.311 | -0.323 | 0.170 | 0.445 |
| | (0.992) | (0.923) | (0.028) | (1.011) | (1.011) |
| Baseline Bullying Index | 0.007 | -0.013 | -0.020 | 0.011 | -0.006 |
| | (0.647) | (0.677) | (0.003) | (0.607) | (0.668) |
| Baseline Grit Index | 0.009 | -0.054 | -0.062 | 0.047 | 0.089 |
| | (0.664) | (0.679) | (0.006) | (0.667) | (0.649) |
| Female | 0.484 | 0.505 | 0.021 | 0.494 | 0.508 |
| | (0.500) | (0.500) | (0.002) | (0.500) | (0.500) |
| Special Education | 0.139 | 0.153 | 0.014 | 0.109 | 0.101 |
| | (0.346) | (0.360) | (0.002) | (0.311) | (0.301) |
| URM | 0.817 | 0.842 | 0.025 | 0.756 | 0.651 |
| | (0.386) | (0.365) | (0.011) | (0.429) | (0.477) |
| English Learner | 0.381 | 0.324 | -0.057 | 0.315 | 0.167 |
| | (0.486) | (0.468) | (0.009) | (0.465) | (0.373) |
| Poverty | 0.828 | 0.813 | -0.016 | 0.739 | 0.598 |
| | (0.377) | (0.390) | (0.010) | (0.439) | (0.491) |
| Students | 276,553 | 12,326 | | 3,539 | 1,171 |

*Notes:* This table provides summary statistics for LAUSD students as observed in the LAUSD student micro-data. Index variables (i.e., bullying and grit) are normalized such that larger values denote better outcomes. Columns 1 and 2 report averages for in-person and remote students, respectively. Column 3 reports the corresponding difference in average characteristics. We recruited a sample of survey respondents by randomly contacting 100,000 families through the LAUSD's internal communication system in April 2022. Column 4 reports averages for every family who completed at least one question on our survey. Column 5 reports averages for every student who completed the hypothetical choice experiment questions within the survey. Baseline test scores are measured in the 2018–2019 school year, and baseline non-cognitive outcomes are measured in the 2020-2021 school year. In Columns 1, 2, 4, and 5, standard deviations for each measure are reported in parentheses. In Column 3, standard errors clustered at the school level from a regression of each measure on a remote indicator are reported in parentheses.

sections. The first, a descriptive component, asked parents about their experiences with remote learning during the pandemic and their demand for remote options going forward; we use these responses to document motivating facts in Section 4. The second component was a series of hypothetical choice experiments that we use to measure household preferences over remote learning, described in detail in Section 6. A sample of 3,539 parents completed the descriptive component, and 1,171 parents completed the choice experiment—a group we will sometimes refer to as the "conjoint sample." Respondents consented to have their responses linked to administrative records.[14] Appendix Section C reproduces the full survey instrument.

Table 1 provides summary statistics for two comparisons relevant to the rest of the paper. Columns 1–3 compare remote and in-person students in 2022. Remote students performed significantly worse on standardized exams in 2019, ranging between 0.24 and 0.32$\sigma$ lower baseline

[14]Surveys with choice experiments embedded in them are sometimes referred to as conjoint survey experiments in other fields. To the best of our knowledge the term conjoint is from Luce and Tukey (1964) who were studying how to recover measurements from judgments involving combinations of two or more components. Green and Rao (1971) later asked survey respondents to evaluate product profiles composed of several characteristics, allowing them to infer implicit preferences for the characteristics. Louviere and Woodworth (1983) helped develop the modern choice-based approach by combining conjoint-style experimental designs with discrete choice frameworks, creating a direct connection to our approach developed later. For this reason, throughout the paper we refer to the survey as a conjoint survey and also refer to the choice trials as choice experiments.

test scores. They also have worse non-cognitive outcomes, including grit and bullying. Remote students are also more likely to be female, under-represented minorities, and more likely to have a special education status; they are less likely to be classified as English learners, while their low-income status is similar to that of in-person students. These systematic differences between remote and in-person students underscore that the average effects of remote learning cannot be identified from a simple comparison of observed outcomes—a pattern that motivates the selection-correction strategy at the center of our analysis.

Columns 4 and 5 of Table 1 compare survey respondents and choice experiment respondents to the broader LAUSD population. Both samples are positively selected: families who completed the survey have students performing 17–19 percent of a standard deviation above district averages, and the academic differences are larger for the subset who completed the choice experiment.[15] This selective participation has implications for both the descriptive evidence and the empirical analysis based on our survey, and we adopt several strategies throughout the paper to ensure that our results speak to the broader LAUSD population.

# 4 Descriptive Evidence on Demand

We begin our analysis by using survey responses from LAUSD parents to document empirical patterns that motivate the rest of the paper. The descriptive component of our survey, described in Section 3, asked parents to respond to four statements about remote learning. Three statements speak to household demand: whether the family enjoyed remote learning during the pandemic, whether the family would like the district to expand its remote offerings, and whether the family is likely to opt for remote schooling in the future. The fourth statement asks whether the child excelled academically in remote relative to in-person instruction. Figure 2 reports the mean rates of agreement and disagreement with each statement, weighted to reflect the average LAUSD family.[16]

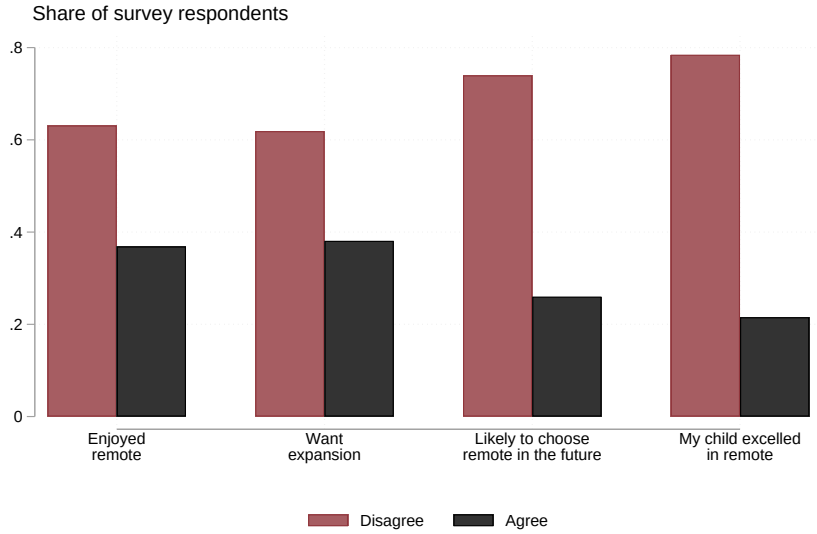
Turning first to the demand-related statements, the responses reveal a clear pattern: most families express low demand for remote learning, but a meaningful minority express high demand along each dimension. Approximately 62 percent of respondents disagreed that their child enjoyed remote learning during the pandemic, consistent with broader evidence that students struggled with virtual schooling during this period (Goldhaber et al., 2022, Jack et al., 2022, Loades et al., 2020). Roughly 62 percent disagreed that the district should expand its remote offerings, and approximately 74 percent indicated they were unlikely to opt for remote schooling in the future. At the same time, a meaningful share of families express positive demand: roughly 38 percent agreed that they would like the district to expand its remote offerings, and approximately 26 percent indicated they were likely to opt for remote schooling in the future. The two patterns taken together point to substantial heterogeneity in household preferences for remote learning.

A similar pattern appears in parents' assessments of their child's academic experience. Ap-

[15] These differences do not appear to be driven by geographic differences in response rates. Appendix Figure A.6 shows that respondents represent all school district regions.

[16] Appendix Figure D.1 presents summary statistics for students enrolled in remote learning, which indicate that a plurality indicated that both safety and academic concerns were equally important factors in their decision.

Figure 2: Experiences and Demand for Remote Learning




*Notes:* This figure reports survey results on the share of respondents ($N = 3,539$) who agree with four statements on their experiences and demand for remote learning. Individual responses are weighted to produce means that correspond to the average family in LAUSD. Specifically, we define the weight for each observation as $w_i \equiv P(Survey = 1)/p(Survey = 1|X_i)$, where $p(Survey = 1|X_i)$ is the estimated propensity to respond to the survey based on student characteristics $X_i$ using the full sample of LAUSD students, and $P(Survey = 1)$ is the share of all LAUSD families with survey responses. Appendix Section C reports the complete text for the survey questions (see question 5).

proximately 78 percent of respondents disagreed that their child excelled academically in remote learning relative to in-person instruction, consistent with the broader evidence on pandemic learning loss (Goldhaber et al., 2022, Jack et al., 2022, Singh et al., 2022). At the same time, roughly 22 percent agreed: one in five parents report that their child performed better academically in remote learning. The heterogeneity in stated academic experience is striking. Parents within the same district, often whose children attended the same schools and were exposed to the same district-wide remote infrastructure, report markedly different academic experiences for their children.

We emphasize that the patterns documented here are descriptive: they reflect parental perceptions and stated preferences rather than rigorous estimates of demand or causal effects. However, these facts motivate the theoretical and empirical framework we develop in Sections 5 and 6, which summarizes household demand for remote learning in a scalar index (measured using choice experiments) that both serves as a control for selection and allows the causal effect of remote learning to vary across students.

# 5 Conceptual and Empirical Framework

This section develops the conceptual framework underlying our main analysis of remote learning's effects on student achievement. The descriptive patterns documented in Section 4 point to two forms of heterogeneity that motivate our framework: heterogeneity in household demand for remote learning, and heterogeneity in achievement effects across students. We formalize these patterns through a discrete choice model that captures latent household heterogene-

ity in preferences over remote learning enrollment and a treatment-effects model that allows causal effects of remote learning to vary across students. The choice structure implies a scalar preference-based selection index $v_i$ and its propensity-score transformation $p_i$ that summarizes the household's relative valuation of remote enrollment given its preferences over schooling attributes. Our framework shows how conditioning on $v_i$ (equivalently, $p_i$) yields a transparent selection-on-observables representation for outcomes, separating systematic differences in baseline achievement across the propensity to enroll in remote learning from treatment-effect heterogeneity.

### 5.1 Remote vs. Traditional In-Person Schooling Enrollment

We model each household's enrollment decision as a choice between remote schooling ($R$) and the traditional in-person option ($T$). Households differ in latent preferences over the attributes of the two options, such as the academic quality of each schooling option, travel costs associated with in-person enrollment, and mode-specific factors that affect the desirability of remote learning.

Utility from schooling option $j \in \{R, T\}$ for student $i$ is:

$$U_{ij} = V_{ij} + \xi_{ij},$$

where $V_{ij}$ is the systematic component of indirect utility—a function of the attributes of option $j$ and the household's preferences over them.[17] The model includes $\xi_{ij}$ as an idiosyncratic taste shock, and we assume that $\xi_{iR}$ and $\xi_{iT}$ are independently and identically distributed Type-I extreme value shocks, independent of the systematic components of utility, with scale normalized to one.

The latent utility difference between remote and in-person enrollment satisfies:

$$u_i \equiv U_{iR} - U_{iT} = v_i + \varepsilon_i, \tag{1}$$

where $v_i \equiv V_{iR} - V_{iT}$ is the systematic component of relative indirect utility and $\varepsilon_i \equiv \xi_{iR} - \xi_{iT}$ is the residual difference in taste shocks. Importantly, $v_i$ includes systematic preferences for relative school quality, expected bullying exposure, and school composition, among numerous other factors. The remote enrollment decision is $D_i = 1\{u_i \geq 0\} = 1\{v_i + \varepsilon_i \geq 0\}$. Under our Type-I distributional assumption, $\varepsilon_i$ is logistic and the propensity score is defined as:

$$p_i \equiv \mathbb{P}(D_i = 1 \mid v_i) = \Lambda(v_i) \equiv \frac{\exp(v_i)}{1 + \exp(v_i)}, \tag{2}$$

which is a monotone transformation of $v_i$. Since utility in discrete choice models is identified only up to the scale of the idiosyncratic taste shocks, the propensity score $p_i$ provides a natural unit-free summary of the household's desire to enroll remotely, which motivates its use as a conditioning object when we discuss the outcome equation in the next section.

[17] Of course, the specific functional form is determined by the particular application. Section 6 describes the parameterization specific to our remote learning setting and the conjoint survey data.

### 5.2 Academic Potential Outcomes and Key Assumptions

We now link the enrollment choice model to a potential-outcomes framework for achievement. Let $Y_i(1)$ denote student $i$'s achievement outcome under remote enrollment and $Y_i(0)$ the outcome under traditional in-person enrollment. The observed outcome satisfies:

$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0).$$

We allow potential outcomes to depend on observed baseline characteristics $X_i$ and on the same selection-relevant heterogeneity captured by the scalar index $v_i$. In particular, we can write:

$$Y_i(d) \;=\; m_d(X_i, v_i) \;+\; \eta_{id}, \qquad d \in \{0, 1\}, \tag{3}$$

where $m_d(\cdot)$ is a flexible function of baseline observables and $\eta_{id}$ is an idiosyncratic component of achievement. Equation (3) allows $v_i$, the scalar index that drives remote enrollment, to shape potential outcomes. This is a substantive modeling choice. It rules out selection-relevant heterogeneity affecting outcomes through channels other than the selection index $v_i$, and the credibility of any empirical application of this framework rests on empirical evidence that this restriction is consistent with the data. In particular, two kinds of evidence are central: balance tests which demonstrate that accounting for $v_i$ achieves comparability on observable baseline characteristics, and out-of-sample validation comparing the framework's predictions against causal effects estimated from credible external variation. We provide both kinds of evidence in our remote learning application in Sections 7.4 and 8.5.

The key issue for identification is that enrollment decisions depend on $v_i$ through the selection rule $D_i = 1\{v_i + \varepsilon_i \geq 0\}$, so untreated and treated students may differ in outcome-relevant dimensions even after conditioning on observables alone. Our framework therefore imposes a condition under which accounting for the preference-based index via its propensity-score transformation is sufficient to remove selection bias.

**Assumption 1** (Outcome shocks are selection-irrelevant conditional on preferences)**.** *For each $d \in \{0, 1\}$, the idiosyncratic components of potential outcomes are independent of the enrollment shock conditional on $(X_i, v_i)$:*

$$(\eta_{i0}, \eta_{i1}) \;\perp\; \varepsilon_i \;\mid\; (X_i, v_i), \qquad \textit{and} \qquad \mathbb{E}[\eta_{id} \mid X_i, v_i] = 0.$$

Assumption 1 states that after conditioning on baseline observables and the selection-relevant index, the remaining idiosyncratic component that tips a household into (or out of) remote enrollment—$\varepsilon_i$ in Equation 1—captures decision noise or idiosyncratic administrative frictions and is not systematically related to achievement. The credibility of Assumption 1 depends on directly measuring $v_i$ in a manner that captures the selection-relevant heterogeneity. As we detail in Section 6, we use choice experiments to obtain new data that allow us to characterize $v_i$.[18]

[18] Measuring $v_i$ directly is also what distinguishes our strategy from standard approaches that rely on instrumental variables and exclusion restrictions. Absent a direct measure of $v_i$, the selection-relevant heterogeneity would enter potential outcomes only through the unobservables, where its influence on enrollment would cor-

Assumption 1 delivers an ignorability result: remote enrollment is as-good-as-random with respect to potential outcomes after conditioning on the preference-based index. In particular, Rosenbaum and Rubin (1983) show that

$$(Y_i(0), Y_i(1)) \perp D_i \mid (X_i, v_i), \qquad \text{and hence} \qquad (Y_i(0), Y_i(1)) \perp D_i \mid (X_i, p_i), \tag{4}$$

where $p_i = \Pr(D_i = 1 \mid v_i)$ is the propensity score defined in Equation 2 (a monotone transformation of $v_i$).[19] In practice, the propensity score is convenient for empirical applications since it is invariant to the scale normalization that is otherwise necessary to identify $v_i$ within a standard discrete choice model.

### 5.3 Student Outcomes Model

Our conceptual framework allows for heterogeneous treatment effects: because potential outcomes are individual-specific, $Y_i(1) - Y_i(0)$ varies across students. A key question for our empirical analysis is whether this variation is systematically related to the selection index—that is, whether students whose households have higher demand for remote enrollment experience different effects than students whose households have lower demand. We now derive a conditional mean decomposition that provides a framework for studying this question.

Equation (4) implies that for each $d \in \{0, 1\}$, we have:

$$\mathbb{E}[Y_i \mid D_i = d,\ X_i,\ p_i] \ = \ \mathbb{E}[Y_i(d) \mid X_i,\ p_i].$$

It is convenient to define the reduced-form conditional mean functions as:

$$g_d(x, p) \ \equiv \ \mathbb{E}[Y_i(d) \mid X_i = x,\ p_i = p], \qquad d \in \{0, 1\}.$$

We then define the conditional treatment effect function as:

$$\begin{aligned} h(x, p) &\equiv g_1(x, p) - g_0(x, p) \\ &= \ \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x,\ p_i = p]. \end{aligned}$$

With this notation, the observed conditional mean admits the decomposition:

$$\mathbb{E}[Y_i \mid D_i,\ X_i,\ p_i] \ = \ g_0(X_i, p_i) \ + \ h(X_i, p_i)\, D_i. \tag{5}$$

Both components of Equation 5 are indexed by the propensity score $p$, so moving along $p$ traces how (i) baseline outcome levels and (ii) treatment effects vary with students' demand to enroll remotely. In particular, $g_0(x, p) = \mathbb{E}[Y_i(0) \mid X_i = x,\ p_i = p]$ captures how expected in-person achievement changes across the propensity distribution, while $h(x, p)$ captures how the causal

relate treatment with potential outcomes—a correlation that instrumental-variables methods are designed to break using a valid instrument that shifts $D_i$ but is excluded from potential outcomes. Because we measure $v_i$ and condition on it directly, this correlation is absent in our framework: under Assumption 1, the residual $\varepsilon_i$ that governs the remaining variation in enrollment is unrelated to $\eta_{i0}$ and $\eta_{i1}$, which leaves no selection-relevant heterogeneity in the unobservables for an instrument to address.

[19] Conditional on $(X_i, v_i)$, $D_i$ depends only on $\varepsilon_i$ (Equation 1) and $Y_i(d)$ depends only on $\eta_{id}$ (Equation 3). Assumption 1 establishes that $\varepsilon_i$ and $\eta_{id}$ are conditionally independent, which yields the ignorability result.

effect of remote enrollment varies across the same propensity distribution. We refer to $g_0(x,p)$ varying with $p$ as selection on levels and to $h(x,p)$ varying with $p$ as selection on gains, in the spirit of Roy (1951). Crucially, $p$ is constructed from a direct measure of the selection-relevant heterogeneity $v_i$. This is a substantive departure from standard models of essential heterogeneity, which—absent a direct measure of the selection-relevant heterogeneity—index the treatment effect by the unobserved component of selection and characterize how effects vary along that latent dimension (Heckman et al., 2006). In our framework, the measured preference index $v_i$ captures systematic heterogeneity that can jointly influence choices and outcomes, and under Assumption 1, the residual enrollment shock $\varepsilon_i$ reflects decision noise or administrative frictions that affect enrollment but are unrelated to potential outcomes. We therefore need not index treatment effects by $\varepsilon_i$.

Equation (5) clarifies the connection between our approach and selection-on-observable approaches that rely on propensity-score stratification. A common approach (e.g., Dehejia and Wahba, 2002) is to partition the support of $p_i$ into bins $\{B_a\}_{a=1}^{A}$ (e.g., quantiles) and compare treated and untreated outcomes within bins. Within bin $a$, one can form the within-bin contrast:

$$h_a(x) \equiv \mathbb{E}[Y_i \mid D_i = 1,\ X_i = x,\ p_i \in B_a] - \mathbb{E}[Y_i \mid D_i = 0,\ X_i = x,\ p_i \in B_a],$$

which is a step-function approximation to $h(x,p)$. A regression analog replaces $p_i$ by bin indicators (and their interactions with $D_i$), while flexibly controlling for $X_i$:

$$Y_i = \alpha(X_i) + \sum_{a=1}^{A} \theta_a \, \mathbf{1}\{p_i \in B_a\} + \sum_{a=1}^{A} \psi_a \, D_i \cdot \mathbf{1}\{p_i \in B_a\} + \nu_i, \tag{6}$$

where the coefficients $\psi_a$ correspond to treated–control differences within propensity bins.

Rather than specifying $g_0(\cdot)$ and $h(\cdot)$ as step functions, one can approximate these functions with a low-dimensional basis in the propensity score and flexible controls for $X_i$. Let $f(\cdot)$ denote a chosen basis (e.g., linear, polynomial, or spline). We focus on models of the form:

$$Y_i = \alpha(X_i) + \theta \, f(p_i) + \big(\beta + \psi \, f(p_i)\big) D_i + \nu_i, \tag{7}$$

which corresponds to approximating $g_0(x,p)$ and $h(x,p)$ in Equation (5) as $g_0(x,p) = \alpha(x) + \theta f(p)$ and $h(x,p) = \beta + \psi f(p)$. This parameterization characterizes how treatment effects vary across the propensity distribution through a small number of interpretable parameters, while still recovering canonical average treatment effects as derived quantities. We turn to these derivations next.

### 5.4 Average Treatment Effect Parameters

Under the linear approximation introduced in Section 5.3, the conditional treatment effect function takes the form $h(X_i, p_i) = \beta + \psi f(p_i)$. Conventional treatment effect parameters are obtained by averaging $h(\cdot)$ over the relevant distribution of $p_i$ in the population of interest.

Three parameters are particularly relevant for evaluating remote schooling policy:

$$\begin{aligned}
\text{ATE} &\equiv \mathbb{E}[h(X_i, p_i)] = \beta + \psi\, \mathbb{E}[f(p_i)]\,, \\
\text{ATT} &\equiv \mathbb{E}[h(X_i, p_i) \mid D_i = 1] = \beta + \psi\, \mathbb{E}[f(p_i) \mid D_i = 1]\,, \\
\text{ATU} &\equiv \mathbb{E}[h(X_i, p_i) \mid D_i = 0] = \beta + \psi\, \mathbb{E}[f(p_i) \mid D_i = 0]\,.
\end{aligned}$$

The three parameters share the common slope $\psi$ and intercept $\beta$ but differ in the average value of $f(p_i)$ in the relevant subpopulation. The ATE averages $f(p_i)$ over the full population. Since $p_i$ is the probability of enrolling, students with high $p_i$ are overrepresented among remote enrollees and underrepresented among in-person students. Given this, when treatment effects increase with $p_i$ (i.e., there is positive selection on gains), we have the ordering ATT $>$ ATE $>$ ATU. In words, the students who most value remote enrollment are also those for whom it is most beneficial.

A convenient normalization simplifies the mapping from regression coefficients to these parameters: when the basis $f(p_i)$ is centered so that $\mathbb{E}[f(p_i)] = 0$ (for example, by demeaning $f(p_i)$ before entering it into the regression), the ATE is given directly by $\beta$, and the ATT and ATU differ from the ATE by $\psi$ times the conditional expectation of $f(p_i)$ in the relevant subpopulation. We adopt this normalization in our empirical analysis, so the coefficient on $D_i$ is directly interpretable as the ATE.

This mapping clarifies the connection between our framework and conventional treatment effect parameters. The regression coefficients $\hat{\beta}$ and $\hat{\psi}$ summarize the conditional treatment effect function $h(\cdot)$; conventional parameters are then obtained by averaging $h(\cdot)$ across the empirical distribution of $\hat{p}_i$ in the relevant subpopulation. We report the regression coefficients in our main tables and the implied ATE, ATT, and ATU in subsequent results.

# 6 Choice Experiments

Our conceptual framework establishes the scalar selection index $v_i$ as the central object for identifying treatment effects: under Assumption 1, conditioning on $v_i$ removes the confound of preference-driven selection into remote learning. The credibility of this approach depends on having a measurement of $v_i$ that captures the relevant dimensions of household preference heterogeneity.

We address this measurement challenge by collecting new data through a series of hypothetical choice experiments. Respondents are presented with a sequence of schooling options and asked to make trade-offs between attributes that vary randomly across choice scenarios. This approach builds on a literature using hypothetical choice surveys to identify preferences over workplace attributes, product characteristics, and other settings where observed choices are limited or subject to selection (Mas and Pallais, 2017, Moshary et al., 2022, Wiswall and Zafar, 2018).

Our approach is particularly well-suited to measuring the preference heterogeneity that governs schooling choices for two reasons. First, the experimental variation in attributes is, by construction, uncorrelated with any confounding factors driving selection into remote enroll-

ment. This is a key advantage over approaches that infer preferences from observed enrollment decisions, which require the strong assumption that observed choice attributes are independent of unobservable confounders—an assumption that can fail in plausible ways, for instance if families sort residentially based on the same motivations that drive school choice. Second, families had already experienced remote instruction for an extended period during the pandemic, giving them an opportunity to learn about persistent dimensions of their child's relative suitability for remote learning before making the hypothetical choices we use to measure preferences.

The remainder of this section describes how we operationalize this approach in our setting. We begin by detailing the choice experiment design and the survey instrument, including the parametric utility specification that the design is built around. We then describe how we recover preference parameters from the resulting choice data. Both the descriptive survey evidence and the structural WTP estimates that follow characterize the household preferences summarized by $v_i$. We then turn to reporting willingness-to-pay estimates from the choice experiment, which reveal substantial heterogeneity across families and highlight the important role of bullying-related considerations. We close by validating the choice experiment.

### 6.1 Design

Our conceptual framework defines $v_i$ as a scalar index summarizing the household's relative valuation of remote enrollment. To empirically measure this object, we designed a series of hypothetical choice experiments that implement a willingness-to-pay design with travel time as the numeraire. The household trades off the focal treatment (remote vs. in-person) against the numeraire, allowing us to recover $v_i$ in interpretable units. Travel time is a natural numeraire in our setting: public schools do not transact on a market price dimension, and introducing a tuition or stipend attribute would feel artificial and potentially less credible. Moreover, our focus on travel time follows a long tradition in the empirical school choice literature where travel time is a central determinant of choices (Abdulkadiroğlu et al., 2020, Allende, 2019, Beuermann et al., 2022, Burgess et al., 2015, Campos and Kearns, 2024, Hastings et al., 2005, Neilson, 2021).

We also vary academic quality across choice scenarios. Holding quality fixed would risk respondents forming their own priors about the relative quality differences between remote and in-person options—a particular concern given the salience of pandemic-era evidence on remote schooling effects (Goldhaber et al., 2022, Jack et al., 2022). By varying quality explicitly in each scenario, we hold respondents' mental judgments about quality fixed and isolate the household's preference for instructional mode itself. Varying multiple attributes allows us to pin down remote preferences faster. A willingness-to-pay design with only the focal choice (i.e., remote learning) would identify preferences by varying the numeraire across many trials to pin down the value at which each respondent switches between accepting and rejecting the choice with the characteristic of interest—a process that requires many trials and risks choice fatigue. By varying multiple attributes simultaneously, our design extracts information about preferences from each comparative judgment across scenarios, reducing the number of trials needed and the repetitiveness of the task.

Formally, our survey design parameterizes the systematic component of indirect utility from

Section 5.1, as:

$$V_{ij} = \omega_{Qi} Q_{ij} + \omega_{Ri} \mathbf{1}(j = R) + \omega_{ti} t_{ij}, \tag{8}$$

where $Q_{ij}$ is the academic quality of school option $j$, $\mathbf{1}(j = R)$ is an indicator for whether option $j$ is the remote schooling option, and $t_{ij}$ is the travel time from the household's residence to school $j$ (set to zero for the remote option). The vector $\omega_i = (\omega_{Qi}, \omega_{Ri}, \omega_{ti})$ collects the household-specific preference parameters that govern how the household values the three attributes. The selection index implied by this specification is $v_i = V_{iR} - V_{iT} = \omega_{Ri} + \omega_{Qi}(Q_{iR} - Q_{iT}) - \omega_{ti} t_{iT}$: the relative utility a household derives from remote enrollment given the academic quality and travel time of their alternatives.

The choice experiment is designed to identify the vector $\omega_i$ for each respondent. Each respondent is presented with ten choice scenarios. In each scenario, the respondent is shown three hypothetical schooling options characterized by the three attributes in Equation 8: instructional mode, academic quality, and travel time. Academic quality corresponds to school proficiency rates on standardized exams (in percentage terms) and is drawn from the set $\{30, 50, 60, 70, 90\}$, while travel time is measured in minutes and drawn from $\{15, 30, 45, 60, 75\}$, with attribute values randomized independently across options and scenarios. Travel time is set to zero for the remote option by construction. For each scenario, the respondent identifies which option they would most prefer and which they would least prefer for their child in the upcoming academic year, generating a ranking over the three options.[20]

Two design choices help ensure that respondents' choices reflect preferences over the attributes in Equation 8. First, the instructions ask parents to assume that "pandemic-related safety issues are as they were in 2019 before COVID," so that responses reflect preferences over the schooling attributes rather than time-specific concerns about transmission risk. Importantly, this framing is intended to remove COVID-related safety concerns, while preserving what families learned from their prior experience with remote instruction about their child's suitability for this learning modality. Section 6.4 provides further discussion and evidence that respondents followed this instruction. Second, the instructions ask parents to assume that, apart from the attributes shown, the three options are otherwise identical in terms of academic instruction and quality, so that responses reflect the randomized attributes rather than imagined differences along unspecified dimensions. This framing parallels the design used in other studies that elicit preferences over alternatives with multiple unspecified dimensions (Mas and Pallais, 2017, Wiswall and Zafar, 2018).

Together, our design choices give a specific interpretation to the household-specific preference parameter $\omega_{Ri}$. In the simplest specification that we could have implemented—varying only remote schooling and the numeraire—the parameter $\omega_{Ri}$ would absorb the full set of household-specific reasons to demand remote schooling, including preferences for the remote modality itself, student-specific match quality, expected bullying exposure, and perceptions of school quality. By randomizing academic quality across choice scenarios, our design separately identifies $\omega_{Qi}$ from $\omega_{Ri}$, so quality-related considerations no longer enter $\omega_{Ri}$. The resulting interpretation of $\omega_{Ri}$ is therefore an omnibus preference for remote schooling, net of differential distance

[20] Inducing a rank-ordered preference list also helps with efficiency by providing an additional "choice" for identification.

costs and experimentally varied school achievement, that absorbs the remaining unobserved considerations including match quality and bullying exposure.[21]

### 6.2 Estimation

We estimate household preferences using the rank-ordered logit framework (Beggs et al., 1981, Hastings et al., 2005, Hausman and Ruud, 1987). For each of the ten scenarios, a respondent identifies a most-preferred option and a least-preferred option among the three available, generating an implied complete ranking of the three options. Under the Type I extreme value assumption on $\xi_{ij}$ introduced in Section 5.1, the probability that respondent $i$ produces a particular ranking $R_{ik} = (R_{i1k}, R_{i2k}, R_{i3k})$ in scenario $k$ takes the closed form

$$\Pr(R_{ik} \mid Q_{ik}, t_{ik}, \omega_i) = \frac{\exp(V_{iR_{i1k}})}{\sum_{m \in \{R_{i1k}, R_{i2k}, R_{i3k}\}} \exp(V_{im})} \cdot \frac{\exp(V_{iR_{i2k}})}{\sum_{m \in \{R_{i2k}, R_{i3k}\}} \exp(V_{im})}, \tag{9}$$

where $V_{ij}$ takes the parametric form in Equation 8, with $Q_{ij}$ and $t_{ij}$ taking the values corresponding to option $j$ in scenario $k$. The vectors $Q_{ik}$ and $t_{ik}$ collect the academic-quality and travel-time attributes of the three options in scenario $k$. The two terms in Equation 9 correspond to the choice of the most-preferred option from the full set and the choice of the second-ranked option from the two remaining alternatives; the final alternative is then mechanically assigned the lowest rank.

Aggregating across a respondent's ten scenarios, the likelihood for respondent $i$ is

$$\mathcal{L}_i(\omega_i \mid R_i, Q_i, t_i) = \prod_{k=1}^{10} \Pr(R_{ik} \mid Q_{ik}, t_{ik}, \omega_i), \tag{10}$$

where $R_i = (R_{i1}, \ldots, R_{i,10})$ and $Q_i$ and $t_i$ collect the attribute values across all ten scenarios. We estimate $\omega_i$ by maximizing this likelihood respondent by respondent. In principle, we can estimate $\omega_i$ for each respondent; however, we can also explore subgroup heterogeneity by grouping respondents into groups $g \in \mathcal{G}$. Therefore, we can estimate preferences that are common to respondents in group $g \in \mathcal{G}$ by aggregating Equation 10:

$$\mathcal{L}_g(\omega_g \mid R_g, Q_g, t_g) = \prod_{g(i)=g} \prod_{k=1}^{10} \Pr(R_{ik} \mid Q_{ik}, t_{ik}, \omega_g), \tag{11}$$

where $R_g$, $Q_g$, and $t_g$ are vectors containing all group $g$ respondents' choices across their scenarios.[22] In the next section, we use this pooled estimator to characterize how preferences vary across demographic and stated-demand subgroups, providing structural counterparts to the descriptive heterogeneity documented in Section 4.

[21] Given the role of bullying in our empirical analysis, a natural consideration is why bullying was not also included as an attribute in the choice trials. We chose a parsimonious design to limit respondent burden and preserve the interpretability of the estimated tradeoffs. This concern is consistent with behavioral models of bounded rationality, in which decision makers simplify high-dimensional choice problems by attending to a sparse subset of variables (Gabaix, 2014), and with evidence from choice experiments that respondents often use attribute non-attendance as a simplifying heuristic (Hole et al., 2013). We therefore study bullying as a downstream outcome rather than as an experimentally varied attribute.

[22] If we let $g$ correspond to the set of all respondents, then we estimate a preference vector that is common to all students in that group.

As noted in Section 5.1, the parameters $\omega_i$ (or $\omega_g$) are identified only up to a scale normalization. The experimental variation in attributes identifies the ratios of the parameters — for example, the marginal rate of substitution between travel time and academic quality — but not their levels. We follow standard practice and normalize the variance of $\xi_{ij}$ to $\pi^2/6$, the variance implied by the Type I extreme value distribution.

### 6.3 Experimental Preference Estimates

We now report average willingness-to-pay (WTP) estimates from applying the rank-ordered logit estimator to the entire conjoint sample. These estimates are the key input to our empirical strategy: they characterize the household preferences that the framework treats as the selection-relevant heterogeneity $v_i$, and the heterogeneity in WTP across families is what generates variation in $v_i$ that makes the conditioning strategy in our framework empirically informative. The estimates also offer a more rigorous characterization of the demand heterogeneity documented descriptively in Section 4.

The first row of Table 2 reports two WTP estimates for the average family in the conjoint sample. The first is the willingness to travel for an additional 10 percentage point increase in the share of students meeting state academic standards. The average family is willing to travel an additional 13 minutes for this increase in academic quality, consistent with established findings on family preferences over school quality (Abdulkadiroğlu et al., 2020, Burgess et al., 2015, Hastings et al., 2005). The second WTP measure is more directly relevant to our application: the increase in academic proficiency that would make a family indifferent between in-person and remote schooling. The average family requires a 46 percentage point increase in proficiency at the remote option to be indifferent between the two modes, indicating that the average family has a strong distaste for remote learning relative to in-person enrollment.[23]

These averages mask substantial heterogeneity. The patterns of heterogeneity provide structural counterparts to the descriptive evidence on demand reported earlier, where roughly a quarter of families indicated they were likely to opt for remote schooling in the future. Re-estimating the rank-ordered logit on subsamples defined by these same stated demand measures, Panel A shows that families currently enrolling their children in remote learning, and families who indicate they plan to enroll in the future, require essentially no compensation to switch to remote: their estimated achievement compensation is statistically indistinguishable from zero. The structural estimates therefore reproduce, with the precision of an explicit model, the heterogeneity in demand that the descriptive evidence surfaces. We also find statistically significant differences in willingness to travel for academic quality, with families currently in the remote option having a lower willingness to travel.

[23]We elicited preferences for proficiency rates because the units are more natural to LAUSD families. To draw comparisons to other estimates in the literature, we can also convert our preference estimates to student standard deviation units using publicly available data from the California Department of Education (California Department of Education, 2025). Specifically, we use the relationship between statewide percentile ranks and scale scores and approximate this relationship linearly over the 10th to 90th percentiles to translate changes in proficiency rates into standard-deviation units. Using this approximation, we find that the typical family is willing to travel 4.1 minutes for an additional $0.1\sigma$ increase in a school's academic achievement. Similarly, the average family requires a $1.49\sigma$ increase in school achievement to be indifferent between in-person and remote options. We emphasize that these are approximations as the relationship between proficiency rates and student standard deviations is not linear.

Table 2: Experimental Preference Estimates

| | (1) WTT for 10 p.p. Quality Increase (Minutes) | (2) Remote Compensation (Achievement p.p.) |
|---|---|---|
| Panel A: By Enrollment Status and Future Plans | | |
| All | 13.02 | 46.32 |
| | (0.55) | (2.29) |
| Currently in Remote | 10.87 | 0.99 |
| | (1.56) | (7.62) |
| Plans to Enroll in Remote | 10.20 | 3.43 |
| | (1.25) | (5.55) |
| Not Currently in Remote | 13.27 | 49.44 |
| | (0.59) | (2.40) |
| $H_0$ : No Heterogeneity ($p$-value) | 0.042 | 0.000 |
| Panel B: Student Characteristics | | |
| Low Baseline Achievement | 16.84 | 30.36 |
| | (3.08) | (6.48) |
| High Baseline Achievement | 13.27 | 46.76 |
| | (0.94) | (3.85) |
| Female | 12.98 | 44.74 |
| | (0.78) | (3.22) |
| Special Education | 9.40 | 51.57 |
| | (1.73) | (11.69) |
| URM | 12.73 | 42.66 |
| | (0.76) | (3.21) |
| White | 13.28 | 56.97 |
| | (1.01) | (4.24) |
| English Learner | 11.62 | 54.82 |
| | (1.87) | (9.62) |
| Poverty | 12.88 | 41.95 |
| | (0.79) | (3.16) |
| Elementary/Middle School | 12.69 | 47.92 |
| | (0.59) | (2.64) |
| High School | 14.42 | 40.25 |
| | (1.50) | (4.45) |
| $H_0$ : No Heterogeneity ($p$-value) | 0.387 | 0.009 |
| Panel C: Baseline Bullying | | |
| Lower Half (More Bullying) | 14.35 | 40.24 |
| | (1.30) | (4.00) |
| Upper Half (Less Bullying) | 12.27 | 54.84 |
| | (1.06) | (5.61) |
| $H_0$ : No Heterogeneity ($p$-value) | 0.215 | 0.034 |
| Observations | 22,338 | 22,338 |

*Notes:* This table reports experimental preference estimates derived from maximizing the group-specific likelihood shown in Equation 11. Each row reports two estimates: (1) marginal willingness to travel for a ten percentage point increase in school proficiency rates expressed in minutes; (2) the percentage points increase in school proficiency required to compensate for remote schooling. Column (1) reports -$10\omega_Q/\omega_t$ and Column (2) reports $-\omega_R/\omega_Q$, using the preference coefficients defined in Equation 8. Panel A reports estimates for different students based on their enrollment status at the time of the survey. The "All" row is estimated by letting $g$ from Equation 11 correspond to the set of all respondents; estimates for remaining rows are found by limiting the sample to respondents in the given sub-group. Panel B reports estimates for different groups based on student characteristics. Last, Panel C reports estimates for students with different baseline bullying outcomes. The last row of each panel reports $p$-values testing the null hypothesis of no heterogeneity across estimates in the given column. In Panel A, we exclude the all group from the joint equality test. Standard errors are reported in parentheses below each estimate and are estimated via the delta method.

Preferences also vary systematically with baseline characteristics that the descriptive analysis did not examine directly. Panel B of Table 2 shows that White families require an average compensation in quality that is 33 percent larger than that of URM families to switch to remote. Families with lower-achieving students also require lower compensation to be indifferent between remote and in-person schooling, a 16 percentage point difference between families with low- and high-achieving students. We also find that families with younger children require more compensation to enroll their children in remote schooling than families with older children. Notably, this heterogeneity is concentrated in the compensation required for remote schooling. We fail to reject equality across demographic groups in their willingness to travel for academic quality (Column 1, $p$-value=0.387), while compensation for remote schooling varies significantly across groups (Column 2, $p$-value=0.009). Taken together, the results suggest that demographic heterogeneity is concentrated in the remote-versus-quality tradeoff rather than in the valuation of academic quality relative to travel time.

A final dimension of heterogeneity is particularly important for interpreting our subsequent treatment-effect estimates. Panel C of Table 2 shows that families with children in the bottom two quartiles of the baseline bullying distribution (i.e., those whose children experienced more bullying at baseline) require approximately three-quarters the compensation to switch to remote than families with children in the top quartiles ($p$-value = 0.034). We complement this with observational estimates of families' marginal rates of substitution, finding that families are willing to forgo 17 percentage point improvement in school academic proficiency rates (SE=7.12) to obtain a school whose typical student is one standard deviation above average in terms of the no bullying index. These patterns indicate that bullying-related considerations are a substantial determinant of household demand for remote schooling, a finding we will return to when we examine treatment effects on non-cognitive outcomes in Section 8.

### 6.4 Choice Experiment Validation Exercises

This subsection presents two pieces of evidence supporting the choice experiment as a credible measurement instrument for household demand. The first addresses whether respondents followed the design's framing instructions on pandemic conditions. The second addresses whether the elicited preferences reflect determinants of real-world enrollment behavior.

A potential concern with the choice experiment is that respondents may have brought pandemic-era safety concerns to bear on hypothetical schooling decisions, despite instructions to assume pre-COVID safety conditions. If respondents anchored on local pandemic conditions during their responses, the elicited preferences would reflect a mix of preferences over the schooling attributes of interest and time-specific concerns about transmission risk. Appendix F addresses this concern by relating zip-code-level preference estimates to several measures of local COVID conditions, including the COVID-19 Vulnerability and Recovery Index produced by Los Angeles County and zip-code-level case and death rates. Across these measures, we find no systematic relationship between local COVID conditions and either willingness to travel for academic quality or the compensation required to switch to remote schooling. This is consistent with respondents following the framing instruction and indicates that the elicited preferences are not substantially contaminated by pandemic-specific concerns.

Another concern is that hypothetical survey choices may not reflect real-world enrollment behavior. To assess this, we combine the conjoint preference estimates with each student's own choice attributes to construct propensity scores, $\hat{p}_i$. We then estimate the relationship between actual enrollment $D_i$ and the propensity scores. The estimated slope coefficient is 1.19 (SE = 0.236), and the propensity score explains approximately 8 percent of the residual variation in real-world enrollment within cells. The fact that the slope is statistically indistinguishable from one indicates that the propensity scores are approximately forecast-unbiased for actual enrollment. This is a strong validation: the choice experiment recovers preferences that can explain real-world remote enrollment decisions.

# 7 From Choice Experiments to Treatment Effects

This section operationalizes the framework developed in Section 5 for estimation in the LAUSD setting. The framework's outcome equation depends on the selection index $v_i$ and its propensity-score transformation $p_i$. Section 6 measured these objects using the choice experiment, yielding credible preference estimates for the LAUSD households who completed it. The first task of this section is to extend these estimates to the full LAUSD sample using an extrapolation procedure based on baseline characteristics and assess the validity of this extrapolation. The second task is to specify the operational version of the outcome equation and the estimation strategy for the treatment effect parameters of interest. Finally, the third task is to present balance tests that assess the credibility of the conditioning strategy. As previewed above, these tests are the empirical counterpart to the framework's identification assumption: if conditioning on the propensity score is sufficient to eliminate selection bias, then observable baseline characteristics should be balanced across enrollment status within the propensity distribution.

## 7.1 Estimating Preferences for All LAUSD Students

As described above, the choice experiments in our survey allow us to obtain credible preference estimates for the subset of students with parents who completed our survey. To maximize statistical power, we use the full sample of LAUSD students in our analysis. As highlighted by Table 1, one challenge with this approach is that our sample of LAUSD respondents differs from the general population of LAUSD students. To account for these differences and extend the preference estimates to the broader LAUSD population, we use an extrapolation approach that allows preferences to vary flexibly with baseline student characteristics.[24]

Formally, our extrapolation approach constructs $\hat{v}_i$ for each LAUSD student by applying a baseline covariate cell-specific preference vector to the student's own choice attributes:

$$\hat{v}_i = \hat{\omega}_{Q,c(X_i)}\Delta Q_i + \hat{\omega}_{R,c(X_i)} - \hat{\omega}_{t,c(X_i)}t_i, \tag{12}$$

where the parameters $\hat{\omega}_{Qc(X_i)}$, $\hat{\omega}_{Rc(X_i)}$, and $\hat{\omega}_{t,c(X_i)}$ are allowed to vary flexibly by cell, $c(X_i)$,

[24]The results based on only students who participated in the survey are qualitatively similar to our headline estimates but are not estimated precisely. Appendix Figure E.4 shows that our preferred estimates based on the extrapolation method lie within the confidence intervals of estimates using only the conjoint sample. Our preferred estimates are more conservative in magnitude and qualitatively similar across the propensity score distribution.

defined by the complete interaction of a poverty status indicator, an indicator for being an underrepresented minority, a high-school indicator, and baseline ability indicators (three terciles plus a separate indicator for missingness). This specification makes the treatment of preference heterogeneity explicit, motivated by the patterns documented in Table 2. This cell-based approach to modeling preference heterogeneity is similar to Abdulkadiroğlu et al. (2020).

In Equation 12, $\Delta Q_i$ is the academic quality of the remote option relative to student $i$'s outside option, a traditional in-person school, while $t_i$ is the travel time to the in-person alternative. The implied propensity score is $\hat{p}_i = \Lambda(\hat{v}_i)$, the logistic transformation established in Section 5. Our approach effectively assumes that the experimentally identified taste parameters $\hat{\omega}_{c(X_i)}$ are sufficient to pin down the correct marginal rates of substitution *within cell*, converting the within-cell variation in $\Delta Q_i$ and $t_i$ into student-specific propensity scores. If these propensity scores are accurate, Assumption 1 allows us to use them to account for selection into treatment in the spirit of Rosenbaum and Rubin (1983).

We emphasize that preference extrapolation is common in the empirical school choice literature. For example, estimation procedures that rely on stability properties of centralized matches (Fack et al., 2019) require an extrapolation of preferences of individuals with larger feasible choice sets to those with smaller choice sets (Agarwal and Somaini, 2020). Equipped with these extrapolated preferences, there is precedent in using these within-cell extrapolated preferences inferred from observed choices to construct control functions to characterize selection into treatment (Abdulkadiroğlu et al., 2020, Otero et al., 2021). Our approach is similar but instead uses choice experiments to help predict choices that subsequently characterize selection into treatment, bridging literature using choice experiments to better understand demand (Mas and Pallais, 2017, Wiswall and Zafar, 2018) and literature using observed choices to characterize selection (Abdulkadiroğlu et al., 2020, Heckman, 1979).

The cell-by-cell estimation produces a wide distribution of preferences across covariate cells. Table 3 provides a basic summary of the distribution of taste parameters across covariate cells, while Appendix Figure D.2 shows the densities for all preference parameter estimates.[25] Across the cells, students prefer high achieving peers and dislike both remote learning and travel time. This dispersion in tastes for remote learning is notably wide: across the covariate cells, the compensation in peer academic proficiency required to make a family indifferent between remote and in-person offerings ranges from roughly 10 to 78 percentage points (i.e., the 5th to 95th percentile of the cell distribution), a spread of roughly 68 percentage points.

### 7.2 Extrapolation Validity

The credibility of the extrapolation procedure rests on two empirical questions. First, do the conjoint respondents and the broader LAUSD population overlap on baseline characteristics, so that cell-specific preferences are estimated on representative respondents? Second, does the cell-based extrapolation accurately predict preferences for households not included in the estimation sample? We address each question in turn.

[25] Appendix Table D.1 reports more detailed summary statistics, including the minimum and maximum of each parameter across cells. The bounds show that every cell-level estimate has the expected sign, with families consistently valuing academic quality and disliking both remote learning and travel time.

Table 3: Summary Statistics for Preference Estimates

| | (1) Mean | (2) SD | (3) p5 | (4) p95 |
|---|---|---|---|---|
| Academic Quality ($\omega_Q$) | 0.04 | 0.02 | 0.01 | 0.08 |
| Remote ($\omega_R$) | -1.70 | 1.00 | -3.27 | -0.24 |
| Travel Time ($\omega_t$) | -0.03 | 0.02 | -0.07 | -0.01 |
| WTT for 1 p.p. Quality Increase ($-\omega_Q/\omega_t$) | 1.45 | 0.91 | 0.53 | 3.79 |
| Remote Compensation ($-\omega_R/\omega_Q$) | 43.88 | 24.93 | 9.65 | 77.93 |

*Notes:* This table reports summary statistics for preference parameters that were estimated separately for each covariate cell, where covariate cells are defined by the complete interaction of a poverty status indicator, an indicator for being an underrepresented minority, a high-school indicator, and baseline ability indicators (three terciles plus a separate indicator for missingness). Columns (1)–(4) report the mean, standard deviation, 5th percentile, and 95th percentile of the respective row variable. The last two rows report the willingness to travel (WTT) for an additional 1 percentage point increase in academic proficiency and the amount of compensation in achievement units necessary to make respondents indifferent to the remote option. We omit one outlier observation in the statistics presented for the final row as it skews the mean and standard deviation.

Appendix Figure E.1 addresses the first question by plotting the distribution of a summary index of baseline characteristics for the conjoint sample and the broader LAUSD sample. The index is constructed by predicting 2022 ELA test scores from baseline covariates including lagged achievement, URM status, sex, socioeconomic status, English-learner status, and special education status, and using the predicted values as a summary statistic. Appendix Figure E.2 reports overlap for baseline test scores in isolation.[26] Across both figures the respondent and non-respondent distributions overlap substantially across the support of the index and baseline test scores, indicating that the conjoint sample's covariate profile is not so distinct from the broader population that extrapolation would require predictions far outside the observed support.

Next, Figure E.3 addresses the second question through an out-of-sample validation exercise. We randomly partition the conjoint sample into an estimation sample and a holdout sample, stratifying to ensure the estimation sample matches the baseline characteristics of the average LAUSD student. Using only the holdout sample, we estimate cell-specific preferences and construct the associated propensity scores for the estimation sample by extrapolation. We then compare these extrapolated propensity scores to those that would have been estimated directly using the estimation sample. The slope of the relationship between extrapolated and direct propensity scores is 1.07, and the intercept is near zero. The mean difference between the two scores is 0.006, with a standard deviation of 0.075. This pattern indicates that the extrapolation procedure is approximately forecast unbiased for preferences of households not used to estimate the cell-specific preference parameters.

[26]Overlap for the remaining individual covariates (URM status, sex, socioeconomic status, English-learner status, and special education status), all of which are discrete, can be observed from the summary statistics contained in Table 1.

### 7.3 Main Outcome Specification

We operationalize the framework's outcome equation using a specification that models propensity scores as a linear function, i.e., $f(p_i) = p_i$. This delivers the regression model:

$$Y_i = \alpha_c + \gamma' X_i + \beta D_i + \theta \tilde{p}_i + \psi \left(\tilde{p}_i \times D_i\right) + \epsilon_i, \tag{13}$$

which includes covariate cell fixed effects $\alpha_c$ and the vector $X_i$ of the remaining baseline student characteristics (indicators for English learner status with a separate dummy for missing data, female, and special education status).[27] We emphasize that the covariate cell fixed effects are necessary to ensure that the variation in the propensity score leveraged for identification is driven entirely by the way students trade-off academic achievement and travel time as implied by the preferences estimated in the conjoint, and *not* from differences in the estimated preferences across students with different characteristics. The latter source of variation will be mechanically related to the student-level characteristics used for extrapolation and hence could create an avenue for confounding, which we avoid by including cell-fixed effects.

The outcome $Y_i$ is a cognitive or non-cognitive measure observed in 2022, and $D_i$ is an indicator for remote enrollment. The variable $\tilde{p}_i$ is the demeaned, rescaled propensity score: $\tilde{p}_i = (\hat{p}_i - \bar{\hat{p}})/0.10$, where $\hat{p}_i = \Lambda(\hat{v}_i)$ is the logistic transformation of the household preference index defined in Section 5, demeaned around the sample mean and rescaled so that one unit of $\tilde{p}_i$ corresponds to 10 percentage points of propensity. We report robust standard errors clustered at the school level to account for correlation within schools.[28]

As highlighted in Section 5, the parameter $\beta$ is the average treatment effect of remote enrollment for the average student. Because $\tilde{p}_i$ is demeaned, averaging the conditional average treatment effect $\beta + \psi\tilde{p}_i$ over the full sample implies that the ATE=$\beta$. The parameter $\theta$ governs how expected outcomes under in-person schooling vary across the propensity distribution after conditioning on cell fixed effects and baseline characteristics. The parameter $\psi$ governs selection on demand intensity: how the treatment effect varies across the propensity distribution. Both $\theta$ and $\psi$ are substantively informative about the patterns of sorting that drive the distribution of treatment effects across students.

The framework distinguished three average treatment effect parameters: the average treatment effect (ATE), the average treatment effect on the treated (ATT), and the average treatment effect on the untreated (ATU). Under Equation 13, each is a simple linear combination of $\beta$ and $\psi$ as follows:

$$\begin{aligned}
\text{ATE} &= \beta, \\
\text{ATT} &= \beta + \psi \cdot \mathbb{E}[\tilde{p}_i \mid D_i = 1], \\
\text{ATU} &= \beta + \psi \cdot \mathbb{E}[\tilde{p}_i \mid D_i = 0].
\end{aligned}$$

[27] Recall that the cell fixed effects defined above are the complete interaction of poverty status, underrepresented minority status, a high-school indicator, and four indicators for baseline ability. The vector $X_i$ includes indicators for English learner status (with a separate dummy for missing data), female, and special education status (see Appendix Table G.1 for the associated coefficients); however, these additional controls are only included for completeness. Our substantive conclusions are not sensitive to omitting these additional controls (see Appendix Table G.2).

[28] We discuss estimates that account for estimation error in the preferences in Section 9.

The ATT reflects the average effect of remote enrollment on the students who actually enrolled, who have higher propensity scores than the average student by construction; if $\psi > 0$ (positive selection on demand intensity), the ATT exceeds the ATE. The ATU reflects the effect of remote enrollment for the typical in-person student, who has a lower propensity score; if $\psi > 0$, the ATU lies below the ATE. We report all three parameters in our main results.

### 7.4 Balance Tests and Common Support

Next, we provide an initial test of the validity of our empirical approach by assessing balance on baseline student characteristics using our propensity-score based method. Specifically, we use measures of baseline student characteristics as dependent variables in specifications based on Equation 13. Figure 3 reports estimates of the coefficient on a remote-learning indicator from these balance tests.

As a benchmark, the left-most bars report estimated differences adjusting for a standard set of demographic controls: poverty status, English learner status (with a separate indicator for missing status), special education status, an indicator for female, and under-represented-minority status. As dependent variables, we consider three baseline student characteristics: lagged average achievement (the mean of lagged ELA and math scores), an overall bullying index; and a grit index. The bullying and grit indices are constructed from the student survey data. After adjusting for demographic controls, remote students score approximately 27 percent of a standard deviation lower on baseline achievement and have worse bullying and grit outcomes.

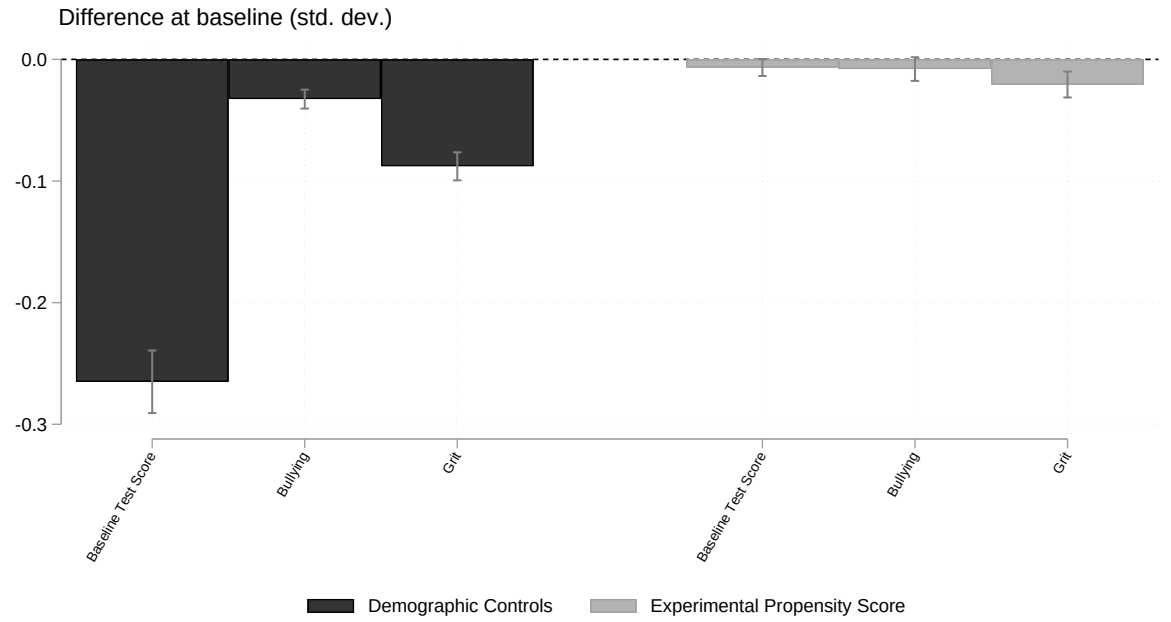
In contrast, the right-most bars show that the experimental propensity score strategy strongly eliminates differences in baseline achievement between students who do and do not enroll in remote learning. Specifically, we find that the estimated remote-in-person gap in baseline achievement falls to essentially zero at $-0.007$ standard deviations, and the corresponding differences in bullying and grit are similarly small ($-0.008$ and $-0.021$, respectively). To formalize the contrast between the two approaches, we test and reject that the observational and propensity score-based differences are equal at the one-percent level for all three outcomes. In sum, moving from standard demographic controls to the experimentally derived propensity score significantly changes the estimated baseline imbalance between remote and non-remote students.

Finally, we further probe the empirical support for our strategy by plotting the distribution of the propensity score $\hat{p}_i$ across treatment groups. Appendix Figure E.5 shows the density of $\hat{p}_i$ separately for students who enroll in remote learning and those who do not. The two distributions overlap across the full support of $\hat{p}_i$. At the same time, the remote distribution is shifted toward higher values, consistent with the experimental propensity scores capturing meaningful variation in who selects into remote learning.

## 8 Main Results

We provide a comprehensive assessment of remote learning's impacts on cognitive and non-cognitive outcomes from the 2021-2022 academic year. Our cognitive outcomes are standard-

Figure 3: Balance Tests on Baseline Student Characteristics




*Notes:* This figure reports balance tests for three baseline student characteristics: the average of lagged (2019) standardized math and ELA achievement scores, a bullying index, and a grit index. The bullying and grit indices are normalized so that larger values denote better outcomes. Each bar reports the estimated coefficient on an indicator for remote enrollment from a regression in which the baseline characteristic is the dependent variable. The results from the demographic controls model are based on specifications that adjust for a standard set of demographic controls: poverty status, English learner status (with a separate indicator for missing data status), special education status, an indicator for female, and under-represented-minority status. Results from the experimental propensity score are based on specifications that control for the propensity score from our preferred model. Whiskers denote 95 percent confidence intervals based on robust standard errors clustered at the school level.

ized test scores, and the non-cognitive are from the School Experience Survey. We begin by estimating the average effects of remote learning on these outcomes. We then turn to study heterogeneous impacts using the selection-on-levels and selection-on-gains parameters to characterize how outcomes vary across the remote-learning propensity distribution. The latter allows us to assess the empirical importance of match effects and whether families sort into remote schooling on the basis of relative gains. We then broaden the heterogeneity analysis to examine how treatment effects vary across observable student characteristics. Finally, we use lottery variation to provide an experimental validation of the heterogeneous treatment effects implied by our approach.

## 8.1 Mean Test Score Impacts

We begin by examining the average effects of remote learning on cognitive outcomes. Panel A in Table 4 reports average effects for achievement in ELA and math. Column (1) reports results from a conventional value-added approach where we estimate remote impacts using a specification that conditions on student attributes and lagged achievement. These estimates imply negative remote-learning effects ranging from 23 to 24 percent of a standard deviation. These effects are consistent with other studies employing alternative quasi-experimental methods that also find negative selection into remote learning (Bueno, 2020). By contrast, the estimates from

our preferred approach based on Equation 13, reported in Column (2), are more modestly negative. ELA and math effects for the average student are $-0.17\sigma$ and $-0.18\sigma$, respectively. As robustness, we also formally test that the average treatment effects estimated in the full sample equal the average treatment effects estimated in the conjoint sample, and we fail to reject equality.[29] Moreover, Appendix Figure E.4 also reports the full distribution of treatment effects across the two samples, underscoring the qualitatively similar evidence in the conjoint sample, albeit with substantially less precision. These results corroborate recent evidence suggesting that remote learning tends to produce adverse outcomes for the average student (Bueno, 2020, Goldhaber et al., 2022, Jack et al., 2022, Singh et al., 2022). Panel A in Table 4 reports the mean effects in Column 2, along with other selection parameter estimates, which we will discuss shortly.

Table 4: Causal Effects of Remote Learning on Key Outcomes

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | VAM | Preferred Model | | | | | ATE Equality $p$-values | |
| | ATE | ATE | ATT | ATU | Levels | Gains | VAM=Pref. | Full=Conj. |
| Panel A: Cognitive Outcomes | | | | | | | | |
| ELA ($N = 209,339$) | -0.231 | -0.167 | -0.135 | -0.171 | -0.273 | 0.111 | $< 0.01$ | 0.490 |
| | (0.010) | (0.007) | (0.007) | (0.007) | (0.013) | (0.005) | | |
| Math ($N = 210,367$) | -0.238 | -0.180 | -0.148 | -0.183 | -0.267 | 0.112 | $< 0.01$ | 0.259 |
| | (0.010) | (0.007) | (0.007) | (0.007) | (0.013) | (0.006) | | |
| Panel B: Non-Cognitive Outcomes | | | | | | | | |
| No Bullying Index ($N = 179,586$) | 0.128 | 0.142 | 0.140 | 0.143 | -0.035 | -0.006 | $< 0.01$ | – |
| | (0.004) | (0.004) | (0.005) | (0.004) | (0.005) | (0.003) | | |
| No Physical Bullying ($N = 179,496$) | 0.242 | 0.264 | 0.258 | 0.264 | -0.041 | -0.014 | $< 0.01$ | – |
| | (0.005) | (0.005) | (0.005) | (0.005) | (0.006) | (0.003) | | |
| No Online Bullying ($N = 175,159$) | 0.144 | 0.162 | 0.154 | 0.164 | -0.016 | -0.021 | $< 0.01$ | – |
| | (0.004) | (0.005) | (0.005) | (0.005) | (0.005) | (0.003) | | |
| Grit Index ($N = 181,849$) | 0.004 | 0.003 | 0.023 | -0.000 | -0.051 | 0.053 | 0.750 | – |
| | (0.005) | (0.006) | (0.006) | (0.006) | (0.006) | (0.003) | | |

*Notes:* This table reports estimates of the effects of remote learning on cognitive and non-cognitive outcomes measured in 2022. Panel A reports ELA and math achievement; Panel B reports bullying outcomes and grit. Index variables are normalized so that larger values denote better outcomes. Column (1) reports the ATE from a canonical value-added specification. In addition to a remote-enrollment indicator, this model conditions on the demographic controls (poverty status, English learner status, special education status, an indicator for female, and under-represented-minority status) together with lagged (2019) standardized measures of ELA and math achievement, along with separate indicators for whether a student is missing either of the baseline scores. Columns (2)–(6) report estimates from our preferred model in Equation 13: the ATE, ATT, and ATU in Columns (2)–(4), and selection on levels ($\theta$) and gains ($\psi$) parameters in Columns (5) and (6). Because the propensity score is demeaned, the ATE equals the coefficient on remote enrollment ($\beta$); the ATT and ATU equal $\beta + \psi E[\tilde{p}_i \mid D_i = 1]$ and $\beta + \psi E[\tilde{p}_i \mid D_i = 0]$, respectively. Propensity scores are scaled so that one unit corresponds to 10 percentage points. Column (7) reports the $p$-value from a test that the VAM ATE in Column (1) equals the preferred-model ATE in Column (2). Column (8) reports $p$-values from tests that the preferred-model ATE estimated in the full sample equals the corresponding ATE estimated using only the conjoint sample. These tests are reported only for the cognitive outcomes because few conjoint respondents can be linked to the School Experience Survey. Standard errors are robust and clustered at the school level; standard errors for the ATT and ATU are calculated using the delta method.

[29] We conduct this comparison only for the cognitive outcomes because the overlap between the conjoint sample and the School Experience Survey sample is too small to support corresponding tests for the non-cognitive outcomes.

### 8.2 Mean Non-Cognitive Impacts

Impacts on academic outcomes shed light on one side of the story but potentially paint an incomplete picture. Bacher-Hicks et al. (2022) find that reductions in bullying during the pandemic-induced remote learning experience contributed to mixed evidence on students' mental health and well-being, suggesting that students who were bullied pre-pandemic may have benefited from the remote experience along non-academic margins. The existing evidence motivates our focus on bullying and other non-cognitive outcomes of interest, such as grit, which has also been shown to be affected by schools (Jackson et al., 2020).

Panel B of Table 4 shows a substantial mean improvement, roughly $0.14\sigma$, in the bullying index and a precise mean null impact on the grit index. The bullying index incorporates several dimensions of students' bullying experiences, including physical and online bullying. The reduction in physical bullying is partly mechanical, as remote students have less frequent encounters with their peers. Non-physical bullying materializes online through social media channels, which has also been linked to recent increases in adolescent depression (Twenge, 2017, Twenge et al., 2022, 2020). We find improvements in both physical and online bullying, with physical bullying mean impacts greater than the magnitude of online-bullying impact, amounting to $0.26\sigma$ and $0.16\sigma$ improvements, respectively. This suggests that the reduction in bullying operated in ways that potentially improved the overall mental well-being of students and that may complement academic outcomes.

### 8.3 Selection and Treatment Effect Heterogeneity

Our measures of preference intensity allow us to characterize both selection on levels into remote learning and also heterogeneity in its causal effect. For models with academic achievement as an outcome, the parameter $\theta$ governs selection on levels. It serves as a summary measure of the academic preparedness of students with varying degrees of remote learning demand. For models with bullying as the outcome, $\theta$ is a summary measure of students' proclivity to be bullied independent of the type of school in which they enroll. These summary measures governing sorting patterns allow us to paint a more complete picture of the factors governing sorting into the remote sector in the post-pandemic landscape. We begin by characterizing students sorting into remote schooling based on their academic achievement and bullying-related outcomes.

Starting with academic potential, Column 5 of Panel A in Table 4 demonstrates that students with higher estimated demand for remote learning perform more poorly on standardized exams than students with lower demand regardless of the school in which they enroll. Similarly, Panel B shows that students with higher demand for remote learning have worse bullying-related outcomes regardless of their enrolled school. The difference between a student at the 90th and 10th percentile of the propensity-score distribution—holding constant other covariates in the model—is roughly $-0.11\sigma$, a strong indication that students with worse bullying-related outcomes are substantially more likely to sort into virtual schooling. This is consistent with the evidence in Table 1 showing that remote-learning students have worse baseline test scores and bullying outcomes; however, the results using the experimentally derived measure of demand reveal more specific sorting patterns linked to remote-learning demand.

Next, we turn to our main analysis of the heterogeneous effects of remote learning. As

motivated in our framework, the experimentally derived propensity score allows us to assess how treatment effects vary with the estimated demand for remote learning. The parameter $\psi$ summarizes match effects, where $\psi > 0$ is an indication of positive sorting on gains. Table 4 reports point estimates from our preferred specification, while Figure 4 summarizes these results by plotting the mean treatment effects (i.e., $\hat{\beta} + \hat{\psi}\tilde{p}$), calculated separately for 12 bins of the demeaned propensity score.[30] The upward slope shown in the figure reflects that the interaction coefficient $\hat{\psi}$ is positive at around 0.111 and 0.112 for ELA and math, respectively. The match effects are sizable, generating positive remote effects for a small share of students with large estimated remote-learning demand. The selection patterns imply an ATT that is greater than the ATE, reported in Column 3 of Panel A in Table 4.[31] Reassuringly, Appendix Tables G.5 and G.6 show that our estimated coefficients and the implied average treatment effects (ATE, ATT, and ATU) remain qualitatively similar when we relax the linearity assumption in our approach and estimate a specification that includes quadratic terms for the propensity score and its associated interactions.[32]

For bullying outcomes, Panel B of Table 4 reports a more nuanced set of results. The omnibus bullying index, including both online and physical bullying and reported in the first row, reveals mild negative selection on gains relative to a sizable $0.142\sigma$ main effect of remote learning. The subsequent two rows differentiate between physical and online bullying. In all cases, the match effects are fairly small. As a consequence, the ATTs for bullying outcomes reported in Column 3 of Panel B in Table 4 are similar to the corresponding ATE.

Turning to another non-cognitive outcome, grit, we do not find meaningful mean impacts of remote learning. However, this mean impact masks heterogeneity. The difference in treatment effects between students at the 90th and 10th percentiles of the propensity score distribution is $0.17\sigma$. This evidence is consistent with prior work showing grit is malleable in schools (Alan et al., 2019, Jackson et al., 2020). Match effects are positive, pointing to another outcome dimension along which families appear to sort on gains.

How do we interpret this collection of results? Taking the estimates for math achievement in Figure 4 at face value suggests that students in the top decile of the estimated propensity score distribution experience little to no academic loss, and that some even have positive treatment effects. For a student at the 95th percentile of the propensity score distribution, the estimated effect of remote learning on math achievement is approximately $0.035\sigma$. These may be students for whom self-paced learning is better suited (Armstrong-Mensah et al., 2020) or who benefit from reduced social pressure or bullying (Bacher-Hicks et al., 2022).[33] Table 5 characterizes the baseline characteristics of these high demand students. Compared with students in the

[30] Appendix Table G.4 reports the corresponding estimated effects at selected percentiles of the propensity-score distribution.

[31] The ATT is still negative in both subjects ($-0.135\sigma$ in ELA and $-0.148\sigma$ in math). In terms of a purely achievement-maximizing benchmark, the remote sector is therefore larger than it would be if enrollment were limited to students with positive academic effects. We turn below to studying the non-academic benefits that may rationalize this choice for many families.

[32] Specifically, we estimate the following model: $Y_i = \alpha_c + \gamma' X_i + \beta D_i + \theta_1 \tilde{p}_i + \theta_2 \tilde{p}_i^2 + \psi_1 (\tilde{p}_i \times D_i) + \psi_2 \left(\tilde{p}_i^2 \times D_i\right) + \epsilon_i$.

[33] We caution that our design does not identify the mechanism behind the cognitive gains at the top of the propensity distribution: because reduced bullying is one of several remote-specific factors bundled into $v_i$, we can show that achievement gains accrue to high demand students, but this pattern does not establish that reduced bullying, or any other single channel, is what produces them.

bottom 90 percent of the propensity-score distribution, those in the top decile have substantially lower baseline math and ELA achievement, report worse bullying and grit outcomes, are more likely to receive special education services, be members of underrepresented minority groups, be English learners, and be economically disadvantaged. Thus, the students predicted to benefit most from remote instruction are not conventionally advantaged students; rather, they are disproportionately disadvantaged across several academic, social, and demographic dimensions.

The preceding results identify a group of high-demand students for whom remote learning is estimated to produce gains along both academic and social dimensions. For most students, however, the results point to a tradeoff between academic achievement and bullying outcomes. The estimated effects on physical and online bullying outcomes are positive throughout the demand distribution, including for students who experience academic losses from remote instruction. This pattern is consistent with remote schooling generating a tradeoff between academic achievement and a safer or more comfortable social environment for many students. To investigate this further, Appendix Figure G.1 reports the share of students for whom remote learning is estimated to reduce achievement while improving bullying outcomes. In the full sample, most students are estimated to experience negative learning effects alongside improvements in bullying outcomes.[34] Among students who actually enrolled in remote schooling, this share falls to approximately 84 percent, consistent with the positive selection on achievement gains that we document. Thus, improvements in bullying provide a potentially important compensating benefit for a majority of students who select into remote schooling, while a nontrivial share are estimated to experience positive effects along both academic and social dimensions.[35]

### 8.4 Treatment Effect Heterogeneity by Student Subgroup

The preceding analysis demonstrated how treatment effects vary with preferences. We now assess treatment effect heterogeneity with respect to other observables by focusing on four margins: socioeconomic status, baseline achievement, gender, and grade-level.

Appendix Tables G.7 and G.8 report estimates of our baseline model fully interacted with poverty status and baseline achievement. Appendix Table G.7 shows that average achievement losses are roughly two times as large for non-poverty students—students not flagged as economically disadvantaged by the district—as for students in poverty (e.g., $-0.298\sigma$ vs. $-0.15\sigma$ in ELA). Non-poverty students also see larger improvements in bullying ($0.181\sigma$ vs. $0.136\sigma$). Appendix Table G.8 shows that low-achieving students—those with pre-pandemic achievement below the district average—experience achievement effects that are negative, but small, which stands in contrast to much larger negative effects for high-achieving students. Taken together, these results show that academic losses from remote instruction are larger among non-poverty students and among students with higher baseline achievement. This may be due to the fact that non-poverty and higher baseline achievement students tend to be drawn into remote learning from higher performing schools.[36]

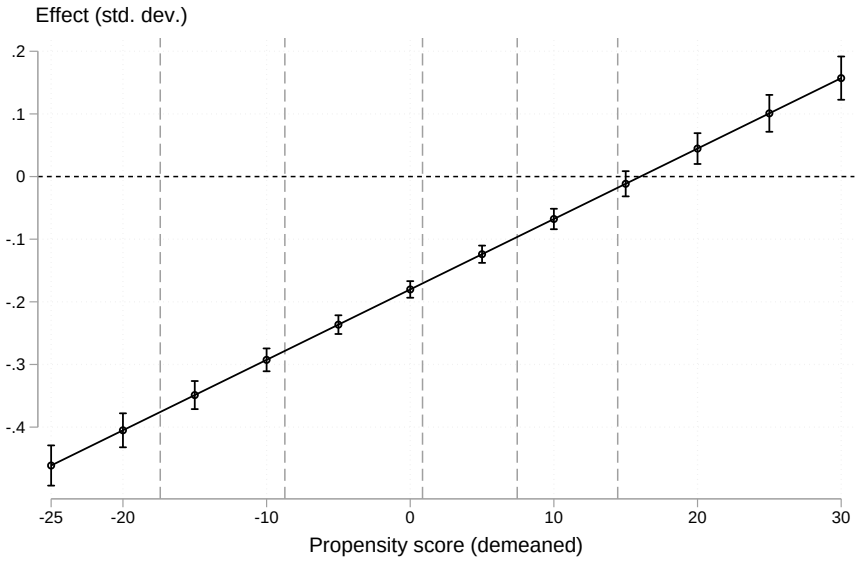
[34]Note that we do not estimate negative bullying effects for any student in our sample.

[35]Notably, Appendix Table G.3 reports qualitatively similar impacts on 2023 outcomes, showing that remote enrollment in 2022 has persistent effects.

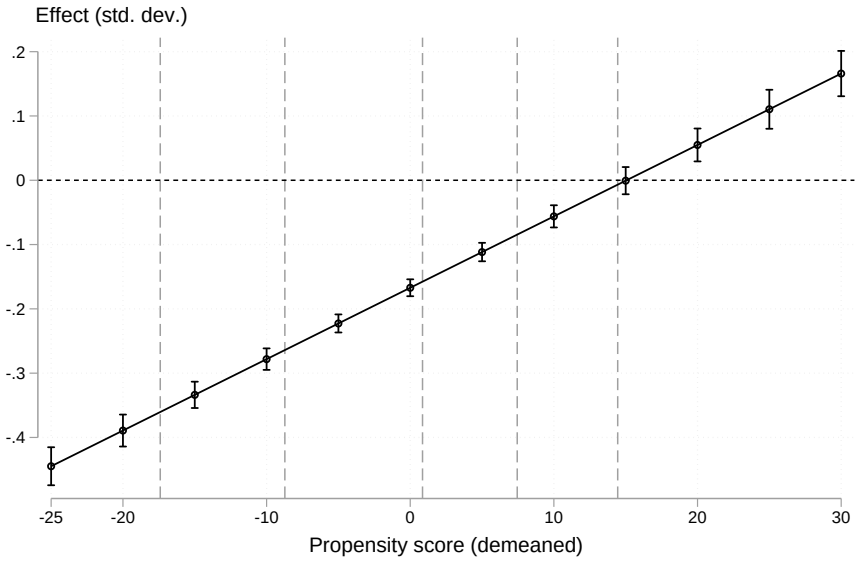
[36]In our sample, non-poverty students who are not enrolled in remote instruction attend schools where average test scores are roughly 44% of a standard deviation higher than their lower-SES peers. Similarly, the school

Figure 4: Estimated Match Effects on Post-Pandemic Test Score Achievement

(a) Math




(b) ELA




*Notes:* This figure reports estimates of treatment effects on post-pandemic math and ELA test scores for 12 bins of estimated propensity scores. The points in black are means for each bin based on Equation 13 and are constructed using the estimated conditional treatment effect (i.e., $\hat{\beta} + \hat{\psi}\tilde{p}$ where $\tilde{p}$ is the demeaned propensity score scaled so that one unit corresponds to 10 percentage points). The horizontal axis reports the corresponding demeaned propensity score in percentage points. Because the propensity score is demeaned, the estimate at zero corresponds to the average treatment effect. The five dashed, gray vertical lines correspond to the 10th, 25th, 50th, 75th, and 90th percentiles of the propensity score distribution. Standard errors are robust and clustered at the school level. Bars surrounding the mean estimate for each bin are estimates of the 95 percent confidence intervals. Sample sizes for both panels are reported in Table 4.

Table 5: Baseline Characteristics by Remote-Learning Propensity

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | | Remote-Learning Propensity | | |
| | Full Sample | Top Decile | Bottom 90% | Diff. |
| Baseline ELA Scores | 0.019 | -0.764 | 0.106 | -0.870 |
| | (0.660) | (0.720) | (0.592) | (0.028) |
| Baseline Math Scores | 0.022 | -0.916 | 0.126 | -1.042 |
| | (0.663) | (0.624) | (0.581) | (0.032) |
| Baseline Bullying Index | 0.008 | -0.070 | 0.018 | -0.088 |
| | (0.631) | (0.786) | (0.606) | (0.009) |
| Baseline Grit Index | 0.015 | -0.199 | 0.044 | -0.243 |
| | (0.664) | (0.699) | (0.654) | (0.011) |
| Female | 0.488 | 0.478 | 0.490 | -0.012 |
| | (0.500) | (0.500) | (0.500) | (0.004) |
| Special Education | 0.125 | 0.254 | 0.110 | 0.144 |
| | (0.331) | (0.435) | (0.314) | (0.007) |
| URM | 0.809 | 0.952 | 0.793 | 0.159 |
| | (0.393) | (0.213) | (0.405) | (0.012) |
| English Learner | 0.350 | 0.584 | 0.324 | 0.261 |
| | (0.477) | (0.493) | (0.468) | (0.017) |
| Poverty | 0.820 | 0.981 | 0.802 | 0.179 |
| | (0.384) | (0.137) | (0.398) | (0.011) |
| Students | 210,367 | 21,035 | 189,332 | |

*Notes:* This table reports summary statistics for students based on their estimated demand for remote schooling. Column (1) reports means for the main analysis sample used for math achievement. Column (2) reports means for students in the top decile of the estimated propensity score distribution, and Column (3) reports means for students in the bottom 90 percent of the distribution. Column (4) reports mean differences. Columns (1)-(3) report standard deviations below each mean, while Column (4) reports the associated standard error from the regression-based difference in means. Standard errors are robust and clustered at the school level.

Appendix Tables G.9 and G.10 report estimates separately by gender and grade level. Achievement effects on ELA are roughly gender-neutral, whereas for math, remote instruction appears to be less harmful for girls on average. Turning to non-cognitive outcomes, we observe that while both genders experience gains on bullying, these improvements are significantly larger for boys. This aligns with a broad body of evidence that boys' behavioral development is quite responsive to the schooling environment (Autor et al., 2019, Bertrand and Pan, 2013, Jacob, 2002). For the grit index, boys experience negative average effects, while girls experience the opposite. Looking at heterogeneity by grade level, the achievement costs of remote learning are largest among younger students and smallest in high school. Elementary school students experience the largest losses in math, while elementary and middle school students experience similarly large losses in ELA. At the same time, improvements in bullying are larger in middle school than in high school, which is consistent with the fact that bullying tends to peak in middle school.[37]

### 8.5 Lottery-Based Validation

While the balance analysis in Section 7.4 is reassuring, this section explores another way to assess the credibility of our main empirical approach. Specifically, we use lottery variation to experimentally validate the heterogeneous treatment effect estimates generated from our conjoint-based model of achievement. As we detail below, following Deming (2014), we test whether the model-implied effects are forecast unbiased: whether, on average, they accurately predict the local average treatment effects for lottery compliers whose fall-back option is remote learning. We fail to reject this hypothesis. Because we observe multiple lotteries, we also use overidentification tests to assess whether the lottery-specific IV variation is consistent with a common relationship between model-predicted and observed achievement (Angrist et al., 2017).

LAUSD has a large portfolio of offerings beyond neighborhood schools such as magnet programs, affiliated charter schools, and schools with selective criteria. Oversubscribed schools use lotteries to determine offers, and we use this lottery variation to validate our empirical approach. Intuitively, some students may hold remote learning as their most preferred alternative to a given oversubscribed school. For these students, randomly failing to receive an offer at their most preferred school exogenously increases the likelihood they attend remotely. Thus, if we find that our conjoint-based estimates of the student level heterogeneous effects from our main analysis predict the lottery-based causal effect for students who are shifted into remote, then this constitutes strong evidence in favor of the validity of our main approach.

For the 2021-2022 academic year, there were 32 oversubscribed programs where students had remote learning as a fallback option. The random variation induced by losing the lottery allows us to assess both the average predictive validity of our approach and whether the lottery-specific variation in observed achievement is consistent with a common proportional relationship to the predictions implied by our preferred model. Let $Y_i$ correspond to the observed outcome of individual $i$ and let $\tilde{Y}_i$ be the *predicted* outcome based on the parameter estimate results from

test score gap between high achieving and low-achieving students not enrolled in remote instruction is 22% of a standard deviation.

[37] Nationally representative U.S. data show substantially higher rates of reported bullying in grades 6–8 than in high school (Nansel et al., 2001, National Center for Education Statistics, 2024).

Equation 13, the observed decision to go remote, and the associated propensity score estimates. Formally, we estimate the following model via two-stage least-squares:

$$Y_i = \phi_0 + \phi_1' W_i + \phi_2 \tilde{Y}_i + \epsilon_{i2}, \tag{14}$$

$$\tilde{Y}_i = \pi_0 + \pi_1' W_i + \sum_{\ell} \pi_\ell Z_{i\ell} + \epsilon_{i1}, \tag{15}$$

where $W_i$ is a vector of school-by-grade lottery strata so that lottery offers (denoted by $Z_{i\ell}$ for each of the $\ell$ lotteries) are random conditional on $W_i$. Appendix Table G.14 provides summary statistics for the lottery sample, as well as associated tests for balance and differential attrition. We fail to reject equality between lottery offers and no-offer students in terms of baseline characteristics and the likelihood of attrition. We interpret this as reassuring evidence of internal validity.

The parameter of interest from Equation 14 is $\phi_2$. A coefficient estimate of unity ($\phi_2 = 1$) indicates that the estimated treatment effect heterogeneity from our preferred model is forecast unbiased for the actual causal effects implied by the lottery. To interpret this, consider the case of standardized math scores. The finding of $\hat{\phi}_2 = 1$ shows that variation in conjoint-based predicted math achievement based on the lottery variation from $Z_{i\ell}$ modeled in the first-stage accurately predicts the *observed* variation in test scores in the second-stage. Tests of this nature are common in the education literature (e.g., Angrist et al., 2017).

Appendix Table G.15 reports the results. For both math and ELA, the first stage is well-powered, with the $F$-stats in excess of 10. For math, the forecast coefficient is 1.01, indicating that the lottery-induced variation in observed achievement closely matches the variation predicted by our preferred model. For ELA, the forecast coefficient is 0.68. We fail to reject the null hypothesis that the forecast coefficient equals one for either math or ELA; we likewise fail to reject forecast unbiasedness in the pooled specification. In addition, over-identification tests in the spirit of Angrist et al. (2017) fail to reject that the lottery-specific IV estimates are consistent with a common forecast coefficient (over-ID $p$-values of 0.45, 0.16, and 0.30 for the math, ELA, and pooled specifications, respectively). Thus, taken together, these results provide reassuring evidence that the heterogeneous treatment effects predicted by our preferred model are consistent with the causal variation generated by the school-choice lotteries.

Finally, we note here that an important feature of this validation is that it is not restricted to the conjoint estimation sample. The predictions evaluated in the lottery exercise are constructed for the broader student sample and therefore rely on extrapolated preferences for students outside the conjoint sample. If selection into the conjoint sample or the extrapolation of preferences materially distorted these predictions, we would expect the model to perform poorly in the lottery-based validation. The fact that it does not provides complementary evidence that concerns about extrapolation validity are not driving our results. More generally, the lottery results provide reassuring evidence for the overall validity of our empirical approach.

# 9 Additional Robustness Exercises

Having validated our preferred model using lottery variation, this section concludes the empirical analysis by examining the robustness of our findings to alternative inference procedures and preference specifications. First, we account for estimation uncertainty in the preference estimates used to construct the experimentally-derived propensity score. Second, we examine sensitivity to alternative parameterizations of household utility and to the choice model likelihood used to recover preferences.

**Accounting for preference-estimation uncertainty:** The main estimates and inference do not account for estimation error introduced in the preference estimation stage. To account for this error in the first-stage, we sample from the asymptotic joint distribution of the preference estimates 250 times. Equipped with these estimates, we construct the implied propensity score for each iteration and estimate Equation 13. We then report the mean estimate across iterations along with the 95 percent confidence region. Appendix Table G.13 and Appendix Figures G.3 and G.4 show that accounting for estimation error in the propensity scores does not qualitatively affect our estimates or inference.

**Robustness of the preference model:** Our preferred model uses a linear parameterization of preferences for travel time and academic quality and assumes no interactions with preferences over remote learning. As robustness checks, we estimate alternative specifications that allow for non-linear travel costs and various interaction terms with the remote schooling indicator. Panels (a)–(c) of Appendix Table G.11 report estimates that are remarkably similar to our preferred estimates from the more parsimonious model.

We also verify that our preference estimates are not sensitive to the specific choice-model likelihood used to recover them. Our baseline uses a rank-ordered logit; as an alternative, we re-estimate preferences using a conditional logit (Chamberlain, 1980), which recovers the attribute preferences from within-respondent variation across the ten choice scenarios while differencing out a respondent-specific utility component common to all alternatives within a scenario.[38] The resulting treatment-effect estimates are quantitatively very similar (Appendix Table G.12).

# 10 Conclusion: Policy Implications and Future Research

The COVID-19 pandemic accelerated national growth in remote-learning enrollment substantially (see Figure 1). As of 2025, roughly 800,000 U.S. students were engaged in remote instruction, placing the sector at a meaningful scale relative to Catholic and charter school, which enroll 1.7 and 4 million students, respectively. School districts have since continued to expand remote options to satisfy continued parental demand (Musaddiq et al., 2022).

At first glance, it may seem puzzling why parents and students would willingly select into remote schooling options. Before the pandemic, a near consensus suggested that virtual schools negatively affect learning (Bueno, 2020, Fitzpatrick et al., 2020, Raymond et al., 2023). More

[38] Because the conditional logit implementation is designed for sequences of binary choices, we convert each three-option ranking into a series of pairwise choices indicating which of the two alternatives is ranked higher.

recent studies on pandemic-era remote schooling similarly document learning losses (Goldhaber et al., 2022, Jack et al., 2022, Singh et al., 2022).

In this paper, we shed light on the drivers of growth in remote learning by studying selection and the heterogeneous causal impacts of this learning modality. Our evidence is based on original survey data that we combine with a selection-on-observables framework that uses experimentally elicited preferences to measure the selection-relevant heterogeneity governing enrollment into remote schooling. The survey was conducted after schools in Los Angeles returned to offering both traditional in-person and remote schooling options. When responding to our survey, parents and students were uniquely able to draw on their past experience with remote learning and firsthand understanding of this mode of instruction. The experimentally elicited preferences allow us to construct a selection index that we use to account for selection into remote learning and to examine how treatment effects vary with demand for remote schooling. We validate the new approach using school choice lotteries and demonstrate robustness on several margins.

Our analysis provides important evidence of heterogeneous impacts of remote learning that suggest this form of learning can indeed be a preferred schooling modality for two main reasons. First, we demonstrate that, while remote learning reduces achievement on average, there are positive match effects that are sufficiently strong to imply that some students with the highest demand are predicted to experience gains in academic achievement. Second, remote learning delivers an across-the-board improvement in bullying outcomes, including online bullying, relative to in-person learning. Importantly, these improvements in bullying outcomes may provide an important compensating benefit for students who experience academic losses from remote instruction.

The combined findings underscore the nuanced nature of parental preferences. While much of the existing literature has focused on parental preferences for different dimensions of academic quality (Abdulkadiroğlu et al., 2020, Ainsworth et al., 2023, Campos, 2023, Hastings and Weinstein, 2008, Rothstein, 2006), an emerging consensus emphasizes the multi-dimensional nature of the education production function (Beuermann et al., 2022, Jackson, 2018, Jackson et al., 2020). In this paper, we highlight an understudied but increasingly important dimension of the broader schooling environment: bullying (Bacher-Hicks et al., 2022). Evidence linking social media use and online harassment to adolescent depression (Twenge, 2017, Twenge et al., 2022, 2020) underscores why bullying-related experiences may matter for student well-being and, consequently, for families' schooling choices. Our findings show that learning modality affects both physical and online bullying, creating an important margin along which remote schooling may differ from in-person instruction. More broadly, the remote-schooling context reveals a tradeoff between academic achievement and bullying-related outcomes: for many students, remote instruction is estimated to reduce achievement while improving bullying outcomes, and families whose children experience more bullying at baseline exhibit substantially greater demand for remote schooling. More work is needed to better understand how parents trade off academic quality and bullying outcomes.

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# Online Appendix

## Who Benefits from Remote Schooling? Self-Selection and Match Effects

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November 12, 2026

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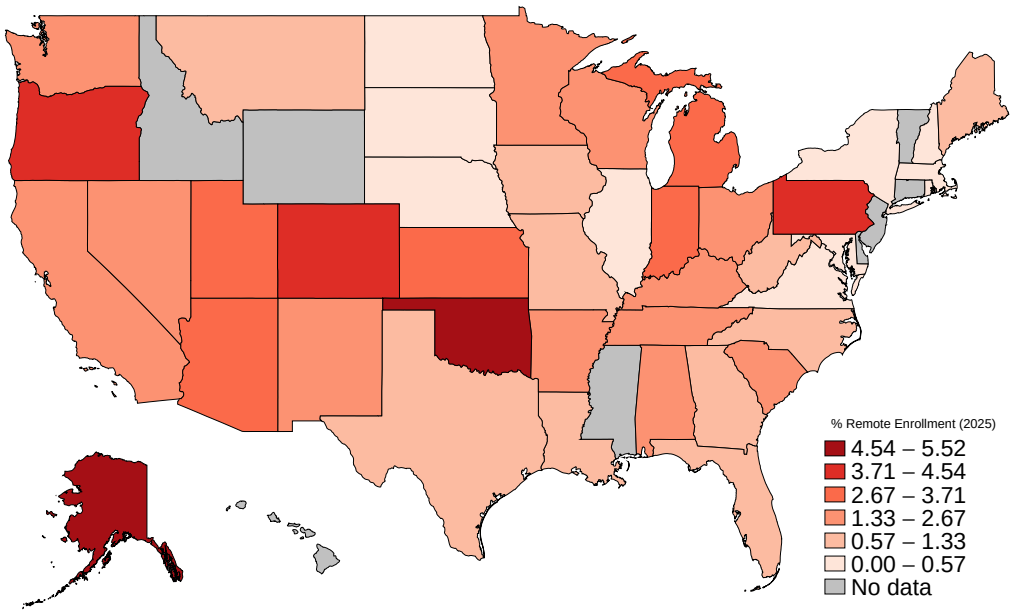
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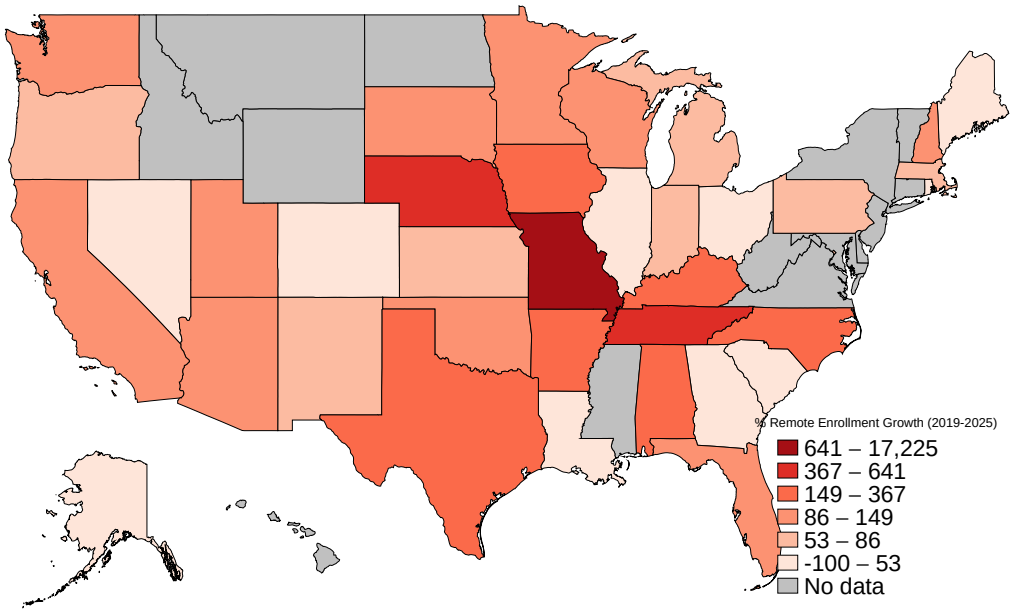
# A National and Descriptive Context

Figure A.1: Remote Schooling Enrollment Shares by State, NCES 2025



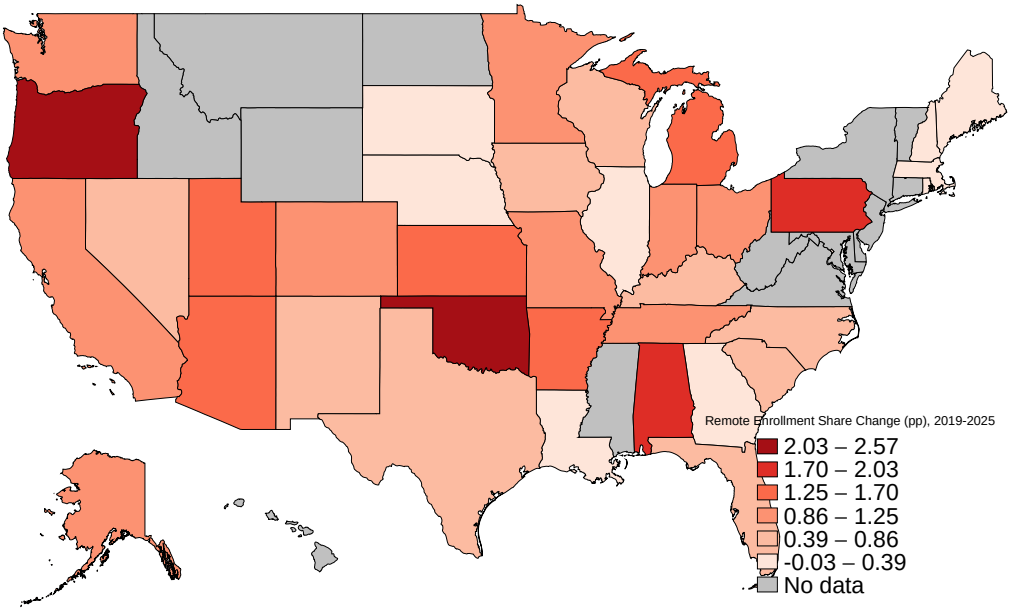

*Notes:* This figure reports exclusively virtual enrollment shares by state reported in the Common Core data provided by the National Center for Education Statistics (NCES). The cuts correspond to the 25th, 50th, 75th, 90th, and 95th percentiles of the state-level distribution. States shaded gray do not have an exclusively virtual enrollment observation in the 2025 NCES files; this may reflect zero enrollment or unreported virtual status.

Figure A.2: Remote Schooling Enrollment Percentage Change by State, NCES 2019-2025



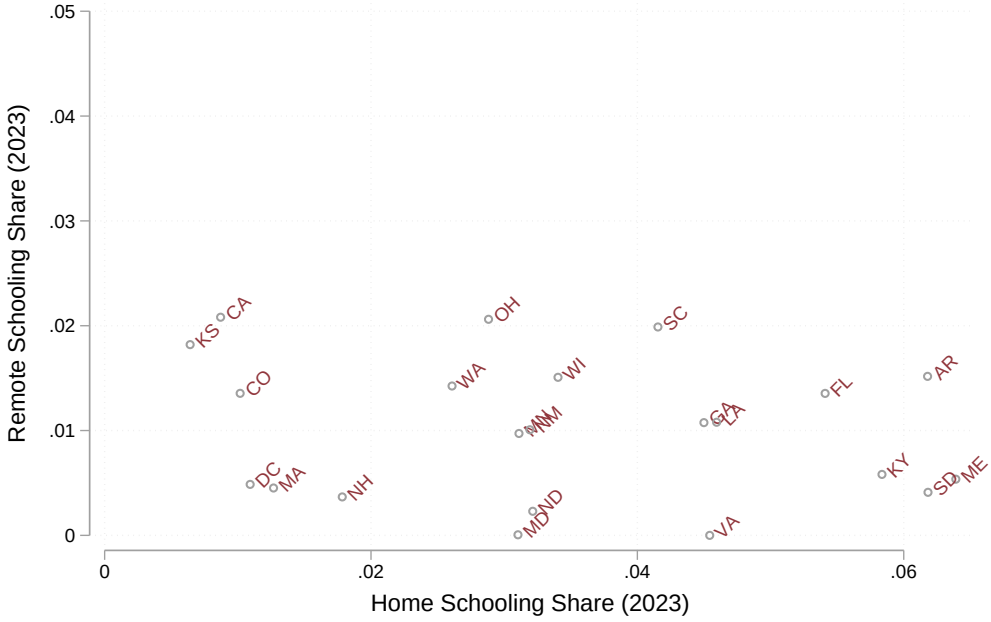

*Notes:* This figure reports 2019-2025 percent changes in exclusively virtual enrollment shares by state reported in the Common Core data provided by the National Center for Education Statistics (NCES). The cuts correspond to the 25th, 50th, 75th, 90th, and 95th percentiles of the state-level distribution. Missouri stands out with rapid enrollment growth: Missouri went from 52 fully virtual students in 2019 to 8780 in 2025. States shaded gray do not have an exclusively virtual enrollment observation in either 2019 or 2025; this may reflect zero enrollment or unreported virtual status.

Figure A.3: Remote Schooling Enrollment Percentage-Point Change by State, NCES 2019-2025




*Notes:* This figure reports 2019-2025 percentage-point changes in exclusively virtual enrollment shares by state using the Common Core data provided by the National Center for Education Statistics (NCES). Percentage-point changes are calculated as 100 times the difference between the 2025 and 2019 enrollment shares. Unlike the preceding figure, which reports percentage changes relative to each state's 2019 share, this figure reports absolute changes in enrollment shares. The cuts correspond to the 25th, 50th, 75th, 90th, and 95th percentiles of the state-level distribution. States shaded gray do not have an exclusively virtual enrollment observation in both 2019 and 2025; this may reflect zero enrollment or unreported virtual status.

Figure A.4: Remote and Homeschooling Shares by State, NCES 2023



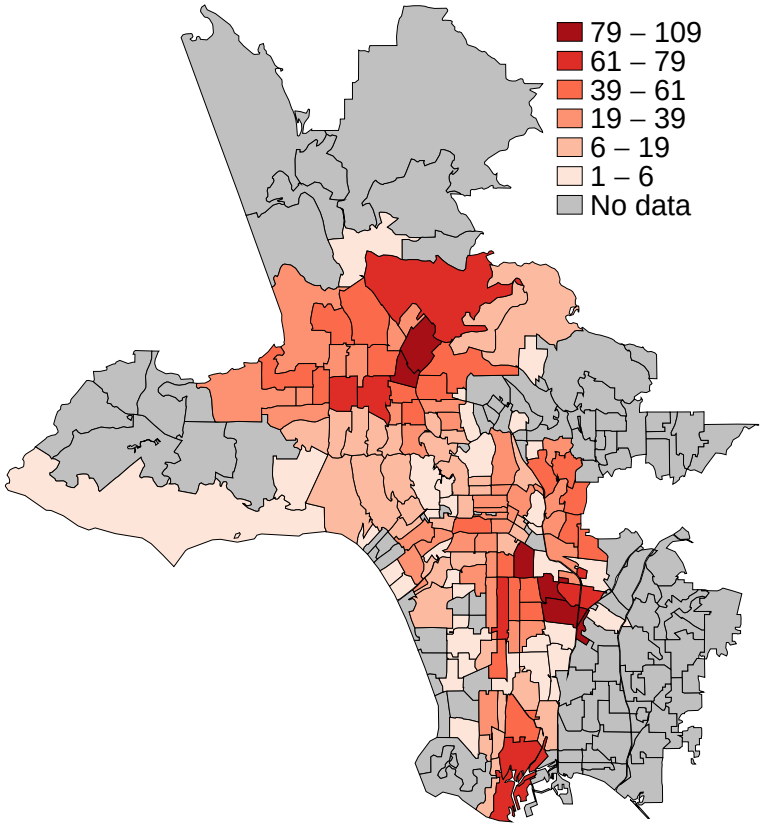

*Notes:* This figure reports the state-level bivariate relationship between remote and homeschooling shares. The remote enrollment share is reported in the Common Core data provided by the National Center for Education Statistics (NCES) and the homeschooling share is reported by the Washington Post. Observations are labeled with their state identifier.

Figure A.5: Remote and Homeschooling Percentage Change by State, NCES 2019-2023




*Notes:* This figure reports the state-level bivariate relationship between 2019-2023 percent changes in remote and homeschooling shares. The remote enrollment share is reported in the Common Core data provided by the National Center for Education Statistics (NCES) and the homeschooling share is reported by the Washington Post. Observations are labeled with their state identifier.

Figure A.6: Spatial Distribution of Remote-Learning Survey Respondents




*Notes:* This figure is a map illustrating the spatial distribution of survey respondents. Each shaded polygon corresponds to a zip code and is shaded according to the number of remote-learning respondents residing in the zip code. Most of the gray areas in the figure are outside the purview of LAUSD. The cuts correspond to the 25th, 50th, 75th, 90th, and 95th percentiles of the zip code-level distribution.

## B School Experience Survey Details

The School Experience Survey (SES) is an annual survey administered by the Los Angeles Unified School District (LAUSD) every academic year since 2010. The survey is administered to parents, students, and staff. Response rates for students and staff are high, while response rates for parents vary substantially: in academic year 2022-23, students had a 95% response rate, teachers had a 98% response rate, and parents had a 69% response rate. The core survey is organized into three broad domains: (i) Academics, (ii) School Climate, and (iii) Social and Emotional Learning. As a point of comparison, the survey elements mirror data collected by Chicago Public Schools (CPS), as studied by Jackson et al. (2020), as well as other large urban school districts. The Social and Emotional Learning section contains questions related to Growth Mindset, Grit, Responsible Decision-Making, Self Awareness, Self-Efficacy, Self-management, and Student Social Awareness. Using the data in the survey, we construct two indices that we use in our analysis: grit and bullying. The specific survey items that we use to construct each index are listed below.

**Bullying Index:** This index consists of six questions. The questions ask how frequently the student experienced a form of bullying during the past 30 days. We recode them so that positive values are interpreted as an improvement in bullying outcomes. The questions are:

1. How many times on school property have you had mean rumors or lies spread about you?
2. How many times on school property have you been teased about what your body looks like?
3. How many times on school property have you been made fun of because of your looks or the way you talk?
4. How many times on school property have you been pushed, shoved, slapped, hit, or kicked?
5. How many times on school property have you had sexual jokes, comments, or gestures made at you?
6. How many times have other students from your school bullied you online?

**Grit Index:** This index consists of 13 questions, each recoded so that positive values are interpreted as having more grit - greater perseverance, classroom engagement, and academic self-efficacy. The first five ask how true each of the following statements is for the student:

1. I came to class prepared.
2. I remembered and followed directions.
3. I got my work done right away instead of waiting until the last minute.
4. I paid attention, even when there were distractions.
5. I stayed calm, even when others bothered or criticized me.

The next four ask how true each of the following statements is for the student:

6. I can change my intelligence with hard work.
7. I can increase my intelligence by challenging myself.
8. I am capable of learning anything.
9. I can do well in a subject even if I am not naturally good at it.

The remaining four ask how confident the student is about each of the following at school:

10. I can earn high marks/grades in my classes.
11. I can do well on all my tests, even when they're difficult.
12. I can master the hardest topics in my classes.
13. I can meet all the learning goals my teachers set.

**Index Construction:** We convert each survey response into a component recoded so that higher values represent better outcomes. We standardize each component within grade level and survey year and construct each index as an average of the available standardized components.

# C Remote-Learning Survey Instrument

**LAUSD Remote Learning Survey**

**(untitled)**

1. Are you a mother, father, or guardian of a K-12 student? *

- Mother
- Father
- Guardian

2. In what grade is your oldest child currently enrolled? *

Kindergarten
1
2
3
4
5
6
7
8
9
10
11
12

3. Is your oldest child currently enrolled in a virtual schooling option?

- Yes
- No

4. Did you choose a remote option mostly for academic or safety (COVID) reasons? *

- Mostly academic reasons
- Mostly safety reasons
- Academics and safety were equally important

**(untitled)**

5. For the following, please tell us if you agree or disagree.*

| | Agree | Disagree |
|---|---|---|
| My child excelled academically with the virtual experience compared to in-person instruction. | | |
| I would like the district to expand its virtual offerings in the future. | | |
| I am likely to opt for virtual schooling in the future. | | |
| I enjoyed the virtual schooling experience during the pandemic. | | |

**(untitled)**

6. You will now see a sequence of scenarios, each with three school options that the school district could offer you in Fall 2022. For each set of three, indicate the one you prefer the most (Best) and the one you prefer the least (Worst).

Recall that a fully remote option is entirely virtual (100% remote) and traditional in-person instruction is 0% remote.

Travel time corresponds to the commute time in minutes from your home to the school. For traditional in-person instruction, students make the trip to school every day.

**Assume pandemic-related safety issues are as they were in 2019 before COVID.**

**Besides the characteristics shown, assume that these schools are otherwise identical in terms of their academic instruction and quality.**

There are no right or wrong answers to these questions. We only want to know which of the options you would most prefer.

*

| | | | |
|---|---|---|---|
| **Type of Instruction** | In Person | In Person | In Person |
| **Percent of students meeting state academic standards** | 50 | 30 | 90 |
| **Travel time to school (minutes)** | 15 | 30 | 45 |
| **Best** | ○ | ○ | ○ |
| **Worst** | ○ | ○ | ○ |

**(untitled)**

7. Do you think your choices will be similar in Fall 2023?*

- Yes
- No

**(untitled)**

8. Thank you for taking the time to answer these questions! We now ask that you let your student in grade 8 through 11 answer the remaining questions, so we can learn more about their experience with remote learning.

Will your child be answering the remaining questions? *

- Yes
- No

**(untitled)**

9. For the following, please tell us if you agree or disagree.*

| | Agree | Disagree |
|---|---|---|
| I am likely to opt for virtual schooling in the future. | | |
| I excelled academically with the virtual experience compared to in-person instruction. | | |
| I would like the district to expand its virtual offerings in the future. | | |

**(untitled)**

10. You will now see a sequence of scenarios, each with three school options that the school district could offer you in Fall 2022. For each set of three, indicate the one you prefer the most (Best) and the one you prefer the least (Worst).

Recall that a fully remote option is entirely virtual (100% remote) and traditional in-person instruction is 0% remote.

Travel time corresponds to the commute time in minutes from your home to the school. For traditional in-person instruction, students make the trip to school every day.

**Assume pandemic-related safety issues are as they were in 2019 before COVID.**

**Besides the characteristics shown, assume that these schools are otherwise identical in terms of their academic instruction and quality.**

There are no right or wrong answers to these questions. We only want to know which of the options you would most prefer.

*

| | | | |
|---|---|---|---|
| **Type of Instruction** | In Person | In Person | In Person |
| **Percent of students meeting state academic standards** | 90 | 60 | 30 |
| **Travel time to school (minutes)** | 75 | 30 | 15 |
| **Best** | ○ | ○ | ○ |
| **Worst** | ○ | ○ | ○ |

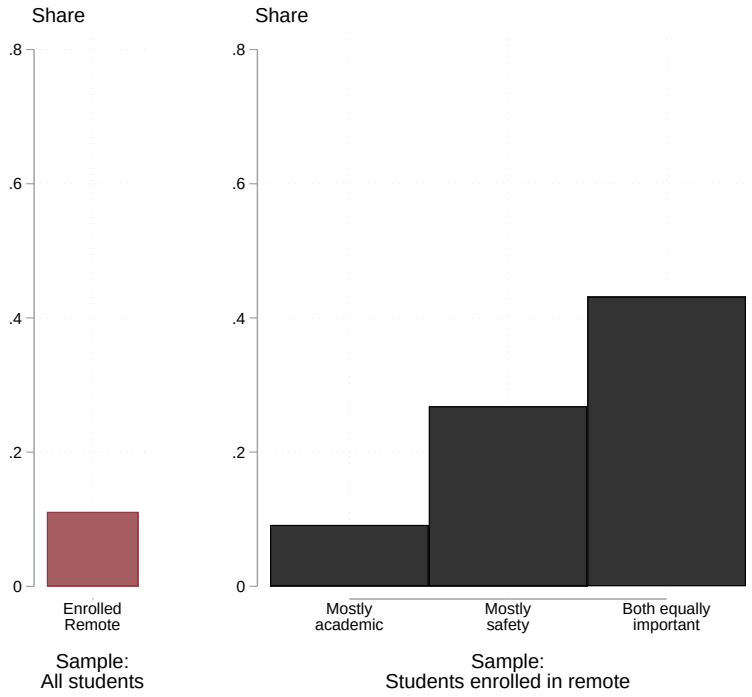
**(untitled)**

11. Do you think your choices will be similar in Fall 2023? *

- Yes
- No

# D Additional Survey Results and Preference Estimates

Figure D.1: Reasons for Enrolling in Remote Learning



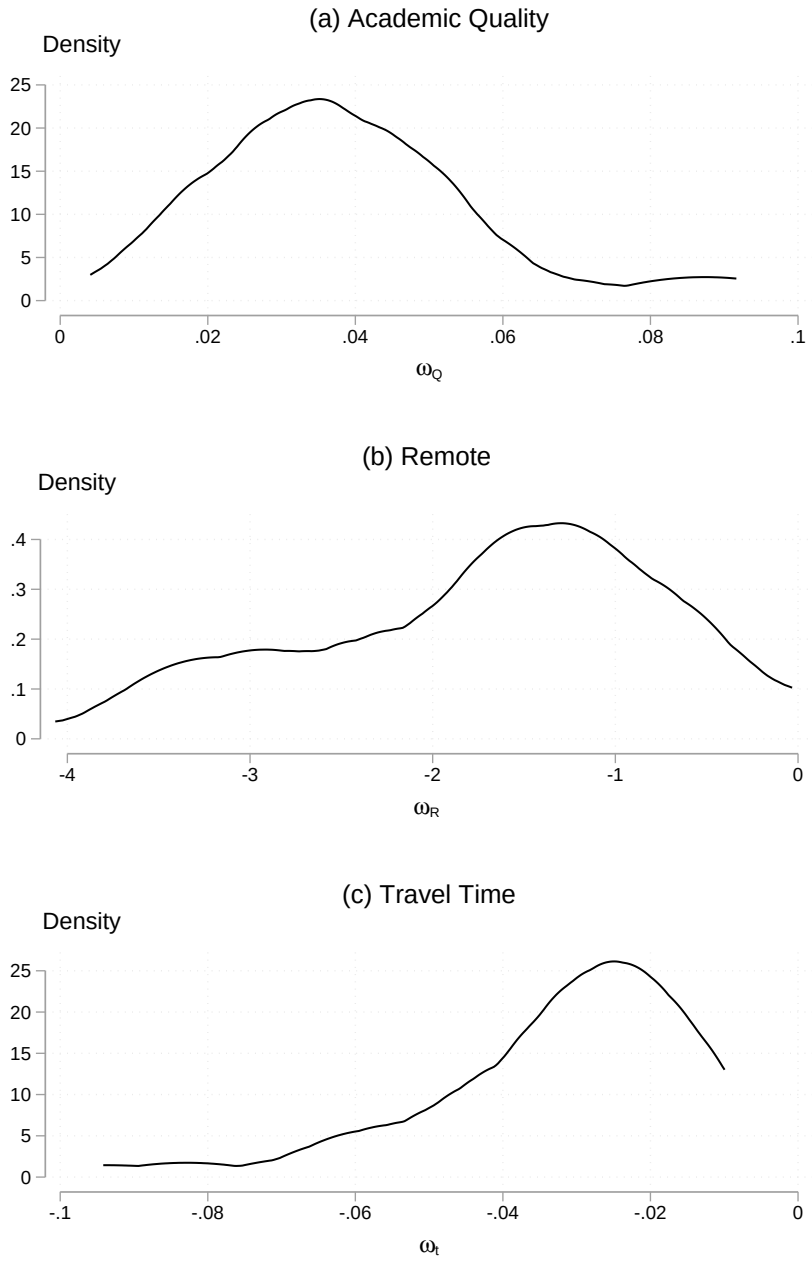

*Notes:* This figure reports various statistics relating to respondents' current remote learning status and their reasons for enrollment. The first red bar reports the share of respondents with students currently enrolled in remote learning. The next three bars report shares of respondents' reason for selecting remote learning, conditional on current remote status. Observations are weighted to produce means that correspond to the average family in LAUSD. In particular, we predict whether we observe a survey response and obtain a propensity score $p_i = p(X_i)$. We weight each observation by $w_i = \frac{P(Survey=1)}{p_i}$, where $P(Survey = 1)$ corresponds to the share of families with survey responses. Appendix Section C reports the actual survey questions.

Table D.1: Distribution of Cell-Level Preference Estimates

| | (1) Academic Quality ($\omega_Q$) | (2) Remote ($\omega_R$) | (3) Travel Time ($\omega_t$) | (4) WTT for 1 p.p. Quality Increase ($-\omega_Q/\omega_t$) |
|---|---|---|---|---|
| Mean | 0.039 | -1.704 | -0.033 | 1.45 |
| SD | 0.019 | 0.997 | 0.019 | 0.91 |
| Min | 0.004 | -4.066 | -0.094 | 0.04 |
| $p5$ | 0.012 | -3.268 | -0.072 | 0.53 |
| $p10$ | 0.020 | -3.130 | -0.058 | 0.60 |
| $p25$ | 0.025 | -2.262 | -0.038 | 0.98 |
| Median | 0.038 | -1.527 | -0.028 | 1.28 |
| $p75$ | 0.046 | -1.061 | -0.022 | 1.68 |
| $p90$ | 0.059 | -0.501 | -0.013 | 2.29 |
| $p95$ | 0.083 | -0.245 | -0.010 | 3.79 |
| Max | 0.092 | -0.032 | -0.010 | 4.60 |

*Notes:* This table summarizes the distribution across covariate cells of preference parameters estimated separately for each cell using the rank-ordered logit model. Covariate cells are defined by the complete interaction of a poverty status indicator, an indicator for being an underrepresented minority, a high-school indicator, and baseline math achievement indicators (three terciles plus a separate indicator for missingness). Columns (1)–(3) report the coefficients on academic quality, remote schooling, and travel time. Column (4) reports the willingness to travel (WTT) for an additional 1 percentage point increase in academic proficiency. Rows report the mean, standard deviation, minimum, selected percentiles, and maximum across cells. Every cell has a positive coefficient on academic quality and negative coefficients on remote schooling and travel time.

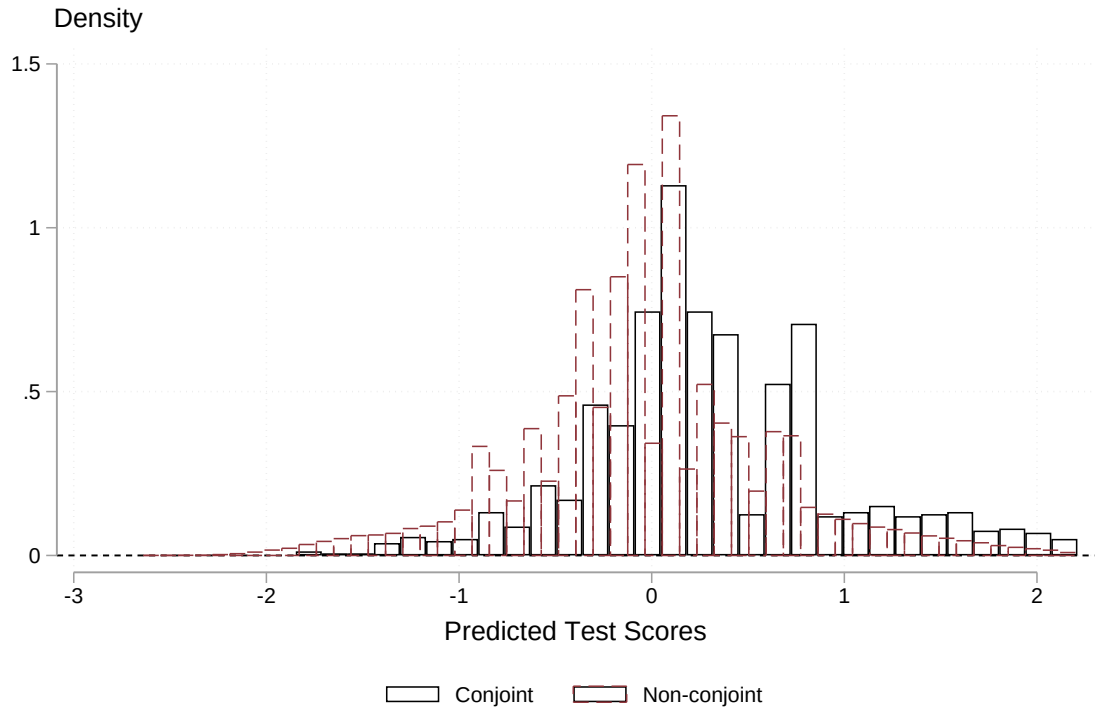
Figure D.2: Densities of Cell-Level Preference Estimates




*Notes:* This figure plots kernel density estimates of the preference parameters across the covariate cells used to estimate the rank-ordered logit model in Equation 9. Panel (a) reports the distribution of the estimated preference for academic quality ($\omega_Q$), Panel (b) reports the distribution of the estimated preference for remote schooling ($\omega_R$), and Panel (c) reports the distribution of the estimated preference for travel time ($\omega_t$). Summary statistics for these distributions, including the minimum and maximum of each parameter, are reported in Appendix Table D.1

# E Extrapolation, Overlap, and Common Support

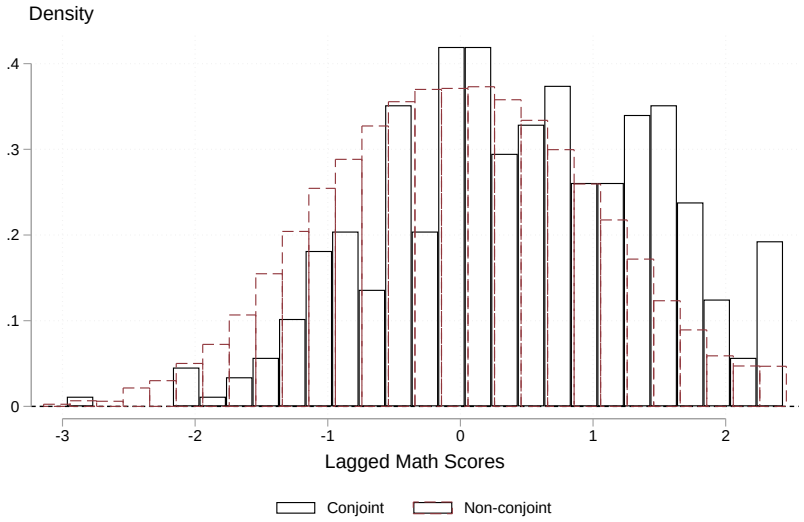
Figure E.1: Distributions of an Index of Baseline Characteristics



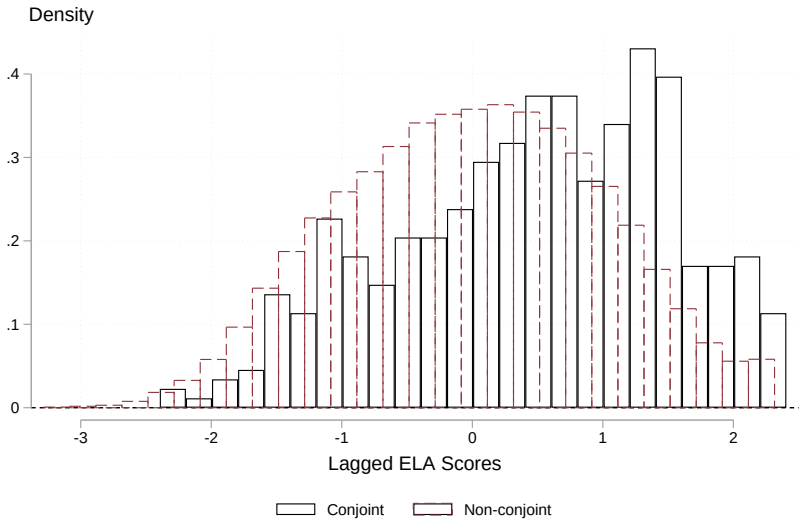

*Notes:* This figure reports the distribution of a summary index measure for the baseline covariates for students in the hypothetical choice and the general student samples. The summary index is constructed by regressing 2022 ELA test scores on an array of student characteristics including lagged (2019) achievement. The summary index corresponds to the predicted values from this regression. The histogram shows there is sufficient overlap between the hypothetical choice and the full LAUSD samples used in the empirical analysis.

Figure E.2: Test Score Overlap Between the LAUSD and Conjoint Samples

(a) Baseline Math Scores

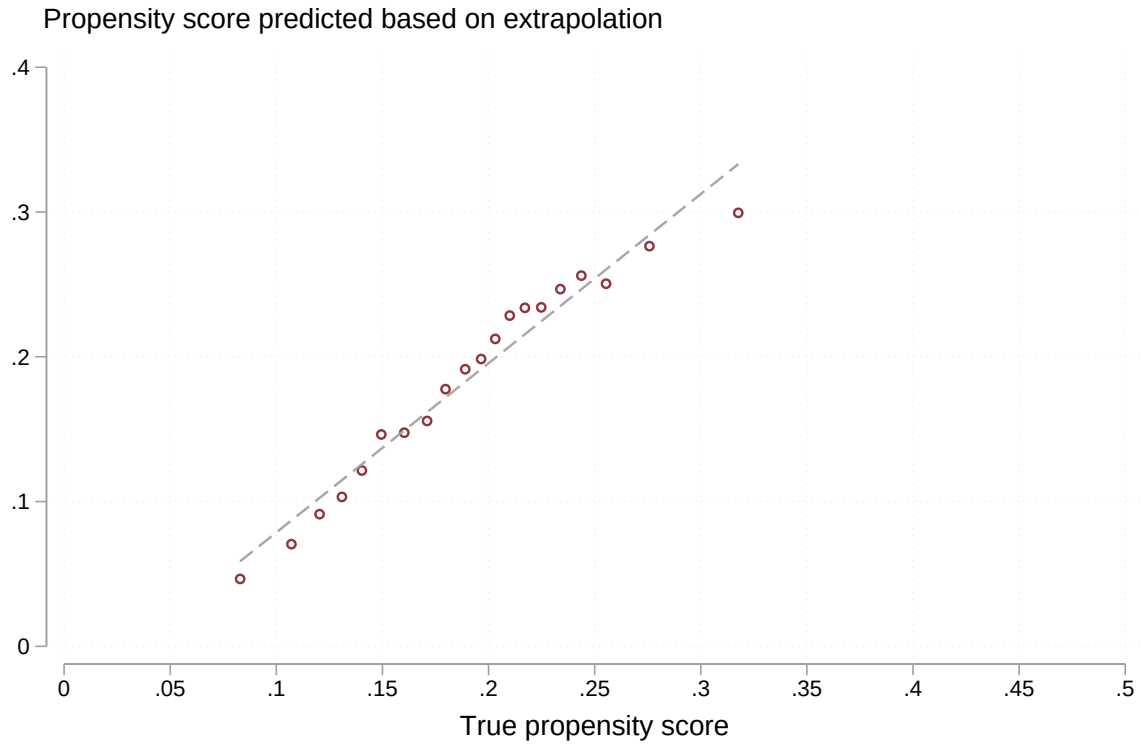



(b) Baseline ELA Scores




*Notes:* This figure reports the distribution of baseline math and ELA scores for all LAUSD students separately for conjoint respondents and non-respondents. The histogram demonstrates a substantial degree of overlap between the hypothetical choice and the full LAUSD samples used in the empirical analysis.

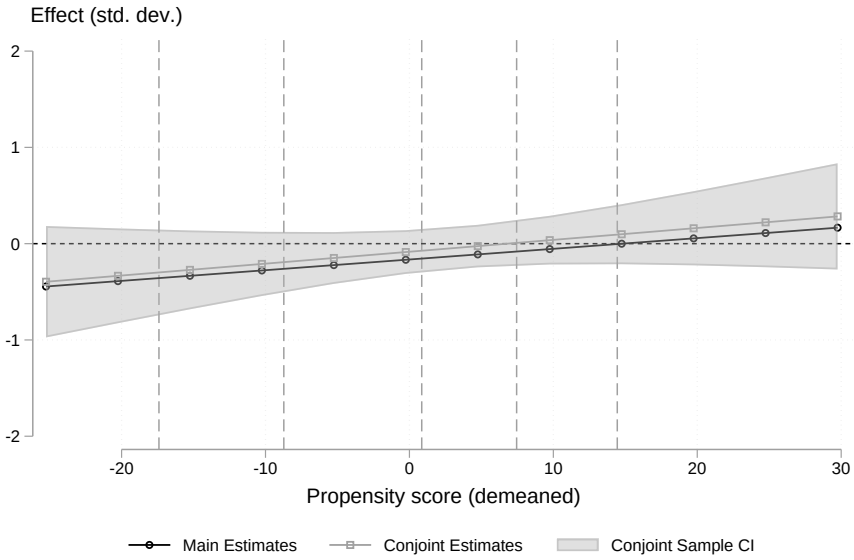
Figure E.3: Correlation Between Directly Estimated Propensity and Extrapolated Propensity Scores



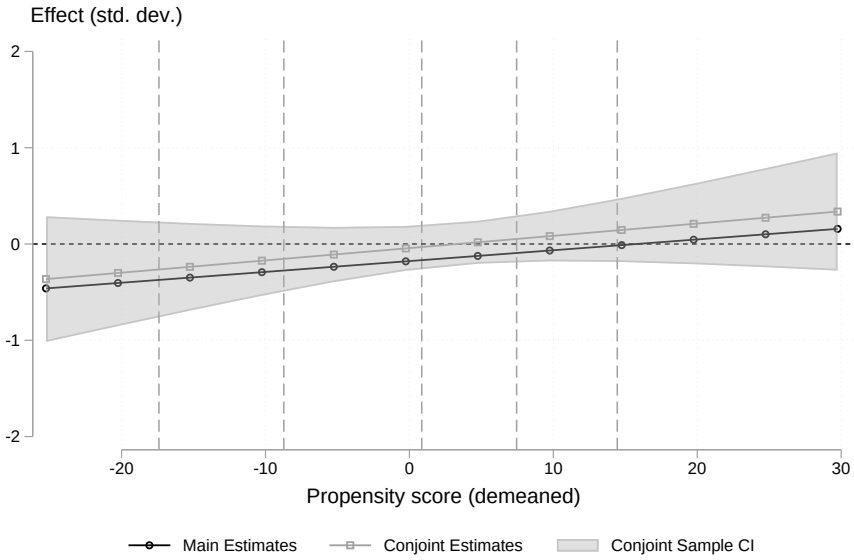

*Notes:* This figure compares two propensity scores that we construct to test the validity of our extrapolation approach. The two scores are estimated as follows. First, we create an estimation sample through stratified random sampling of one-third of the sample of hypothetical choice survey respondents. Our stratification ensures that the resulting estimation sample matches the average student's baseline characteristics. Using this estimation sample, we estimate preference parameters and construct propensity scores. Second, we return to the original survey hypothetical choice sample and use the residual set of respondents who were not included in the estimation sample. In this residual sample, we use our covariate cell approach to create a second set of preference estimates that we extrapolate to the estimation sample. The $x$-axis of the figure shows the directly estimated ("true") propensity scores that we estimate in the first step using the estimation sample. The $y$-axis of the figure shows the "predicted" propensity scores that we estimate for the estimation sample created by extrapolating the preference estimates from the residual sample.

Figure E.4: Robustness to Estimates Using Only the Conjoint Sample

(a) ELA

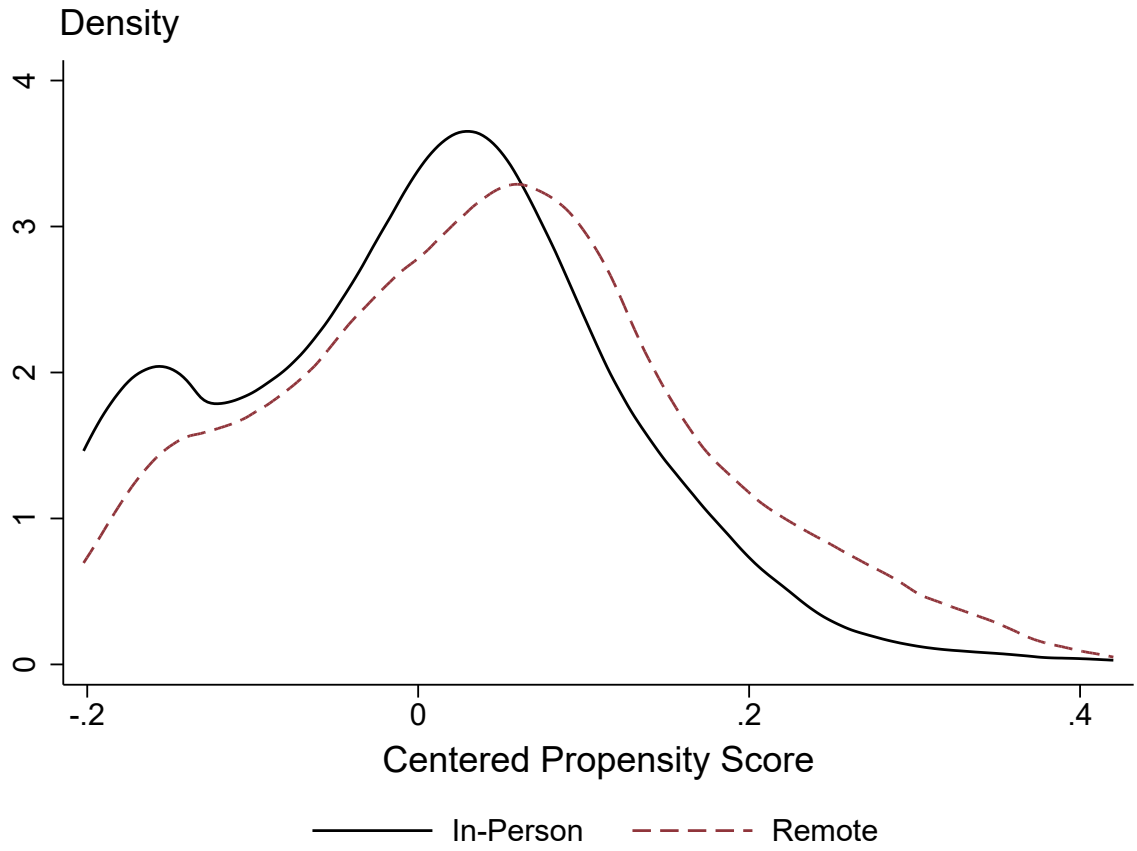



(b) Math




*Notes:* This figure reports separate estimates of treatment effects for 12 bins of estimated propensity scores using our main sample (which extrapolates preferences from the conjoint survey) and the conjoint sample. The confidence interval (shaded region) is for the estimates using only the conjoint sample. Panel (a) reports treatment effects on 2022 ELA and Panel (b) reports treatment effects on 2022 Math. Standard errors are clustered at the school level for the full-sample estimates and are heteroskedasticity robust for the conjoint-sample estimates.

Figure E.5: Common Support: Experimental Propensity Score




*Notes:* This figure reports kernel density estimates of the experimental propensity score $\hat{p}_i = \Lambda(\hat{v}_i)$ by treatment status. The propensity score is constructed using preference parameter estimates from the rank-ordered logit described in Section 6.2 and the cell-specific parameterization described in Section 7.1. The propensity score is demeaned. For legibility, the horizontal axis is truncated at the 99.9th percentile. The distinct distributions for the in-person and remote groups indicate that the experimental propensity score captures the latent demand heterogeneity that drives selection into remote learning.

# F Survey Responses and Covid Experience Heterogeneity

Although we asked survey respondents to remove the influence of Covid-related concerns from their stated choices, our preference estimates could still partly reflect residual COVID-19-related concerns. To assess this possibility, we generated new preference estimates by splitting the sample of choice survey respondents at the zip code level and generating geographic-specific estimates of willingness to pay measures. We correlate these zip-code-level preference estimates with measures from the COVID-19 Vulnerability and Recovery Index produced by Los Angeles County. For each area, the three index measures are intended to measure the risk, severity, and recovery need due to COVID-19.[39] In addition, we correlate the zip-code-level preferences with measures of local area case counts and deaths due to COVID-19.[40]

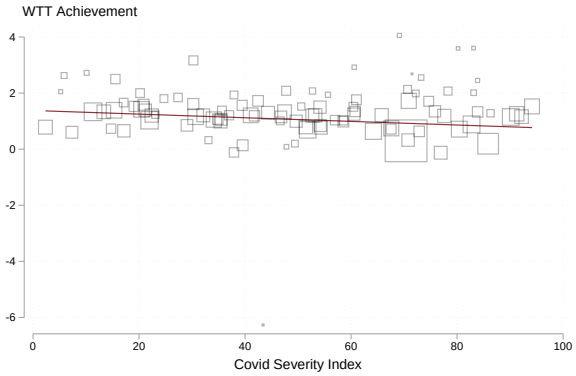
Appendix Figure F.1, Panels (a), (b), and (c) provide scatterplots of each zip code's estimated willingness to travel for academic quality and the three COVID-19 index measures. Each point's size is proportional to the number of respondents used to estimate preference parameters. To supplement these results, Panels (a) and (b) of Appendix Figure F.2 report similar plots for willingness to travel and measures of cases and deaths due to Covid. We report analogous results for estimated measures of preferences for remote schooling in Appendix Figures F.3 and Figure F.4. Overall, there is little visual evidence of a systematic relationship between preference parameters and either the Covid-related index measures or health outcomes at the zip code level. This provides reassuring evidence against the possibility that Covid-related concerns influence respondent choices in our survey.

[39]These measures were defined as follows. The risk measure is based on American Community Survey data from the U.S. Census Bureau on the share of individuals without U.S. citizenship, the share of the population below 200 percent of the federal poverty line, the share of overcrowded housing units, and the share of essential workers. The severity index is based on asthma hospitalization rates, the share of the population below 200 percent of the federal poverty line, the share of seniors aged 75 and over in poverty, the share of the population who is uninsured, heart disease hospitalization rates, and diabetes hospitalization rates. The recovery need index is based on the share of single-parent households, gun injury rates, the share of the population below 200 percent of the federal poverty line, the share of essential workers, the unemployment rate, and the share of the population who is uninsured. The data used for these analyses were downloaded from https://geohub.lacity.org/datasets/lacounty::covid-19-vulnerability-and-recovery-index/about.

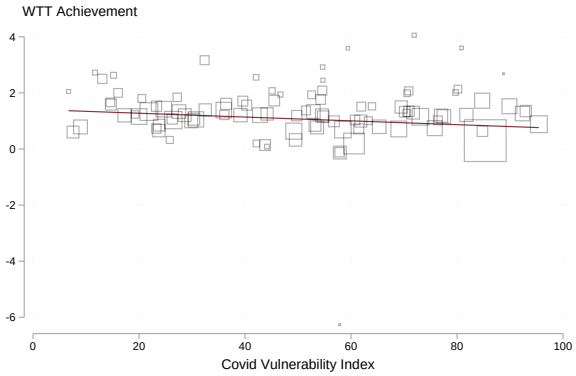
[40]The data used for these analyses were downloaded from http://publichealth.lacounty.gov/media/coronavirus/data.

Figure F.1: Preferences for Academic Quality and Covid Index Measures for Risk, Severity, and Recovery Need

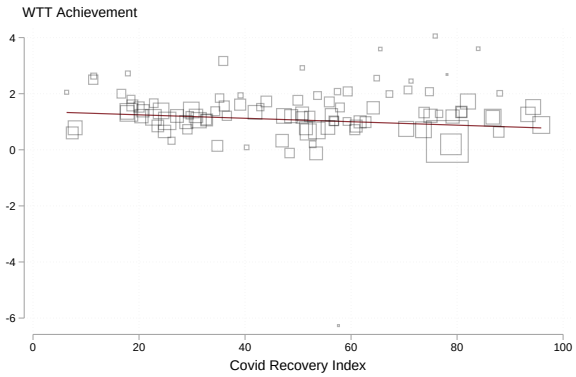
(a) Covid severity index




(b) Covid vulnerability index



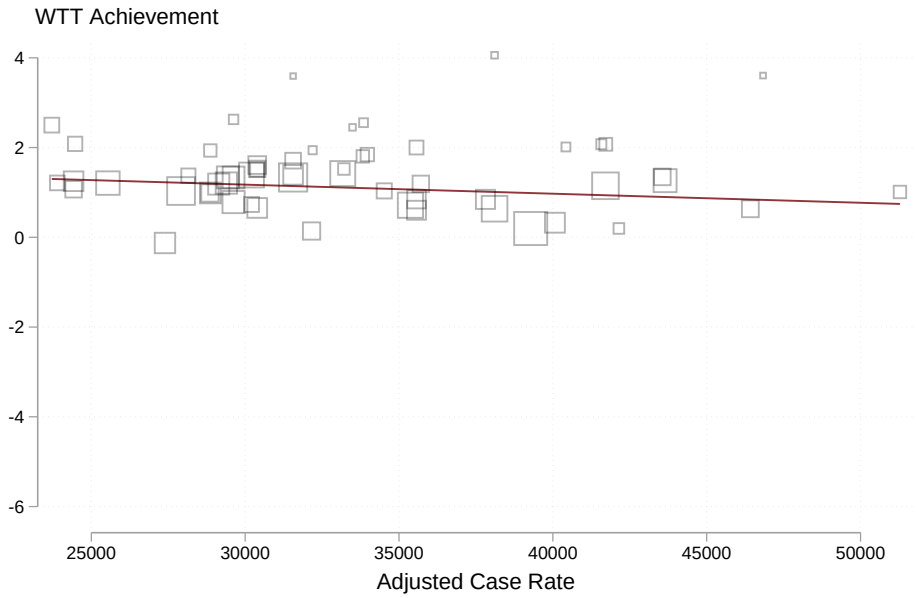

(c) Covid recovery need index



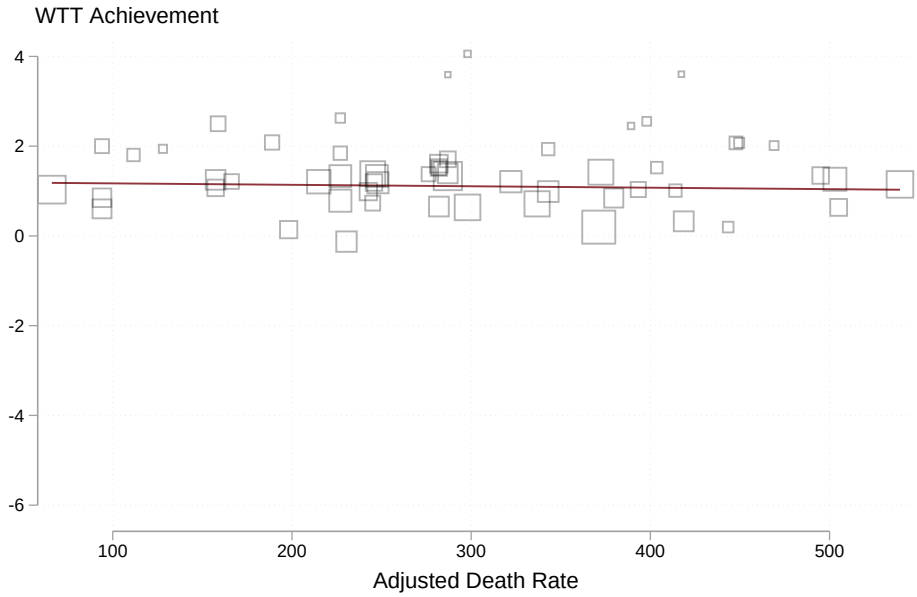

*Notes:* This figure presents scatterplots of zip-code-level mean willingness to travel (in minutes) for a one-percentage point increase in school academic proficiency ($y$-axis) and three measures from the COVID-19 Vulnerability and Recovery Index produced by Los Angeles County ($x$-axis). Panels (a), (b), and (c) present indices for the severity, vulnerability, and recovery need due to COVID-19, respectively. Each point's size is proportional to the number of respondents used to estimate preference parameters.

Figure F.2: Preferences for Academic Quality and Covid-Related Health Outcomes

(a) Covid cases



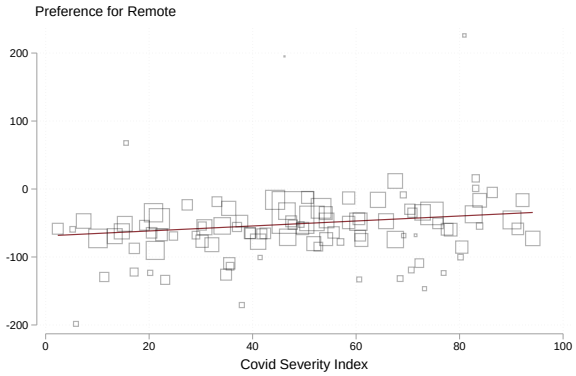

(b) Covid deaths



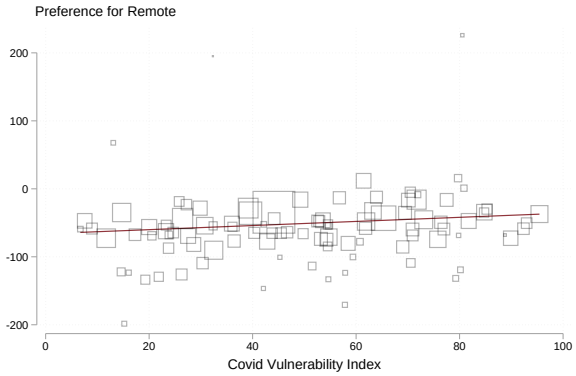

*Notes:* This figure presents scatterplots of zip-code-level mean willingness to travel for academic achievement ($y$-axis) and two measures of the severity of the COVID-19 pandemic on health outcomes in an area ($x$-axis). Panels (a) and (b) measure Covid health impact severity using case count and death measures, respectively. Each point's size is proportional to the number of respondents used to estimate preference parameters.

Figure F.3: Preferences for Remote Learning and Covid Index Measures for Risk, Severity, and Recovery Need

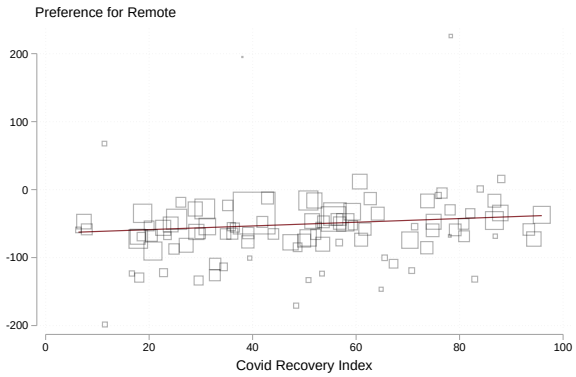
(a) Covid risk index




(b) Covid vulnerability index

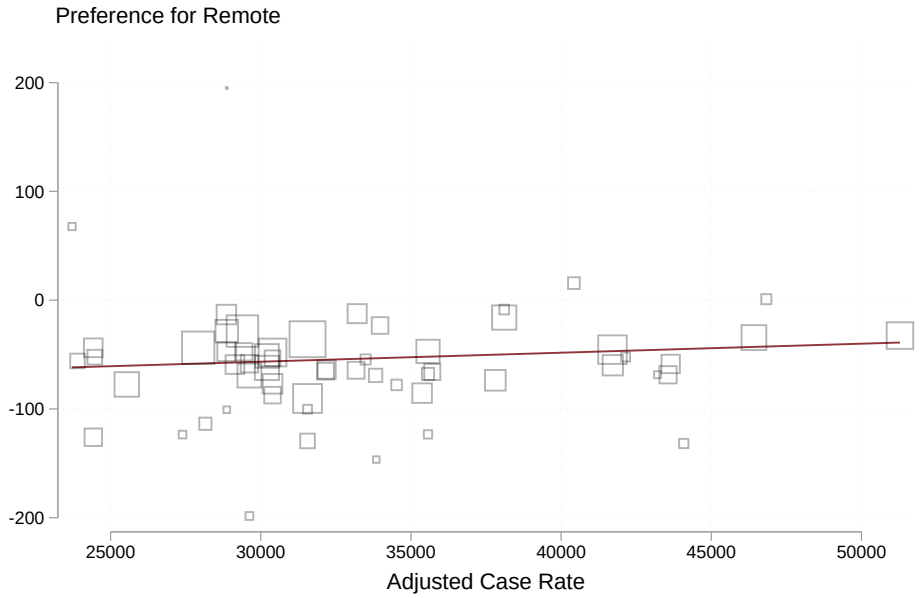



(c) Covid recovery need index

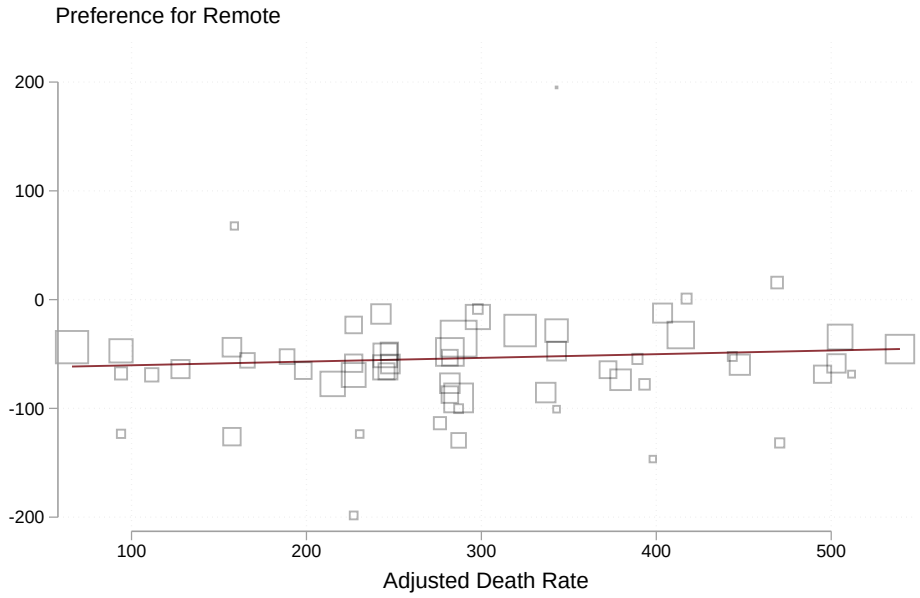



*Notes:* This figure presents scatterplots of zip-code-level measures of mean preferences for remote learning ($y$-axis) and three measures from the COVID-19 Vulnerability and Recovery Index produced by Los Angeles County ($x$-axis). Panels (a), (b), and (c) present indices for the risk, severity, and recovery need due to COVID-19, respectively. Preferences for remote learning are measured as the change in achievement needed to make a family indifferent between the remote and in-person schooling options. Each point's size is proportional to the number of respondents used to estimate preference parameters.

Figure F.4: Preferences for Remote Learning and Covid-Related Health Outcomes




(a) Covid cases




(b) Covid deaths

*Notes:* This figure presents scatterplots of zip-code-level measures of mean preferences for remote learning ($y$-axis) and two measures of the severity of the COVID-19 pandemic on health outcomes in an area ($x$-axis). Preferences for remote learning are measured as the change in achievement needed to make a family indifferent between the remote and in-person schooling options. Panels (a) and (b) measure Covid health impact severity using case count and death measures, respectively. Each point's size is proportional to the number of respondents used to estimate preference parameters.

# G Outcome Model Details, Robustness, and Heterogeneity

## G.1 Outcome Model Details and Additional Results

Table G.1: Detailed Summary of the Outcome Model Parameters

| | (1) ELA | (2) Math | (3) No Bullying Index | (4) Grit Index |
|---|---|---|---|---|
| Panel A: Main Model Estimates | | | | |
| ATE ($\beta$) | -0.167 | -0.180 | 0.142 | 0.003 |
| | (0.007) | (0.007) | (0.004) | (0.006) |
| Levels ($\theta$) | -0.273 | -0.267 | -0.035 | -0.051 |
| | (0.013) | (0.013) | (0.005) | (0.006) |
| Gains ($\psi$) | 0.111 | 0.112 | -0.006 | 0.053 |
| | (0.005) | (0.006) | (0.003) | (0.003) |
| Panel B: Additional Coefficients on Controls | | | | |
| Special Education | -0.634 | -0.576 | -0.056 | -0.162 |
| | (0.010) | (0.013) | (0.006) | (0.006) |
| English Learner | -0.312 | -0.232 | 0.040 | -0.047 |
| | (0.009) | (0.009) | (0.004) | (0.004) |
| Missing Data: English Learner | -0.368 | -0.406 | 0.045 | 0.001 |
| | (0.021) | (0.018) | (0.010) | (0.011) |
| Female | 0.164 | -0.077 | 0.039 | -0.043 |
| | (0.005) | (0.005) | (0.004) | (0.003) |
| Observations | 209,339 | 210,367 | 179,586 | 181,849 |

*Notes:* This table gives a more detailed summary of the estimated parameters from the outcome model. Each column represents a key outcome of interest. Panel A restates the average treatment effect estimates, selection on levels estimates, and selection on gains estimates from table 4. Panel B provides additional detail on the linear controls ($X_i$ from equation 13 in the main text), which include all other baseline student characteristics not used for extrapolating preferences. Our preferred specification includes these additional linear controls for completeness; however, they are not essential (see Appendix Table G.2 for estimates that omit these controls). Robust standard errors are reported in parentheses.

Table G.2: Robustness to Excluding Additional Controls

| | (1) ELA | (2) Math | (3) No Bullying Index | (4) Grit Index |
|---|---|---|---|---|
| Panel A: Main Specification (Includes Controls) | | | | |
| ATE ($\beta$) | -0.167 | -0.180 | 0.142 | 0.003 |
| | (0.007) | (0.007) | (0.004) | (0.006) |
| Levels ($\theta$) | -0.273 | -0.267 | -0.035 | -0.051 |
| | (0.013) | (0.013) | (0.005) | (0.006) |
| Gains ($\psi$) | 0.111 | 0.112 | -0.006 | 0.053 |
| | (0.005) | (0.006) | (0.003) | (0.003) |
| Observations | 209,339 | 210,367 | 179,586 | 181,849 |
| Panel B: Specification Without Controls | | | | |
| ATE ($\beta$) | -0.124 | -0.148 | 0.149 | 0.012 |
| | (0.007) | (0.007) | (0.004) | (0.006) |
| Levels ($\theta$) | -0.307 | -0.291 | -0.032 | -0.054 |
| | (0.014) | (0.014) | (0.005) | (0.006) |
| Gains ($\psi$) | 0.141 | 0.135 | -0.006 | 0.056 |
| | (0.006) | (0.006) | (0.003) | (0.003) |
| Observations | 209,339 | 210,367 | 179,586 | 181,849 |

*Notes:* This table shows robustness to excluding the linearly additive controls ($X_i$) from our baseline outcome model (see Equation 13). For ease of comparison, Panel A reproduces our key results from Table 4. Panel B reports estimates from the same sample using a specification that excludes the linearly additive controls.

Table G.3: Effects of Remote Learning on 2022-2023 Academic Year Outcomes

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Cognitive Outcomes | | | |
| ELA 2023 ($N = 163,075$) | -0.073 | -0.256 | 0.096 |
| | (0.007) | (0.013) | (0.006) |
| Math 2023 ($N = 162,921$) | -0.160 | -0.250 | 0.095 |
| | (0.007) | (0.013) | (0.006) |
| Panel B: Non-Cognitive Outcomes | | | |
| No Bullying Index 2023 ($N = 220,815$) | 0.060 | -0.018 | -0.006 |
| | (0.003) | (0.005) | (0.002) |
| No Physical Bullying 2023 ($N = 220,748$) | 0.105 | -0.021 | -0.005 |
| | (0.003) | (0.006) | (0.002) |
| No Online Bullying 2023 ($N = 215,569$) | 0.040 | -0.009 | 0.005 |
| | (0.003) | (0.005) | (0.003) |
| Grit Index 2023 ($N = 222,931$) | -0.024 | -0.031 | 0.022 |
| | (0.003) | (0.005) | (0.002) |

*Notes:* This table reports estimates of the effects of remote learning on cognitive and noncognitive outcomes as observed in the school year 2022-2023. Estimates are based on versions of the model specified in Equation 13. The remote enrollment treatment indicator is based on 2021-2022 year enrollment, as elsewhere in the paper. Panel (a) provides results on ELA and math achievement, and Panel (b) provides results on bullying-related outcomes and a grit index measured in 2022-2023. Index variables are normalized such that larger values denote better outcomes. Columns 1, 2, and 3 report estimates of the main effect of remote learning ($\beta$), which represent the average treatment effect (ATE), the selection on levels effect ($\theta$), and the selection on gains coefficient ($\psi$), respectively. Propensity scores are in units equal to 10 percent for interpretation reasons. Standard errors are robust and clustered at the school level.

Table G.4: Estimated Match Effects on Cognitive and Non-cognitive Outcomes by Percentile

| Percentile | (1) ELA | (2) Math | (3) No Bullying Index | (4) Grit Index |
|---|---|---|---|---|
| p5 | -0.381 | -0.396 | 0.154 | -0.099 |
| | (0.012) | (0.013) | (0.006) | (0.009) |
| p10 | -0.360 | -0.376 | 0.153 | -0.089 |
| | (0.011) | (0.013) | (0.006) | (0.008) |
| p25 | -0.264 | -0.278 | 0.147 | -0.043 |
| | (0.008) | (0.009) | (0.005) | (0.007) |
| p50 | -0.158 | -0.171 | 0.142 | 0.007 |
| | (0.007) | (0.007) | (0.004) | (0.006) |
| p75 | -0.084 | -0.096 | 0.138 | 0.042 |
| | (0.008) | (0.008) | (0.005) | (0.006) |
| p90 | -0.007 | -0.018 | 0.134 | 0.078 |
| | (0.011) | (0.010) | (0.006) | (0.007) |
| p95 | 0.045 | 0.035 | 0.131 | 0.103 |
| | (0.013) | (0.012) | (0.007) | (0.008) |
| N | 209,339 | 210,367 | 179,586 | 181,849 |

*Notes:* This table reports estimated treatment effects of remote learning on cognitive and non-cognitive outcomes measured in the 2021-2022 academic school year, at selected percentiles of the estimated propensity-score distribution. Effects are computed from the model in Equation 13 as $\hat{\beta} + \hat{\psi}\tilde{p}$, where the propensity score is demeaned so that the estimate at the mean corresponds to the average treatment effect for the average student. These estimates are the numeric counterpart to Figure 4, which plots the cognitive effects across the propensity-score distribution. Standard errors are robust and clustered at the school level.

Table G.5: Estimated Parameters of the Linear and Quadratic Models

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Math | | ELA | |
| Parameter Est. | Linear | Quadratic | Linear | Quadratic |
| $\beta$ | -0.18 | -0.171 | -0.167 | -0.159 |
| | (0.007) | (0.008) | (0.007) | (0.008) |
| $\theta_1$ | -0.267 | -0.304 | -0.273 | -0.305 |
| | (0.013) | (0.016) | (0.013) | (0.016) |
| $\psi_1$ | 0.112 | 0.088 | 0.111 | 0.09 |
| | (0.006) | (0.005) | (0.005) | (0.005) |
| $\theta_2$ | | 0.044 | | 0.037 |
| | | (0.0045) | | (0.0047) |
| $\psi_2$ | | 0.0028 | | 0.002 |
| | | (0.0029) | | (0.003) |
| Obs. ($N$) | 210,367 | 210,367 | 209,339 | 209,339 |

*Notes:* This table reports estimated parameters from our baseline linear specification according to Equation 13 (Column 1 and 3) and in an alternative quadratic specification of the form $Y_i = \alpha_c + \gamma' X_i + \beta D_i + \theta_1 P(\hat{v}_i) + \theta_2 P(\hat{v}_i)^2 + \psi_1 \left(P(\hat{v}_i) \times D_i\right) + \psi_2 \left(P(\hat{v}_i)^2 \times D_i\right) + \epsilon_i$ (Column 2 and 4). Our dependent variables are 2022 math achievement (Column 1 and 2) and 2022 ELA achievement (Column 3 and 4). Row 1 reports the coefficient on remote enrollment ($\beta$). In the linear specification, $\beta$ equals the average treatment effect because the propensity score is demeaned; in the quadratic specification, the average treatment effect is instead obtained by averaging the full treatment-effect function over the propensity-score distribution. Appendix Table G.6 reports the corresponding ATEs. Rows 2 and 3 report the linear selection-on-levels ($\theta_1$) and selection-on-gains ($\psi_1$) coefficients, respectively. Rows 4 and 5 report the corresponding quadratic terms ($\theta_2$ and $\psi_2$). Propensity scores are scaled so that one unit corresponds to 10 percentage points. Standard errors are robust and clustered at the school level.

Table G.6: Robustness of Average Treatment Effects to a Quadratic Specification

| | (1) ATE | (2) ATT | (3) ATU |
|---|---|---|---|
| Panel A: Preferred Linear Propensity Score Model | | | |
| ELA | -0.167 | -0.135 | -0.171 |
| | (0.007) | (0.007) | (0.007) |
| Math | -0.180 | -0.148 | -0.183 |
| | (0.007) | (0.007) | (0.007) |
| No Bullying Index 2022 | 0.142 | 0.140 | 0.143 |
| | (0.004) | (0.005) | (0.004) |
| Panel B: Alternative Quadratic Propensity Score Model | | | |
| ELA | -0.158 | -0.130 | -0.159 |
| | (0.007) | (0.007) | (0.007) |
| Math | -0.170 | -0.142 | -0.170 |
| | (0.007) | (0.007) | (0.007) |
| No Bullying Index 2022 | 0.146 | 0.140 | 0.146 |
| | (0.004) | (0.005) | (0.004) |

*Notes:* This table assesses whether the average treatment effect (ATE), average treatment effect on the treated (ATT), and average treatment effect on the untreated (ATU) are robust to allowing treatment-effect heterogeneity to vary quadratically, rather than linearly, in the propensity score. Panel A reports estimates from our baseline linear specification and reproduces the ATE, ATT, and ATU reported in Table 4 for convenience; Panel B reports the corresponding estimates from the quadratic specification. The underlying coefficient estimates for both specifications are reported in Table G.5. In the linear specification, the treatment-effect function is $\beta + \psi_1 P(\hat{v}_i)$; in the quadratic specification, it is $\beta + \psi_1 P(\hat{v}_i) + \psi_2 P(\hat{v}_i)^2$. Column (1) reports the average of the corresponding treatment effect function over the full analysis sample, Column (2) reports its average among students enrolled in remote learning, and Column (3) reports its average among students enrolled in in-person learning. These correspond to the ATE, ATT, and ATU, respectively. Estimates are shown for math, ELA, and bullying outcomes. Standard errors are robust, clustered at the school level, and computed via the delta method.

## G.2 Heterogeneity by Baseline Student Characteristics

Table G.7: Heterogeneous Impacts by Poverty Status

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Poverty | | | |
| ELA | -0.150 | -0.273 | 0.108 |
| | (0.007) | (0.013) | (0.006) |
| Math | -0.156 | -0.263 | 0.101 |
| | (0.007) | (0.013) | (0.006) |
| No Bullying Index | 0.136 | -0.027 | -0.012 |
| | (0.005) | (0.006) | (0.003) |
| Grit Index | 0.021 | -0.045 | 0.054 |
| | (0.006) | (0.007) | (0.004) |
| Panel B: Non-Poverty | | | |
| ELA | -0.298 | -0.290 | 0.192 |
| | (0.010) | (0.020) | (0.015) |
| Math | -0.332 | -0.310 | 0.214 |
| | (0.011) | (0.021) | (0.015) |
| No Bullying Index | 0.181 | -0.019 | 0.013 |
| | (0.009) | (0.016) | (0.011) |
| Grit Index | 0.005 | -0.050 | 0.114 |
| | (0.008) | (0.011) | (0.008) |

*Notes:* This table reports estimates of the effect of remote learning on cognitive and non-cognitive outcomes, allowing the selection parameters to vary by poverty status. Estimates are based on the model in Equation 13, with the propensity score covariates fully interacted with an indicator for economically disadvantaged status as classified by the district. Panels A and B report estimates separately for students classified in the poverty and non-poverty groups, respectively. Each panel reports results for ELA and math achievement, a no-bullying index, and a grit index, all measured in 2021-2022 and normalized so that larger values denote more beneficial outcomes. For this subgroup analysis, the propensity score is demeaned within each subgroup—relative to the subgroup's mean propensity computed over the population eligible for each outcome. Columns 1, 2, and 3 report the average treatment effect (ATE, equal to the coefficient $\beta$ on the remote indicator), the selection on levels coefficient ($\theta$), and the selection on gains coefficient ($\psi$), respectively. Propensity scores are scaled so that one unit corresponds to 10 percentage points. Standard errors are robust and clustered at the school level.

Table G.8: Heterogeneous Impacts by Baseline Academic Achievement

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Low Achievement Student | | | |
| ELA | -0.077 | -0.165 | 0.058 |
| | (0.010) | (0.014) | (0.007) |
| Math | -0.057 | -0.130 | 0.059 |
| | (0.009) | (0.013) | (0.006) |
| No Bullying Index | 0.140 | -0.022 | -0.047 |
| | (0.006) | (0.007) | (0.004) |
| Grit Index | 0.023 | -0.046 | 0.060 |
| | (0.006) | (0.008) | (0.005) |
| Panel B: High Achievement Students | | | |
| ELA | -0.215 | -0.147 | 0.060 |
| | (0.010) | (0.017) | (0.011) |
| Math | -0.182 | -0.167 | 0.118 |
| | (0.013) | (0.020) | (0.014) |
| No Bullying Index | 0.152 | -0.028 | 0.035 |
| | (0.005) | (0.007) | (0.006) |
| Grit Index | -0.039 | -0.040 | 0.186 |
| | (0.005) | (0.008) | (0.006) |

*Notes:* This table reports estimates of the effect of remote learning on cognitive and non-cognitive outcomes, allowing the selection parameters to vary by baseline achievement status. Estimates are based on the model in Equation 13, with the propensity score covariates fully interacted with an indicator for low baseline achievement, defined as baseline math achievement below the district average. Panels A and B report estimates separately for students classified as being in the baseline low- and high-achieving groups, with the high-achieving group consisting of students at or above the district average. Each panel reports results for ELA and math achievement, a no-bullying index, and a grit index, all measured in 2021-2022 and normalized so that larger values denote more beneficial outcomes. For this subgroup analysis, the propensity score is demeaned within each subgroup—relative to the subgroup's mean propensity computed over the population eligible for each outcome. Columns 1, 2, and 3 report the average treatment effect (ATE, equal to the coefficient $\beta$ on the remote indicator), the selection on levels coefficient ($\theta$), and the selection on gains coefficient ($\psi$), respectively. Propensity scores are scaled so that one unit corresponds to 10 percentage points. Standard errors are robust and clustered at the school level.

Table G.9: Heterogeneous Impacts by Gender

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Boys | | | |
| ELA | -0.170 | -0.273 | 0.132 |
| | (0.007) | (0.013) | (0.006) |
| Math | -0.206 | -0.273 | 0.128 |
| | (0.007) | (0.014) | (0.006) |
| No Bullying Index | 0.218 | -0.016 | 0.002 |
| | (0.005) | (0.006) | (0.003) |
| Grit Index | -0.057 | -0.045 | 0.161 |
| | (0.006) | (0.007) | (0.004) |
| Panel B: Girls | | | |
| ELA | -0.163 | -0.274 | 0.090 |
| | (0.007) | (0.013) | (0.006) |
| Math | -0.154 | -0.259 | 0.096 |
| | (0.007) | (0.013) | (0.006) |
| No Bullying Index | 0.111 | -0.035 | -0.017 |
| | (0.005) | (0.006) | (0.004) |
| Grit Index | 0.074 | -0.046 | -0.010 |
| | (0.006) | (0.007) | (0.004) |

*Notes:* This table reports estimates of the effect of remote learning on cognitive and non-cognitive outcomes, allowing the selection parameters to vary by gender status. Estimates are based on the model in Equation 13, with the propensity score covariates fully interacted with an indicator for female. Panels A and B report estimates for boys and girls, respectively. Each panel reports results for ELA and math achievement, a no-bullying index, and a grit index, all measured in 2021-2022 and normalized so that larger values denote more beneficial outcomes. For this subgroup analysis, the propensity score is demeaned within each subgroup—relative to the subgroup's mean propensity computed over the population eligible for each outcome. Columns 1, 2, and 3 report the average treatment effect (ATE, equal to the coefficient $\beta$ on the remote indicator), the selection on levels coefficient ($\theta$), and the selection on gains coefficient ($\psi$), respectively. Propensity scores are scaled so that one unit corresponds to 10 percentage points. Standard errors are robust and clustered at the school level.

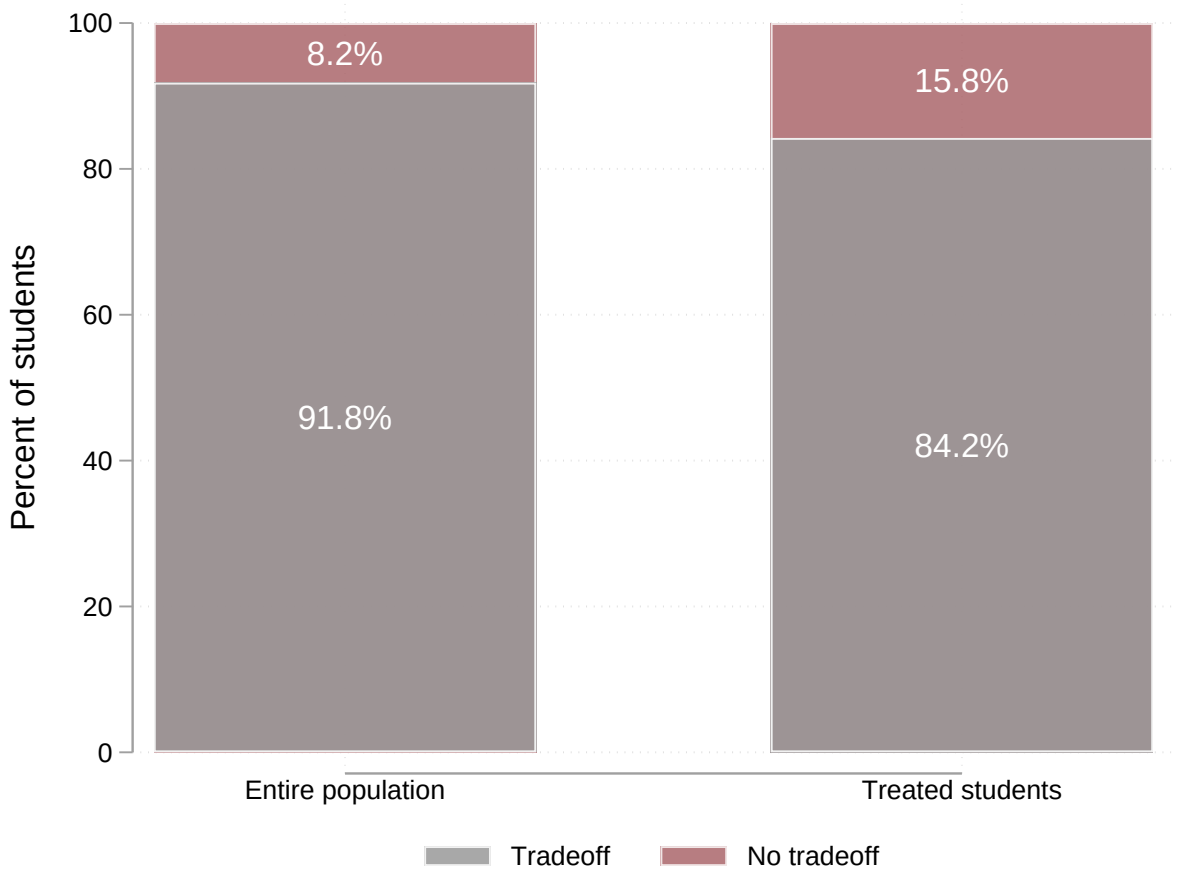
Table G.10: Heterogeneous Impacts by Grade Level

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Elementary | | | |
| ELA | -0.177 | -0.350 | 0.160 |
| | (0.008) | (0.015) | (0.009) |
| Math | -0.235 | -0.355 | 0.136 |
| | (0.008) | (0.016) | (0.009) |
| Panel B: Middle School | | | |
| ELA | -0.174 | -0.218 | 0.075 |
| | (0.011) | (0.016) | (0.007) |
| Math | -0.134 | -0.212 | 0.085 |
| | (0.011) | (0.017) | (0.008) |
| No Bullying Index | 0.174 | -0.031 | -0.039 |
| | (0.007) | (0.010) | (0.004) |
| Grit Index | 0.074 | -0.059 | 0.017 |
| | (0.006) | (0.010) | (0.004) |
| Panel C: High School | | | |
| ELA | -0.084 | -0.221 | 0.084 |
| | (0.019) | (0.023) | (0.011) |
| Math | -0.101 | -0.191 | 0.099 |
| | (0.016) | (0.018) | (0.010) |
| No Bullying Index | 0.135 | -0.022 | -0.003 |
| | (0.006) | (0.006) | (0.003) |
| Grit Index | 0.003 | -0.035 | 0.067 |
| | (0.008) | (0.008) | (0.004) |

*Notes:* This table reports estimates of the effect of remote learning on cognitive and non-cognitive outcomes, allowing the selection parameters to vary by grade groups, elementary, middle, and high school. Estimates are based on the model in Equation 13, with the propensity score covariates fully interacted with grade groups. Non-cognitive outcomes are available for middle and high school students only. Panel A reports estimates for elementary school students; Panel B reports estimates for middle school students; Panel C reports estimates for high school students. Each panel reports results for ELA and math achievement, and Panel B and C report results for a no-bullying index, and a grit index, all measured in 2021-2022 and normalized so that larger values denote more beneficial outcomes. For this subgroup analysis, the propensity score is demeaned within each subgroup—relative to the subgroup's mean propensity computed over the population eligible for each outcome. Columns 1, 2, and 3 report the average treatment effect (ATE, equal to the coefficient $\beta$ on the remote indicator), the selection on levels coefficient ($\theta$), and the selection on gains coefficient ($\psi$), respectively. Propensity scores are scaled so that one unit corresponds to 10 percentage points. Standard errors are robust and clustered at the school level.

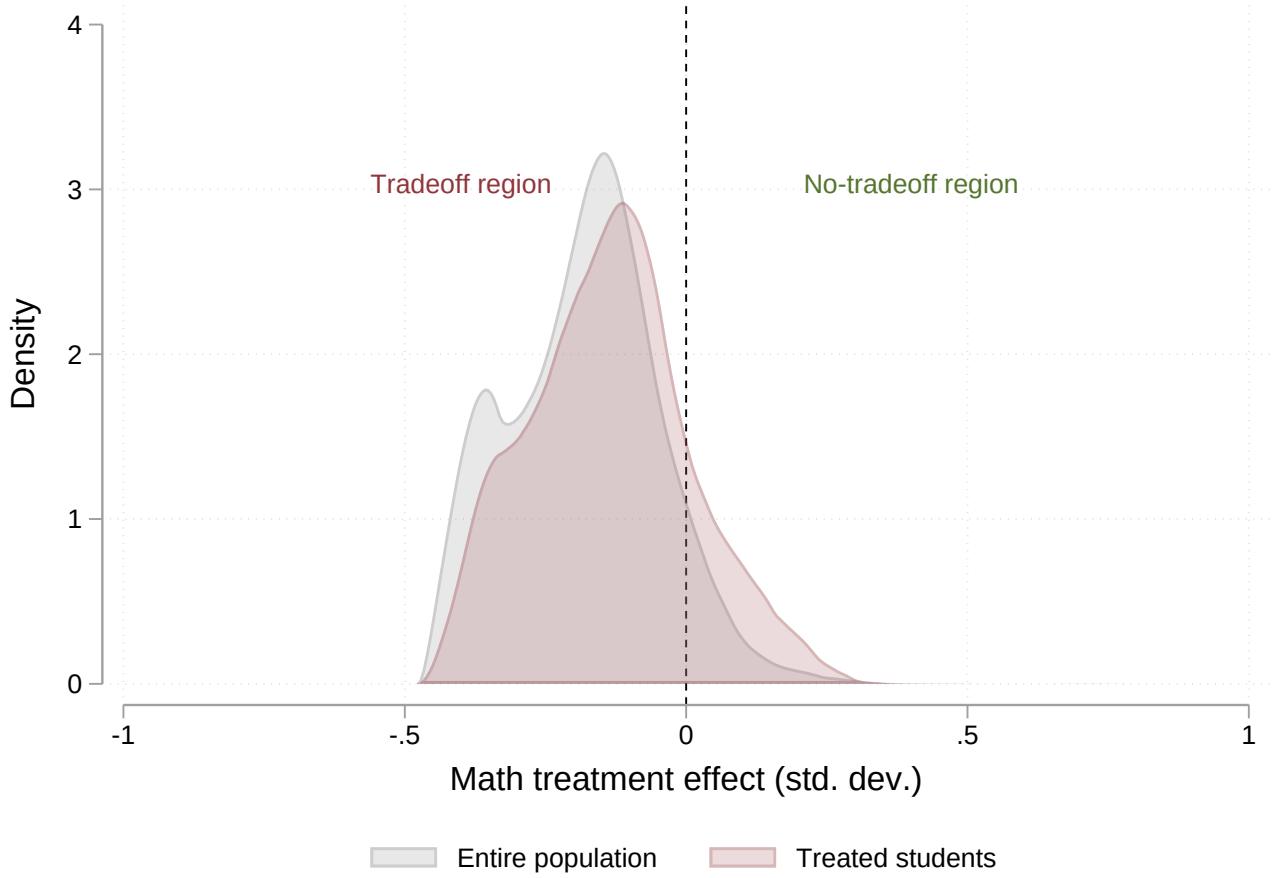
## G.3 Evidence on Compensating Differentials

Figure G.1: Share of Students Facing Tradeoff: Population and Treated




*Notes:* This figure reports the share of students facing a tradeoff between learning and bullying improvements. A student faces a tradeoff if their estimated learning effect (on Math) is negative and their estimated bullying improvement effect is positive. We note that no students in the population are estimated as having a negative bullying effect. The left bar reports the shares in the entire population of students, matching the visualization in Figure 4 showing that just under 10 percent of students have positive remote-learning effects, and roughly 90 percent have negative estimated remote-learning effects. The bar on the right reports the distribution among remote-learning students.

Figure G.2: Distribution of Math Effects and Compensating Differentials




*Notes:* This figure illustrates the density of estimated treatment effects for the entire population of students and remote students separately. Mass to the right of zero corresponds to students who do not face a tradeoff between learning and bullying improvements. Students to the left of zero experience negative academic effects and therefore face a tradeoff, with positive bullying effects potentially serving as a compensating differential.

## G.4 Functional Form Robustness

Table G.11: Robustness to the Functional Form of Preferences

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Non-linear Quality Preferences | | | |
| ELA | -0.170 | -0.239 | 0.108 |
| | (0.007) | (0.012) | (0.006) |
| Math | -0.184 | -0.235 | 0.112 |
| | (0.007) | (0.012) | (0.006) |
| No Bullying Index 2022 | 0.141 | -0.030 | -0.001 |
| | (0.004) | (0.004) | (0.003) |
| Grit Index 2022 | -0.009 | -0.046 | 0.070 |
| | (0.006) | (0.005) | (0.003) |
| Panel B: Non-linear Distance Costs | | | |
| ELA | -0.147 | -0.256 | 0.093 |
| | (0.007) | (0.013) | (0.005) |
| Math | -0.161 | -0.249 | 0.096 |
| | (0.007) | (0.013) | (0.005) |
| No Bullying Index 2022 | 0.145 | -0.029 | -0.005 |
| | (0.004) | (0.005) | (0.002) |
| Grit Index 2022 | -0.004 | -0.047 | 0.042 |
| | (0.006) | (0.005) | (0.002) |
| Panel C: Non-linear Quality Preferences and Distance Costs | | | |
| ELA | -0.151 | -0.224 | 0.093 |
| | (0.007) | (0.011) | (0.005) |
| Math | -0.166 | -0.219 | 0.098 |
| | (0.007) | (0.011) | (0.005) |
| No Bullying Index 2022 | 0.143 | -0.026 | -0.002 |
| | (0.004) | (0.004) | (0.002) |
| Grit Index 2022 | -0.014 | -0.042 | 0.052 |
| | (0.006) | (0.005) | (0.002) |

*Notes:* This table reports estimates of the effects on cognitive and noncognitive outcomes based on versions of the model specified in Equation 13. Each panel reports estimates from models that differ in the underlying model of preferences used to construct propensity scores. Panel (a) provides results from a model that allows for interaction between preferences for academic quality and remote learning. Panel (b) provides results from a model with non-linear (quadratic) distance costs, and Panel (c) provides results from a model that allows for both non-linear preferences for distance costs and interaction between preferences for academic quality and remote learning. Columns 1, 2, and 3 report estimates of the main effect of remote learning ($\beta$), which represent the average treatment effect (ATE), the selection on levels effect ($\theta$), and the selection on gains coefficient ($\psi$), respectively. Propensity scores are in units equal to 10 percent for interpretation reasons. Standard errors are robust and clustered at the school level.

Table G.12: Outcome Model Robustness to Alternative Preference Estimation Methods

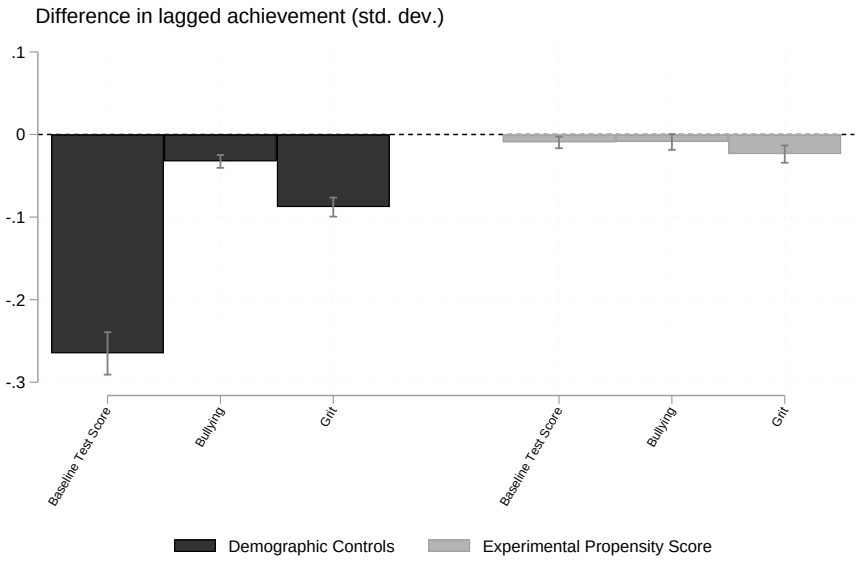
| | (1) Rank-Ordered Logit | (2) Conditional Logit | (3) Top-choice Logit |
|---|---|---|---|
| Panel A: ATE Estimates ($\beta$) | | | |
| ELA ($N$ = 209,339) | -0.167 | -0.166 | -0.160 |
| | (0.007) | (0.007) | (0.007) |
| Math ($N$ = 210,367) | -0.180 | -0.179 | -0.175 |
| | (0.007) | (0.007) | (0.007) |
| No Bullying Index ($N$ = 179,586) | 0.142 | 0.141 | 0.143 |
| | (0.004) | (0.004) | (0.004) |
| No Physical Bullying ($N$ = 179,496) | 0.264 | 0.263 | 0.263 |
| | (0.005) | (0.005) | (0.005) |
| No Online Bullying ($N$ = 175,159) | 0.162 | 0.160 | 0.169 |
| | (0.005) | (0.005) | (0.005) |
| Grit Index ($N$ = 181,849) | 0.003 | 0.005 | 0.010 |
| | (0.006) | (0.006) | (0.005) |
| Panel B: Levels ($\theta$) | | | |
| ELA ($N$ = 209,339) | -0.273 | -0.277 | -0.232 |
| | (0.013) | (0.013) | (0.011) |
| Math ($N$ = 210,367) | -0.267 | -0.271 | -0.226 |
| | (0.013) | (0.014) | (0.011) |
| No Bullying Index ($N$ = 179,586) | -0.035 | -0.037 | -0.030 |
| | (0.005) | (0.005) | (0.004) |
| No Physical Bullying ($N$ = 179,496) | -0.041 | -0.042 | -0.036 |
| | (0.006) | (0.006) | (0.005) |
| No Online Bullying ($N$ = 175,159) | -0.016 | -0.017 | -0.014 |
| | (0.005) | (0.006) | (0.005) |
| Grit Index ($N$ = 181,849) | -0.051 | -0.052 | -0.043 |
| | (0.006) | (0.006) | (0.005) |
| Panel C: Gains ($\psi$) | | | |
| ELA ($N$ = 209,339) | 0.111 | 0.114 | 0.089 |
| | (0.005) | (0.006) | (0.004) |
| Math ($N$ = 210,367) | 0.112 | 0.115 | 0.090 |
| | (0.006) | (0.006) | (0.004) |
| No Bullying Index ($N$ = 179,586) | -0.006 | -0.002 | -0.005 |
| | (0.003) | (0.003) | (0.002) |
| No Physical Bullying ($N$ = 179,496) | -0.014 | -0.012 | -0.008 |
| | (0.003) | (0.003) | (0.002) |
| No Online Bullying ($N$ = 175,159) | -0.021 | -0.013 | -0.025 |
| | (0.003) | (0.003) | (0.002) |
| Grit Index ($N$ = 181,849) | 0.053 | 0.054 | 0.027 |
| | (0.003) | (0.003) | (0.002) |

*Notes:* This table reports estimates from the outcome model using propensity scores constructed from alternative methods of estimating household preferences. Column (1) reports results using the baseline propensity score specification. Column (2) reports results using propensity scores estimated following the conditional logit approach of Chamberlain (1980), implemented using Stata's `clogit` command (StataCorp, 2025). Because the `clogit` implementation is designed for sequences of binary choices, while our survey experiment presented respondents with multiple options, this specification first converts each three-option decision into a series of pairwise choices. Column (3) reports a second robustness check that instead treats each offered option as a separate observation and codes the outcome as equal to one when the option is ranked most preferred. Panel A reports the estimated ATE ($\beta$), Panel B reports selection on levels parameter ($\theta$), and Panel C reports selection on gains parameter ($\psi$). Standard errors are robust and clustered at the school level.

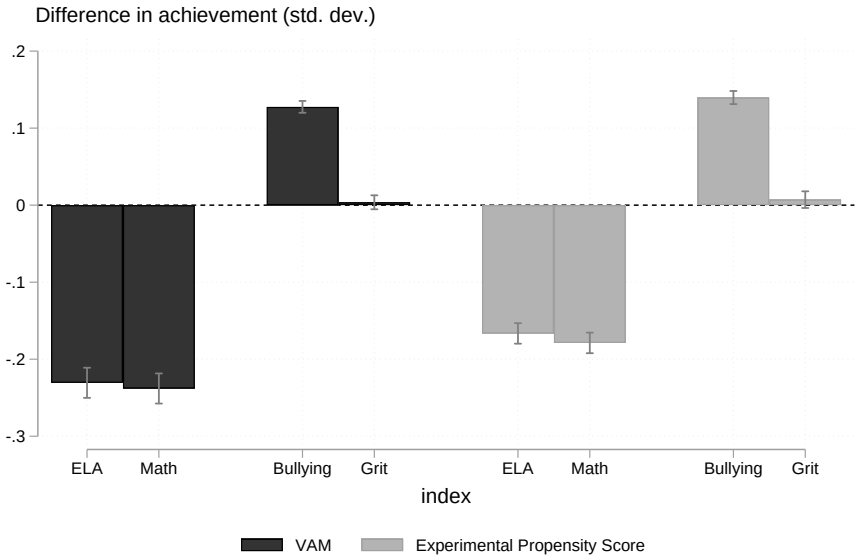
## G.5 Bootstrapped Estimates

Figure G.3: Baseline Balance and the Average Effects of Remote Learning (Bootstrap Version)

(a) Student baseline characteristics

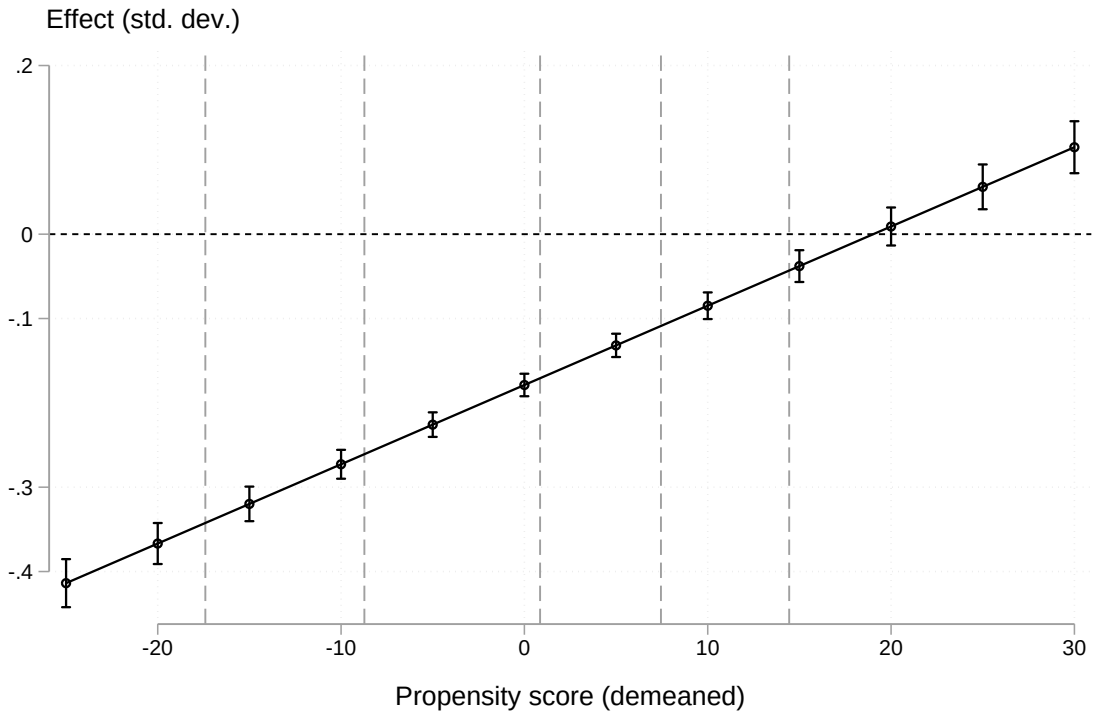



(b) Average effects on post-pandemic (2022) cognitive and non-cognitive outcomes




*Notes:* This figure reports estimates similar to those in Figure 3 and Table 4 but instead provides estimates and confidence intervals obtained through a bootstrapping procedure. To address estimation error in the propensity score estimation, we use a parametric bootstrap. We draw 250 sets of utility weight estimates for each covariate cell from the joint normal distribution with the mean and variance-covariance matrix obtained in the initial estimation step. We then estimate the corresponding regressions 250 times. Finally, we report the mean parameter estimates and the 95 percent confidence region obtained in the bootstrapping procedure.

Figure G.4: Estimated Match Effects on Post-Pandemic Math Achievement (Bootstrap Version)




*Notes:* This figure reports estimates similar to those in Figure 4 but instead provides estimates and confidence intervals obtained through a bootstrapping procedure. To address estimation error in the propensity score estimation, we use the parametric bootstrap. We draw 250 sets of utility weight estimates for each covariate cell from the joint normal distribution with the mean and variance-covariance matrix obtained in the initial estimation step. We then estimate the corresponding regressions and associated linear combination of the parameter estimates 250 times. Finally, we report the mean parameter estimates and the 95 percent confidence region obtained in the bootstrapping procedure.

Table G.13: Effects of Remote Learning (Bootstrap Version)

| | (1) ATE ($\beta$) | (2) Levels ($\theta$) | (3) Gains ($\psi$) |
|---|---|---|---|
| Panel A: Cognitive Outcomes | | | |
| ELA ($N = 209,339$) | -0.167 | -0.258 | 0.093 |
| | (0.007) | (0.013) | (0.005) |
| Math ($N = 210,367$) | -0.179 | -0.253 | 0.094 |
| | (0.007) | (0.013) | (0.005) |
| Panel B: Non-Cognitive Outcomes | | | |
| No Bullying Index ($N = 179,586$) | 0.140 | -0.032 | -0.003 |
| | (0.004) | (0.005) | (0.002) |
| No Physical Bullying ($N = 179,496$) | 0.259 | -0.038 | -0.006 |
| | (0.005) | (0.005) | (0.002) |
| No Online Bullying ($N = 175,159$) | 0.161 | -0.015 | -0.019 |
| | (0.005) | (0.005) | (0.003) |
| Grit Index ($N = 181,849$) | 0.007 | -0.047 | 0.032 |
| | (0.006) | (0.005) | (0.003) |

*Notes:* This table reports estimates similar to those in Table 4 but instead provides estimates and standard errors obtained through a bootstrapping procedure. All outcomes are measured in the post-pandemic school year 2021-2022. To account for estimation error in the propensity score estimation, we use a parametric bootstrap. We draw 250 sets of utility weight estimates for each covariate cell from the joint normal distribution with the mean and variance-covariance matrix obtained in the initial estimation step. We then estimate the corresponding regressions and associated linear combination of the parameter estimates 250 times. Last, we report the mean parameter estimates and the standard errors (in parentheses) obtained in the bootstrapping procedure.

## G.6 Lottery-based Validation Results

Table G.14: Lottery Balance and Attrition

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | | Lottery Sample | | | |
| | Full Sample | All | No Offer | Offer | Mean Diff. |
| Panel A: Balance Measures | | | | | |
| Baseline ELA Scores | 0.00 | 0.16 | 0.22 | 0.15 | -0.07 |
| | | | | | (0.05) |
| Baseline Math Scores | 0.00 | 0.17 | 0.23 | 0.16 | -0.06 |
| | | | | | (0.05) |
| Female | 0.48 | 0.49 | 0.48 | 0.49 | 0.00 |
| | | | | | (0.03) |
| Special Education | 0.14 | 0.09 | 0.07 | 0.09 | 0.02 |
| | | | | | (0.02) |
| URM | 0.82 | 0.63 | 0.66 | 0.63 | -0.03 |
| | | | | | (0.03) |
| Poverty | 0.83 | 0.66 | 0.65 | 0.66 | 0.01 |
| | | | | | (0.03) |
| Observations | 288,879 | 2,654 | 304 | 2,350 | 2,654 |
| Panel B: Post-Lottery Attrition | | | | | |
| Has Math Scores | | | 0.608 | 0.575 | -0.034 |
| | | | | | (0.031) |
| Has ELA Scores | | | 0.615 | 0.574 | -0.041 |
| | | | | | (0.031) |
| Has Either Scores | | | 0.621 | 0.576 | -0.044 |
| | | | | | (0.031) |

*Notes:* This table reports summary statistics for all LAUSD students (Column 1), the lottery sample (Column 2), and the lottery sample split by offer status (Columns 3–4). Because offers are random only within lottery strata, Columns (3) and (4) report adjusted means so that the no-offer and offer groups share the lottery-strata composition of the full lottery sample while preserving the overall sample mean; their difference equals the lottery-strata-adjusted mean difference reported in Column (5). Column (5) reports this regression-adjusted offer-minus-no-offer difference, estimated from regressions of each covariate on an indicator for receiving an offer with lottery fixed effects and standard errors clustered by lottery. Panel A reports balance on baseline characteristics; Panel B reports post-lottery attrition, with Columns (3)–(5) constructed as in Panel A. Stars in Column 5 denote statistical significance: * $p<0.10$, ** $p<0.05$, *** $p<0.01$. Baseline scores are standardized lagged test scores; demographic variables are indicators.

Table G.15: Lottery-Based Tests for Bias in Remote Learning Estimates

| | (1) Math | (2) ELA | (3) Pooled |
|---|---|---|---|
| Forecast Coefficient | 1.01 | 0.68 | 0.85 |
| | (0.18) | (0.20) | (0.14) |
| Forecast-Unbiasedness $p$-value | [0.94] | [0.11] | [0.28] |
| First Stage $F$-statistic | 43.87 | 39.25 | 41.60 |
| Overidentification $p$-value | 0.45 | 0.16 | 0.30 |
| Observations | 1,246 | 1,247 | 2,493 |

*Notes:* This table reports estimates of the lottery-based validation exercise discussed in Section 8.5. The first row reports the estimated forecast coefficient, $\phi_2$, from Equation 14 for both Math and ELA 2021-2022 achievement. A forecast coefficient estimate equal to 1 indicates that the lottery-induced variation in observed achievement matches the variation predicted by our preferred model. The forecast-unbiasedness row reports $p$-values from tests of the null hypothesis $H_0 : \phi_2 = 1$. The table also reports first-stage $F$-statistics and $p$-values from overidentification tests, which assess whether the lottery-specific IV variation is consistent with a common forecast coefficient across lottery instruments. For math, ELA, and pooled specifications, we fail to reject the associated cross-lottery consistency restrictions. Standard errors are reported in parentheses and are heteroskedasticity robust.